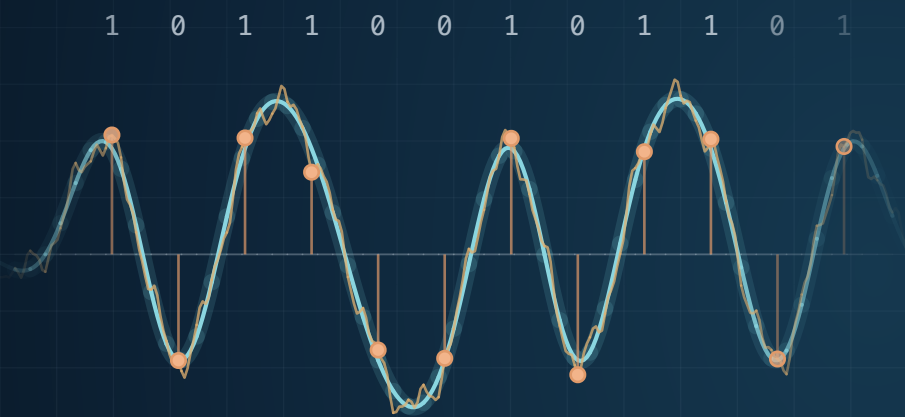




SAMPLING · DETECTION · MODULATION · CODING

Digital Communications

Lecture Notes



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Course conventions

These notes are written for undergraduates. They assume Fourier analysis, probability and random processes. Read the plain explanation before the mathematics. Worked examples use five headings: Given, Find, Method, Solution, and Check.

Noise is white and Gaussian with **two-sided** power spectral density $N_0/2$ watts per hertz. The Gaussian tail is $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-t^2/2} dt = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$. Energy and power use $R = 1 \Omega$.

A **PS** marker points to the related section of Proakis and Salehi, *Fundamentals of Communication Systems*, second edition. The textbook chapter numbers differ from the course chapter numbers.

The transition from analog to digital

A continuous waveform becomes a bit stream in three steps. Sampling makes time discrete. Quantization makes amplitude discrete. Encoding replaces each quantized amplitude with bits. Sampling can be reversed. Quantization cannot.

1.1 Impulse-train sampling

Sampling every T_s seconds is multiplication by a train of impulses. Write $f_s = 1/T_s$ for the sampling frequency.

THE IDEAL SAMPLED SIGNAL

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

$$g_\delta(t) = g(t)p(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s)$$

The second line uses the sampling property $g(t)\delta(t - t_0) = g(t_0)\delta(t - t_0)$, which turns the product under the sum into a number.

The signal g_δ is continuous in time, not a sequence of numbers. Therefore, it has a Fourier transform. A product in time becomes a convolution in frequency:

$$P(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} \delta(f - nf_s) = f_s \sum_{n=-\infty}^{\infty} \delta(f - nf_s)$$

The Fourier-series coefficient of $p(t)$ is the same for every harmonic: $a_k = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi k f_0 t} dt = \frac{1}{T_s}$. Now convolve G with the impulse train:

$$G_\delta(f) = G(f) * P(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} G(f) * \delta(f - nf_s)$$

A shifted impulse shifts the function during convolution. Write the convolution integral and use the even property $\delta(-u) = \delta(u)$:

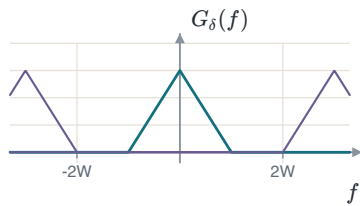
$$\begin{aligned} G(f) * \delta(f - nf_s) &= \int G(\theta) \delta(f - nf_s - \theta) d\theta \\ &= \int G(\theta) \delta(-[\theta - (f - nf_s)]) d\theta \\ &= \int G(\theta) \delta(\theta - (f - nf_s)) d\theta \\ &= G(f - nf_s) \end{aligned}$$

The last line uses the sifting property. Each impulse in $P(f)$ places one copy of G at a multiple of f_s .

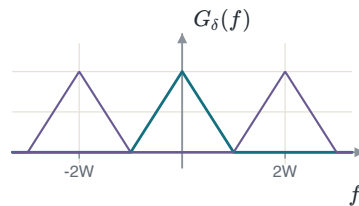
SAMPLING REPLICATES THE SPECTRUM

$$G_{\delta}(f) = f_s \sum_{n=-\infty}^{\infty} G(f - nf_s)$$

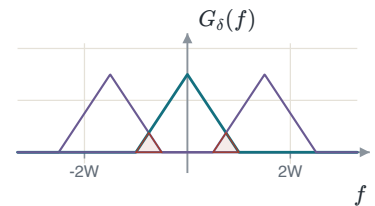
Sampling copies the spectrum to every multiple of f_s and scales it by f_s . Nothing is lost provided the copies do not overlap.



$f_s > 2W$: a gap between the replicas.



$f_s = 2W$: they touch and do not overlap.



$f_s < 2W$: they overlap, and the overlap cannot be undone.

ALIASING

When the replicas overlap, a high message frequency combines with a low frequency from a replica. No filter can separate these components. The samples no longer determine the original signal. A real signal is not strictly bandlimited, so a **guard band** f_g separates the message and the first replica. Then $f_s = 2W + f_g$. An anti-aliasing filter removes frequencies above W before sampling.

A camera samples in space. An image N pixels wide holds at most $N/2$ cycles across its width without aliasing, two pixels a cycle. A finer pattern, such as a striped shirt, folds back into a coarser false pattern called **moiré**. Many cameras put a slight blur in front of the sensor for the same reason an anti-aliasing filter precedes a sampler.

1.2 The sampling theorem and reconstruction

SAMPLING THEOREM

If $G(f) = 0$ for $|f| \geq W$ and $f_s \geq 2W$, then $g(t)$ is determined completely by its samples $g(nT_s)$ and can be recovered from them exactly. If $f_s < 2W$, aliasing occurs and the recovery fails. The rate $2W$ is the **Nyquist rate** and $T_s = 1/(2W)$ the **Nyquist interval**.

An ideal lowpass filter keeps the copy at the origin and rejects the other copies. At $f_s = 2W$:

THE RECONSTRUCTION FILTER AND ITS IMPULSE RESPONSE

$$H_{\text{LPF}}(f) = \begin{cases} \frac{1}{2W}, & |f| \leq W \\ 0, & \text{otherwise} \end{cases}$$

$$h_{\text{LPF}}(t) = \int_{-W}^W \frac{1}{2W} e^{j2\pi ft} df = \frac{\sin(2\pi Wt)}{2\pi Wt} = \text{sinc}(2Wt)$$

The course uses $\text{sinc}(x) = \sin(\pi x)/(\pi x)$. This function is one at zero and zero at every other integer.

RECONSTRUCTION-FILTER GAIN

Sampling scales the spectrum by $f_s = 2W$. The filter must remove this scale factor. A unit-gain filter returns a signal that is $2W$ times too large.

Filtering in frequency is convolution in time. Therefore, convolve the impulse train with h_{LPF} . Use τ as the integration variable:

THE CONVOLUTION INTEGRAL, AND THE SIFTING STEP

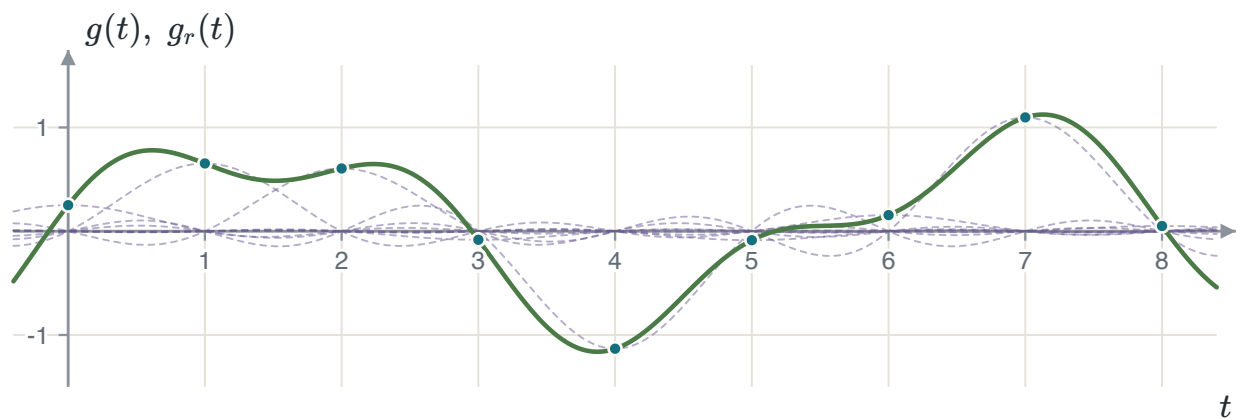
$$\begin{aligned} g_r(t) &= \int_{-\infty}^{\infty} \overbrace{\sum_{n=-\infty}^{\infty} g(nT_s) \delta(\tau - nT_s)}^{= g_s(\tau)} \text{sinc}(2W(t - \tau)) d\tau \\ g_r(t) &= \sum_{n=-\infty}^{\infty} g(nT_s) \underbrace{\int_{-\infty}^{\infty} \text{sinc}(2W(t - \tau)) \delta(\tau - nT_s) d\tau}_{= \text{sinc}(2W(t - nT_s))} \end{aligned}$$

The sum does not depend on τ , so it moves outside the integral. The sifting property then replaces τ with nT_s .

INTERPOLATION

$$\begin{aligned} g_r(t) &= \sum_{n=-\infty}^{\infty} g(nT_s) \text{sinc}(2W(t - nT_s)) \\ g_r(t) &= \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \text{sinc}(2Wt - n) \quad \text{at } f_s = 2W \end{aligned}$$

At $t = kT_s$, every term is zero except the term with $n = k$. Therefore, $g_r(kT_s) = g(kT_s)$. The same sum reconstructs the signal between the sample times.



Each sample contributes one shifted sinc scaled by its own value. Their sum, drawn heavy, passes through every sample because each sinc is zero at all the other sampling instants.

EXAMPLE 1.1 – THREE SAMPLING RATES

GIVEN A signal $x(t)$ bandlimited to $W = 40$ kHz.

FIND Find (a) the Nyquist rate, (b) the rate with a 10 kHz guard band, and (c) the Nyquist rate of $y(t) = x(t) \cos(80000\pi t)$.

METHOD Read (a) and (b) off the geometry of the replicas. For (c) find the bandwidth of y first: the rate follows from that, not from the bandwidth of x .

SOLUTION (a) $f_s = 2W = 80$ kHz. (b) $f_s = 2W + f_g = 90$ kHz. (c) The carrier is at $f_c = 40$ kHz, and multiplication shifts the spectrum both ways and halves it: $Y(f) = \frac{1}{2}X(f - 40k) + \frac{1}{2}X(f + 40k)$. The result occupies $|f| < 80$ kHz, so $f_s = 160$ kHz.

CHECK The highest frequency of y is $f_c + W = 80$ kHz. Therefore, its Nyquist rate is 160 kHz. The rate follows the highest frequency, not the message bandwidth.

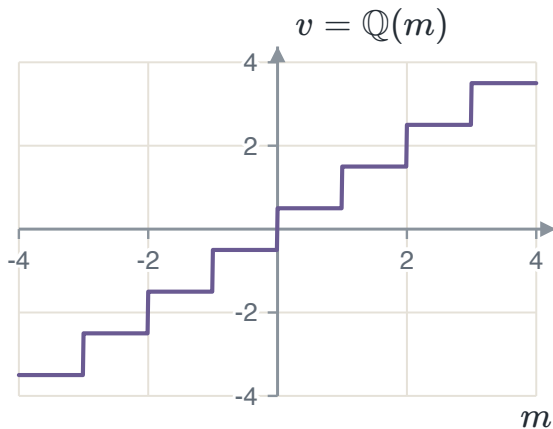
COMMON ERROR IN PART (C)

The rate 80 kHz applies to x , not to y . Modulation moves the highest frequency to 80 kHz. Therefore, y needs a sampling rate of 160 kHz.

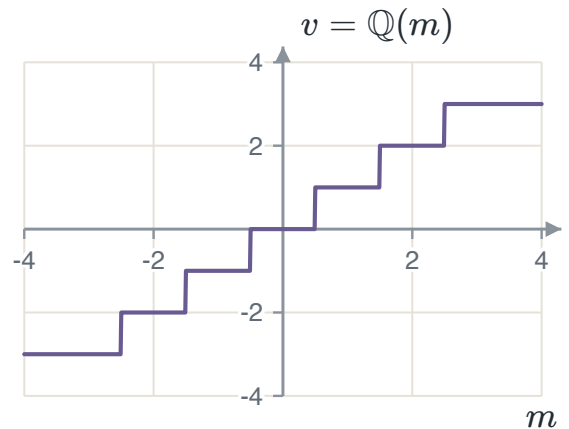
1.3 Quantization

Quantization replaces a sample amplitude with one of L **representation levels**. A **uniform** quantizer uses the same spacing Δ between consecutive levels.

A **mid-rise** quantizer has a decision boundary at zero, so it has no zero output level. A **mid-tread** quantizer has an output level at zero. Thus, a mid-tread quantizer reproduces a small input as zero.



Mid-rise, $L = 8$, $\Delta = 1$. A boundary at the origin.



Mid-tread, $L = 8$, $\Delta = 1$. A level at the origin.

A quantizer divides the input range into L regions $\mathcal{J}_k$. Each region has one output value v_k . Two conditions define an optimal quantizer:

1. Each boundary is the **midpoint** of its two adjacent levels: $m_k = \frac{1}{2}(v_{k-1} + v_k)$. This sends each input to the nearer level.
2. Each level is the **centroid** of its region: $v_k = E[M \mid M \in \mathcal{J}_k]$. This minimizes the mean-square error in that region.

The two conditions depend on each other, so apply them in turn. A uniform quantizer satisfies both conditions for a uniform input.

A quantizer spanning $[-m_{\max}, m_{\max}]$ makes two kinds of error. An input beyond $\pm m_{\max}$ is clipped to the outer level, and its error has no bound. This is **overload**. Inside the range the error stays within $\pm \Delta/2$. This is **granular noise**.

A wider range reduces overload but makes Δ and the granular noise larger. For a Gaussian input and 3 bits, the SQNR is largest at $m_{\max} = 2.34\sigma$, with 14.27 dB. At $m_{\max} = \sigma$ it falls to 6.97 dB, because about 32% of the samples are clipped.

COMMON ERROR

Set $m_{\max}$ from the rare peaks of the signal, not from its typical size. A range equal to one standard deviation clips a third of a Gaussian signal.

1.4 Quantization noise and the signal-to-noise ratio

The quantization error is $Q = M - \mathcal{Q}(M)$. Nearest-level quantization keeps this error between $-\Delta/2$ and $\Delta/2$.

If Δ is small, the density of M is nearly constant across one region. The error can then be modeled as uniform.

THE UNIFORM ERROR MODEL

$$f_Q(q) = \frac{1}{\Delta}, \quad -\frac{\Delta}{2} \leq q \leq \frac{\Delta}{2}$$

$$E[Q^2] = \int_{-\Delta/2}^{\Delta/2} q^2 \frac{1}{\Delta} dq = \frac{\Delta^2}{12}$$

For a zero-mean error, this mean-square value is also the variance and the quantization-noise power.

Substitute $\Delta = 2m_{\max}/L$ and $L = 2^R$. Then divide the signal power by the error power to obtain the SQNR.

SIGNAL-TO-QUANTIZATION-NOISE RATIO

$$E[Q^2] = \frac{1}{12} \left(\frac{2m_{\max}}{L} \right)^2 = \frac{m_{\max}^2}{3 \cdot 2^{2R}}$$

$$\text{SQNR} = \frac{P_M}{E[Q^2]} = \frac{3P_M}{m_{\max}^2} 2^{2R}$$

$$\text{SQNR [dB]} = \underbrace{10 \log_{10} \frac{3P_M}{m_{\max}^2}}_{\alpha} + 6.02R$$

Each extra bit adds $20 \log_{10} 2 = 6.02$ dB. Doubling the level count halves the step size and divides the error power by four.

INPUT-RANGE USE

The term α depends on the signal power and peak amplitude. A full-scale sinusoid gives $\alpha \approx 1.76$ dB. A sinusoid that reaches one quarter of the range loses 12.04 dB.

EXAMPLE 1.2 – A FULL-SCALE SINUSOID

GIVEN	$m(t) = 5 \cos t$ through a uniform quantizer spanning its full range.
FIND	The step size and the SQNR at $R = 3$ and $R = 4$ bits per sample.
METHOD	Average power from Parseval, step size from $\Delta = 2m_{\max}/L$, then $\alpha + 6.02R$.
SOLUTION	The coefficients are $a_{\pm 1} = 5/2$, so $P_M = 2(5/2)^2 = 12.5$ and $m_{\max} = 5$. At $R = 3$: $\Delta = 2(5)/8 = 1.25$ V and $\text{SQNR} = 1.76 + 18.06 = 19.82$ dB. At $R = 4$: $\Delta = 0.625$ V and $\text{SQNR} = 1.76 + 24.08 = 25.84$ dB.
CHECK	The two differ by 6.02 dB, one bit. Independently, $E[Q^2] = \Delta^2/12 = 0.1302$ and $10 \log_{10}(12.5/0.1302) = 19.82$ dB.

MODEL LIMIT

The measured results are 19.09 dB and 25.31 dB. They are below the model by 0.7 dB and 0.5 dB. A sinusoid does not produce a uniform error at low resolution. The gap decreases as R increases.

EXAMPLE 1.3 – WHEN THE MODEL BREAKS

- GIVEN** A zero-mean stationary Gaussian source has $S_X(f) = 2$ for $|f| < 100$ Hz. Its samples enter a five-level quantizer with outputs $-30, -10, 0, 10, 30$ and boundaries at $-40, -20, 20, 40$.
- FIND** The SQNR of the scheme.
- METHOD** The signal power is the area under the spectral density. The noise power must be integrated region by region, because this quantizer is neither uniform nor fine.
- SOLUTION** $P_X = \int_{-100}^{100} 2 df = 400$, and since the mean is zero this is also σ_X^2 . Splitting $P_Q = \int (x - \mathbb{Q}(x))^2 f_X(x) dx$ at the four boundaries gives five contributions of about 7.98, 46.36, 79.50, 46.36 and 7.98, which add to $P_Q = 188.17$, so SQNR = $10 \log_{10}(400/188.17) = 3.28$ dB.
- CHECK** The central region alone contributes 79.50, and it holds the 68% of the mass with $|X| < 20$ and an error of up to 20. That is where a five-level quantizer spends its error, and it is why the answer is a few decibels rather than a few tens.

MODEL LIMIT

This quantizer is coarse, and its outer regions are unbounded. A sample at $x = 120$ has an error of 90. The formula $\Delta^2/12$ predicts 10.8 dB, but direct integration gives 3.28 dB.

Every source gains 6.02 dB a bit. The source sets only the intercept $\alpha = 10 \log_{10}(3P_M/m_{\max}^2)$:

$$\begin{aligned} \text{sinusoid: } 10 \log_{10} \frac{3(A^2/2)}{A^2} &= 1.76 \text{ dB} \\ \text{uniform on } [-1, 1]: 10 \log_{10} \frac{3(1/3)}{1} &= 0 \text{ dB} \\ \text{Gaussian, } m_{\max} = 4\sigma: 10 \log_{10} \frac{3\sigma^2}{16\sigma^2} &= -7.27 \text{ dB} \end{aligned}$$

At equal R the Gaussian source sits $1.76 - (-7.27) = 9.03$ dB below the sinusoid. Its rare large peaks force a wide range, so typical samples use few of the levels.

At a few bits the error follows the signal. On speech it sounds like distortion, not like hiss, and the uniform error model no longer holds. **Dither** is a small random signal, about one step wide, added before the quantizer. Each output still sits on a level, but the error no longer follows the signal. Averaged over time, or by the eye over neighbouring pixels, the output follows the input between the levels.

For example, let an input sit 0.3Δ above a level, with dither uniform on $[-\Delta/2, \Delta/2]$. The next boundary is 0.5Δ up, so the level above is chosen when the dither exceeds 0.2Δ . That happens 30% of the time, and the average output is 0.3Δ above the level. Dither adds noise power. It trades a pattern that the ear or eye notices for a hiss or grain that it ignores.

1.5 Non-uniform quantization

Speech contains small amplitudes more often than large amplitudes. A uniform quantizer gives every amplitude the same step size. Thus, its relative error is larger for small amplitudes.

A non-uniform quantizer uses narrow regions where the probability density is high. It uses wide regions where the density is low. This arrangement reduces distortion without adding levels.

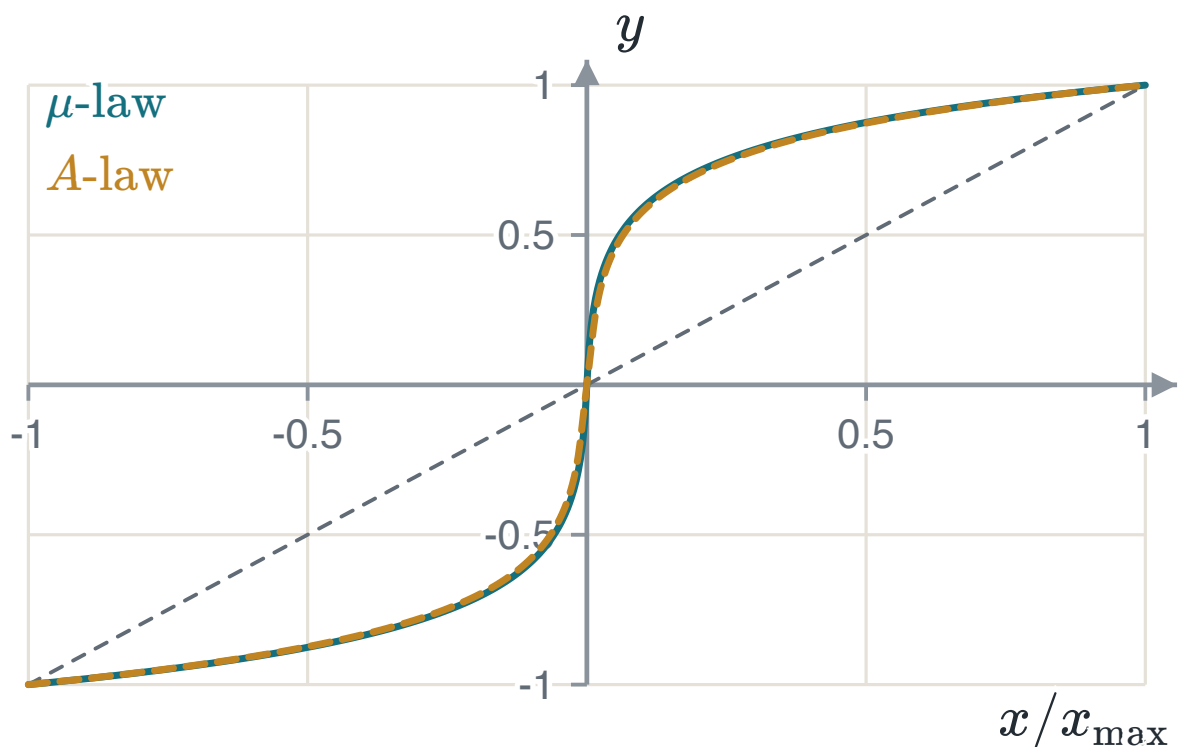
A **compressor** first compresses the signal with a memoryless nonlinearity. It then uses a uniform quantizer. The receiver applies the inverse expansion.

THE TWO STANDARD COMPRESSORS

$$y = \frac{\ln(1 + \mu|x|)}{\ln(1 + \mu)} \operatorname{sgn}(x), \quad |x| \leq 1$$

$$y = \begin{cases} \frac{A|x|}{1 + \ln A} \operatorname{sgn}(x), & 0 \leq |x| \leq 1/A \\ \frac{1 + \ln(A|x|)}{1 + \ln A} \operatorname{sgn}(x), & 1/A < |x| \leq 1 \end{cases}$$

Both laws map $x = \pm 1$ to $y = \pm 1$. The standard parameters are $\mu = 255$ and $A = 87.6$. For the same speech and eight-bit quantizer, their SQNR values differ by less than 0.01 dB.



The identity line shows the result without companding. Both compressors use much of the output range for small input amplitudes.

With a uniform quantizer the noise power $\Delta^2/12$ does not depend on the signal. When a talker drops 20 dB below full scale, the SQNR drops by 20 dB. The μ -law steps shrink with the signal, so its SQNR hardly changes over a wide range of levels.

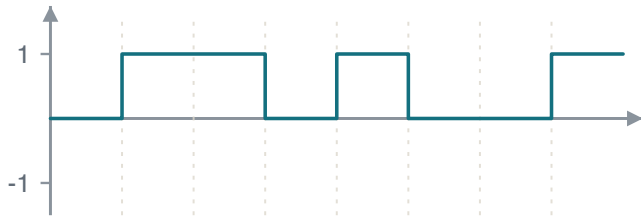
1.6 PCM, DPCM and delta modulation

With $L = 2^R$ levels, each sample needs R bits. At f_s samples per second, the bit rate is $R_b = Rf_s$.

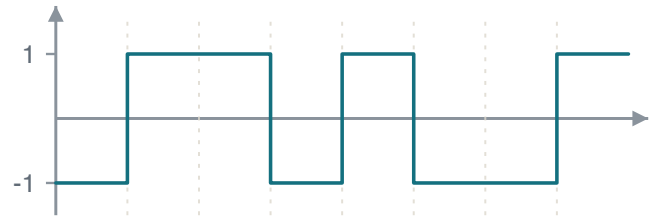
Natural binary coding assigns increasing binary values to the levels. Adjacent **Gray** words differ by one bit. Thus, a small level error changes only one Gray-coded bit.

A **line code** converts the bits into a waveform. Unipolar NRZ has a DC component that causes droop in an AC-coupled stage. Balanced polar NRZ has no DC component.

A long run of equal NRZ bits has no transitions for clock recovery. Manchester coding adds a transition to every bit but uses twice the bandwidth.



Unipolar NRZ.



Polar NRZ, carrying the same eight bits 01101001.

EXAMPLE 1.4 – A COMPLETE PCM STREAM

- GIVEN** $m(t) = 8|\text{sinc}(t - 2)|$, so $0 \leq m(t) \leq 8$, sampled every $T_s = 0.6$ s and quantized by an eight-level uniform quantizer covering $[0, 8]$.
- FIND** The step size, the levels, the code words for $t = 0, 0.6, \dots, 3.6$, and the bit rate.
- METHOD** Calculate Δ from the range and level count. Evaluate each sample. Select the word from the quantization region.
- SOLUTION** $\Delta = 8/8 = 1$ V and the levels sit at 0.5, 1.5, $\dots$, 7.5. The samples are 0, 1.73, 1.87, 7.48, 6.05, 0, 1.51, giving the words 000 001 001 111 110 000 001. With $R = 3$ and $f_s = 1/0.6 = 1.667$, $R_b = 5$ bit/s and $T_b = T_s/3 = 0.2$ s.
- CHECK** At $t = 3$, the sample is $8|\text{sinc}(1)| = 0$. The sinc function is zero at every nonzero integer. The interpolation formula uses the same property.

The bandwidth of PCM

A binary stream of R_b bits a second needs a bandwidth of at least $R_b/2$ hertz. Chapter 2 shows why.

LEAST BANDWIDTH OF PCM

$$B_T \geq \frac{R_b}{2} = \frac{R f_s}{2}$$

$$B_T \geq RW \quad \text{at } f_s = 2W$$

Each extra bit adds 6.02 dB of SQNR and W hertz of bandwidth. Telephone PCM at $f_s = 8$ kHz and $R = 8$ bits sends 64 kb/s and needs at least 32 kHz, eight times the 4 kHz of the voice.

Bit errors

A channel error flips one bit of a code word. With natural binary coding, bit b counted from 0 at the right moves the decoded value by $2^b \Delta$. A wrong first bit, $b = R - 1$, moves the sample by $2^{R-1} \Delta = m_{\max}$, half the range. For $R = 8$ and $\Delta = 1/128$ V that is 1 V. On speech it sounds as a click, far louder than the quantization noise.

Differential PCM

Neighbouring samples of speech are close. **DPCM** quantizes the difference between a sample and its prediction, which has a much smaller range than the sample.

DPCM WITH THE LAST DECODED VALUE AS PREDICTION

$$e[n] = x[n] - \hat{x}[n-1]$$

$$\hat{x}[n] = \hat{x}[n-1] + \mathbb{Q}(e[n])$$

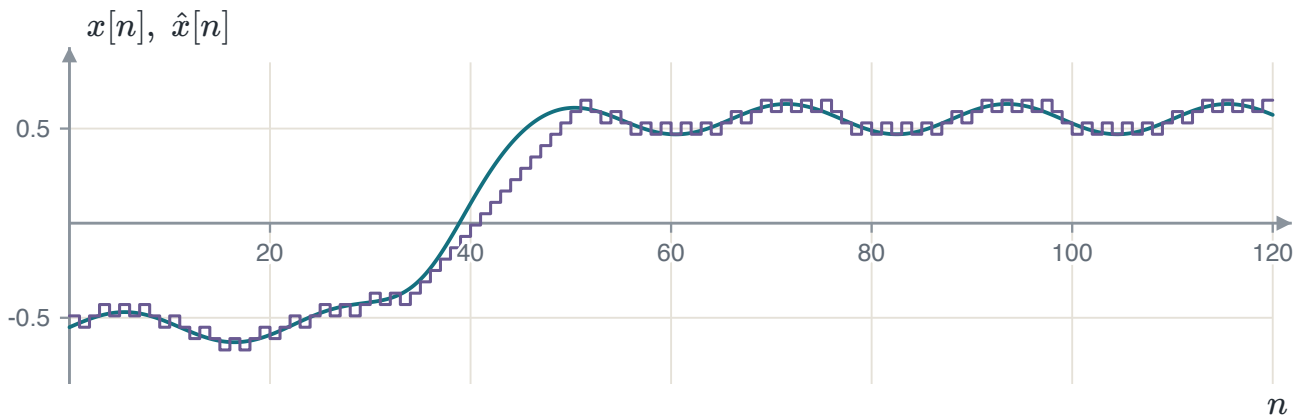
The encoder predicts from $\hat{x}[n-1]$, which the decoder also has. So $x[n] - \hat{x}[n] = e[n] - \mathbb{Q}(e[n])$, and quantization errors do not pile up.

For speech at 8 kHz, DPCM with 4 bits a sample matches PCM with 8. That is 32 kb/s in place of 64 kb/s.

Delta modulation

Delta modulation is DPCM with a two-level quantizer. Each bit moves a staircase up or down by Δ . The staircase climbs at most Δ every T_s , so it follows the signal only while

$$\left| \frac{dx}{dt} \right| \leq \frac{\Delta}{T_s} = \Delta f_s$$



A signal with one steep rise and the delta-modulation staircase. On the rise the staircase falls behind (slope overload). Where the signal is flat it hunts by $\pm \Delta$ (granular noise).

EXAMPLE 1.5 – THE STEP OF A DELTA MODULATOR

- GIVEN** $x(t) = \sin(2\pi 1000 t)$, sampled at $f_s = 64$ kHz by a delta modulator.
- FIND** The smallest step Δ that avoids slope overload.
- METHOD** Find the largest slope of x , then apply $|dx/dt| \leq \Delta f_s$.
- SOLUTION** $dx/dt = 2\pi(1000) \cos(2\pi 1000 t)$, whose largest value is $2\pi(1000) = 6283$ per second. So $\Delta \geq 6283/64\,000 = 0.098$.
- CHECK** A larger Δ avoids overload but raises the granular noise, which grows with Δ .
-

COMMON ERROR

Use the largest slope of the signal, $2\pi f_0 A$, not its peak value A . The staircase fails where the signal changes fastest.

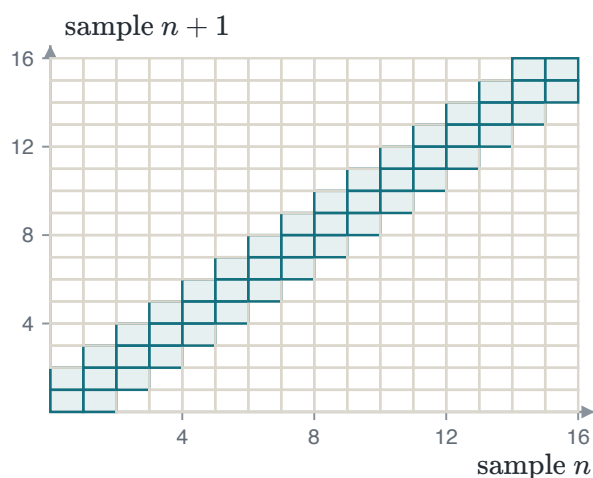
1.7 Vector quantization

Scalar quantization processes one sample at a time. It does not use the dependence between neighboring samples.

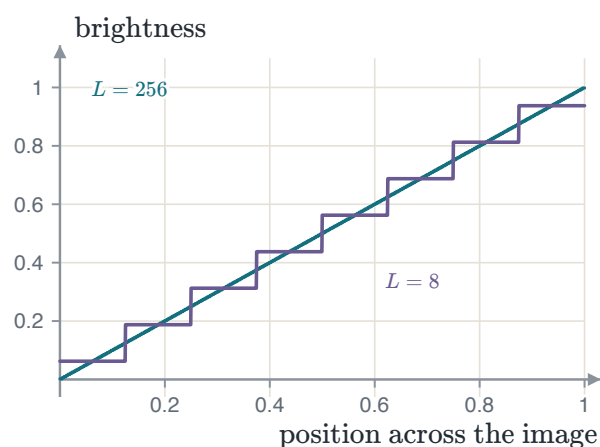
Vector quantization treats n samples as one point in n dimensions. A **codebook** contains the allowed output points. The quantizer selects the nearest codebook point.

EXAMPLE 1.6 – PAIRS OF NEIGHBOURING SAMPLES

- GIVEN** $L = 16$ levels, and a signal smooth enough that a sample never moves more than one step from the one before it.
- FIND** What a pair costs, quantized separately and quantized together.
- METHOD** Count the pairs each scheme has to be able to name, and take the base-two logarithm.
- SOLUTION** Separately, every combination is allowed: $L^2 = 256$ pairs, $\log_2 256 = 8$ bits a pair, or 4 bits a sample. Together, only the pairs with $|i - j| \leq 1$ can occur, and there are $3L - 2 = 46$ of them: $\lceil \log_2 46 \rceil = 6$ bits a pair, or 3 bits a sample.
- CHECK** The cell size and rounding error do not change. The code omits 210 pairs that the signal model cannot produce. The rate decreases from four to three bits per sample.
-



A scalar quantizer represents all 256 pairs. Under the smooth-signal assumption, only the 46 shaded pairs can occur.



The coarse quantizer changes a smooth gradient into flat steps. Each step boundary appears as a false line.

SOURCE DEPENDENCE

Most of the rate reduction comes from dependence between neighboring samples. Better cell shapes can give a smaller gain for independent samples. This course does not develop that case.

EXAMPLE 1.7 – QUANTIZING AN IMAGE

- GIVEN** A 512×512 grayscale image at 8 bits a pixel, so $L = 256$ levels.
- FIND** The size of the file, and the size and the cost at $L = 32$.
- SOLUTION** $512^2(8) = 2\,097\,152$ bits, which is 256 KiB. At $L = 32$ the rate is $R = \log_2 32 = 5$ bits a pixel, so 1 310 720 bits or 160 KiB — 37.5% smaller.
- COST** $\Delta\text{SQNR} = 6.02(8 - 5) = 18.06$ dB. A smooth gradient becomes flat bands with false edges.
- CHECK** The decoder cannot recover the position of a pixel inside its quantization interval. Therefore, this compression is **lossy**. JPEG also quantizes transformed image blocks.

1.8 Speech, audio and image coding

The same three steps, sampling, quantization and encoding, sit under the coders in everyday use. This section shows four of them briefly.

Linear predictive coding

LPC does not send the waveform. Every 20 ms, that is 160 samples at 8 kHz, it sends a model of the voice. The receiver runs the model to make the speech again.

$$x_n = \sum_{i=1}^p a_i x_{n-i} + G w_n$$

w_n is a pulse train for a voiced sound and white noise for an unvoiced one. The coefficients a_i describe the shape of the vocal tract, and the peaks of the filter are the formants.

Each frame sends the a_i , the gain G , the pitch and one voiced bit. A coder with 48 bits a 20 ms frame runs at $48(50) = 2400$ b/s, against 64 kb/s for PCM.

Time-division multiplexing

Many PCM calls share one line by taking turns. Each call gets one slot in every frame, and a frame lasts one sampling interval, $125 \mu\text{s}$ at 8 kHz.

$$24(8) + 1 = 193 \text{ bits a frame}$$

$$193(8000) = 1.544 \text{ Mb/s}$$

The extra bit marks the start of each frame. The 24 calls alone need $24(64) = 1.536$ Mb/s. Four T1 lines make a 6.312 Mb/s line, and seven of those a 44.736 Mb/s line. Europe and Türkiye use E1: 32 slots of 64 kb/s, 2.048 Mb/s.

Oversampling and sigma-delta conversion

At a rate far above $2W$, neighbouring samples are nearly equal, and one bit a sample can carry the change. A sigma-delta converter integrates the error between the input and its one-bit output:

$$\begin{aligned} v[n] &= v[n-1] + x[n] - y[n-1] \\ y[n] &= \text{sgn}(v[n]) \end{aligned}$$

The running average of the bits y stays on the input. A lowpass filter removes the fast swings of the bits and returns the signal. A CD player that oversamples 44.1 kHz audio by $U = 256$ runs its one-bit converter at $256(44.1 \text{ kHz}) = 11.2896 \text{ MHz}$.

Transform coding and JPEG

JPEG cuts a picture into 8×8 blocks and takes the discrete cosine transform of each. Most of the energy of a block lands in a few low-frequency coefficients.

Each coefficient has its own uniform step, larger at high frequencies. Most high-frequency coefficients round to zero and cost almost nothing to send. At a coarse step each block keeps little more than its average, and the picture breaks into visible 8×8 squares.

A 512×512 picture coded at 0.5 bit a pixel takes $512^2(0.5)/8 = 16\,384$ bytes, or 16 KiB. That is sixteen times less than at 8 bits a pixel.

1.9 Summary

RESULT	STATEMENT	ANCHOR
Replication	$G_\delta(f) = f_s \sum_n G(f - n f_s)$	PS CH7.1.1
Sampling theorem	$f_s \geq 2W$ for a message bandlimited to W	PS CH7.1.1
Reconstruction	$g_r(t) = \sum_n g(nT_s) \text{sinc}(2Wt - n)$, filter gain $1/(2W)$	PS CH7.1.1
Step size	$\Delta = 2m_{\max}/L$, $L = 2^R$	PS CH7.2.1
Error power	$E[Q^2] = \Delta^2/12$, when the step is small	PS CH7.2.1
Signal-to-noise	$\text{SQNR [dB]} = \alpha + 6.02R$	PS CH7.2.1
Bit rate	$R_b = R f_s$	PS CH7.3, 7.4.1
PCM bandwidth	$B_T \geq R_b/2 = RW$ at $f_s = 2W$	PS CH7.4.1
DPCM	quantize $e[n] = x[n] - \hat{x}[n-1]$	PS CH7.4.2
Delta modulation	no slope overload while $ dx/dt \leq \Delta f_s$	PS CH7.4.3
T1 line	193 bits every $125 \mu\text{s}$, 1.544 Mb/s	PS CH7.6.1
Vector quantization	round n samples together and use a codebook	PS CH7.2.2

Sampling is reversible and quantization is not. Everything after this chapter takes the bit stream as given and asks what the channel does to it.

CHAPTER 2

Baseband transmission of digital signals

Chapter 1 produced a stream of bits. Each bit now becomes a waveform, and the channel adds noise. The receiver must identify the transmitted waveform. This chapter selects the receiver filter and decision threshold. It also studies pulse overlap in a bandlimited channel.

2.1 The matched filter

Over one bit interval the receiver sees $x(t) = g(t) + w(t)$, the waveform plus white Gaussian noise of two-sided density $N_0/2$. It passes this through a filter and takes one sample at the end of the interval. Because the filter is linear, its output splits the same way, into a signal part $g_0(t) = g * h$ and a noise part $n(t) = w * h$.

The quantity worth maximising is the signal at the sampling instant against the noise power that accompanies it. Call this ratio η .

THE PEAK PULSE SIGNAL-TO-NOISE RATIO

$$\eta = \frac{|g_0(T)|^2}{E[n^2(T)]} = \frac{|\int_{-\infty}^{\infty} G(f)H(f)e^{j2\pi fT} df|^2}{\frac{N_0}{2} \int_{-\infty}^{\infty} |H(f)|^2 df}$$

Scaling H up multiplies the top and the bottom by the same factor, so the answer cannot be "make H large". What is being chosen is the shape of H .

Schwarz's inequality says that $|\int \phi_1 \phi_2|^2 \leq \int |\phi_1|^2 \int |\phi_2|^2$, with equality only when $\phi_1 = k\phi_2^*$. Taking $\phi_1 = H(f)$ and $\phi_2 = G(f)e^{j2\pi fT}$, the factor $\int |H|^2$ cancels between the top and the bottom and a bound appears that contains no H at all.

THE BOUND, AND THE FILTER THAT REACHES IT

$$\eta \leq \frac{2}{N_0} \int |G(f)|^2 df = \frac{2E}{N_0}$$

$$H(f) = k G^*(f) e^{-j2\pi fT} \iff h_{\text{opt}}(t) = k g(T - t)$$

The optimum filter is the transmitted waveform reversed in time and shifted into the interval. It is called the **matched filter**.

TWO THINGS THIS SAYS

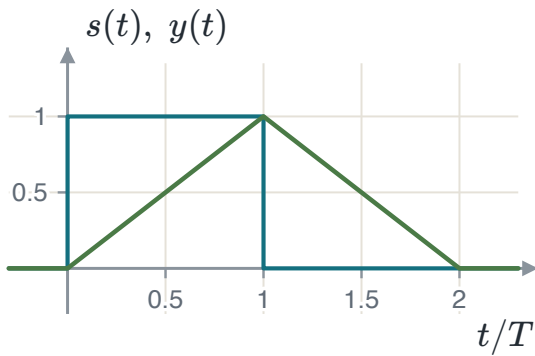
The best achievable ratio is $\eta_{\text{max}} = 2E/N_0$. It depends on the **energy** of the pulse and on the noise density, and on nothing else about the pulse. Two completely different waveforms of the same energy perform identically. The constant k cancels. Therefore, the filter is fixed only up to a gain.

The matched-filter output can be found directly, without going through frequency. Write it as a convolution and substitute $h(t) = g(T - t)$.

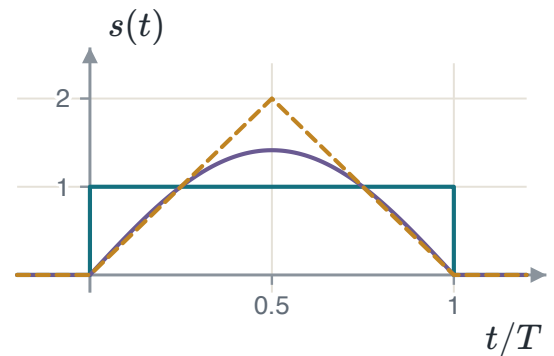
THE MATCHED-FILTER OUTPUT AS A SLIDING OVERLAP

$$y(t) = \int_{-\infty}^{\infty} g(\tau) h(t - \tau) d\tau = \int_{-\infty}^{\infty} g(\tau) g(T - t + \tau) d\tau$$

At $t = T$ the two copies of g line up exactly, so $y(T) = \int g^2(\tau) d\tau = E$. The peak of the matched-filter output is always the pulse energy, whatever the pulse shape.



A rectangular pulse and the output of the filter matched to it. The output peaks at exactly $t = T$, and its peak is the energy of the pulse.



Three pulses of the same energy. Against white Gaussian noise a matched-filter receiver performs identically on all three.

EXAMPLE 2.1 – MATCHED FILTER OF A RECTANGULAR PULSE

GIVEN $s(t) = A$ on $[0, T]$, with $A = 2$ V, $T = 0.5$ ms and $N_0/2 = 10^{-4}$ W/Hz.

FIND The matched filter $h(t)$, the output $y(t)$ and the largest peak SNR $\eta_{\max}$.

METHOD Reverse and shift the pulse to get $h(t)$. Slide it across $s(t)$ to get $y(t)$. Then use $\eta_{\max} = 2E/N_0$.

SOLUTION $h(t) = s(T - t) = A$ on $[0, T]$. Sliding the overlap gives $y(t) = A^2 t$ for $0 \leq t \leq T$ and $y(t) = A^2(2T - t)$ for $T \leq t \leq 2T$. The energy is $E = A^2 T = (2)^2(0.5 \times 10^{-3}) = 2 \times 10^{-3}$ J. Then $\eta_{\max} = 2E/N_0 = 2(2 \times 10^{-3})/(2 \times 10^{-4}) = 20$.

CHECK $10 \log_{10} 20 = 13.0$ dB, and $y(T) = A^2 T = E$, as the general result requires.

COMMON ERROR

Use $N_0 = 2 \times 10^{-4}$ W/Hz in $2E/N_0$, not $N_0/2$. Using $N_0/2$ in place of N_0 gives 40, twice the true value.

2.2 From waveform to number

Both waveforms of a binary baseband system are multiples of one shape, so a single unit-energy function carries both. With $\psi(t) = 1/\sqrt{T_b}$ on the bit interval, polar NRZ is $s_m(t) = s_m \psi(t)$ with $s_0 = -A\sqrt{T_b}$ and $s_1 = +A\sqrt{T_b}$. Both carry energy $E_b = A^2 T_b$, so $A\sqrt{T_b} = \sqrt{E_b}$ and the two waveforms have become two *numbers*, $\pm\sqrt{E_b}$, on one axis.

This axis is a signal space of one dimension, the axis of $\psi(t)$. The point $s_0 = -\sqrt{E_b}$ stands for the waveform $s_0(t)$, and $s_1 = +\sqrt{E_b}$ for $s_1(t)$.

THE TWO SYMBOLS AS POINTS

$$s_m = \int_0^{T_b} s_m(t) \psi(t) dt = \pm\sqrt{E_b}, \quad d = s_1 - s_0 = 2\sqrt{E_b}$$

A larger distance d means noise must move the received point further before it crosses to the wrong symbol.

The demodulator can be built two ways. The **matched filter** convolves with $\psi(T_b - t)$ and samples at T_b . The **correlator** multiplies by $\psi(t)$ and integrates over the interval. Put $t = T_b$ in the matched-filter output, and the convolution integral becomes the correlator integral, for any shape of ψ .

THE TWO DEMODULATORS AGREE AT $t = T_b$

$$h(t) = \psi(T_b - t)$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) \psi(T_b - t + \tau) d\tau \implies y(T_b) = \int_0^{T_b} x(\tau) \psi(\tau) d\tau$$

This equality holds at $t = T_b$ for every basis shape $\psi(t)$, not only for a rectangle.

For the rectangular ψ used in polar NRZ, the two demodulators agree over the whole bit, not only at $t = T_b$. The matched-filter output is the ramp $y(t) = s_m t / \sqrt{T_b}$ for $0 \leq t \leq T_b$, and the correlator output at time t is the same running integral $\int_0^t x(\tau) \psi(\tau) d\tau$.

A DIFFERENT BASIS SHAPE

For a shape such as a half-sine ψ , the running matched-filter output and the running correlator output differ before $t = T_b$. Only at $t = T_b$ do the two demodulators agree, because that is where the convolution and the correlation cover the same interval.

THE DECISION STATISTIC

$$y = \int_0^{T_b} x(\tau) \psi(\tau) d\tau = s_m \underbrace{\int_0^{T_b} \psi^2}_{=1} + \underbrace{\int_0^{T_b} w\psi}_n = s_m + n$$

The noise term n is Gaussian, because it is a linear operation on a Gaussian process.

Find the variance of n from the autocorrelation of white noise. Write n^2 as a double integral, substitute the two-sided density, and use the sifting property of $\delta(\tau - u)$.

THE NOISE VARIANCE, STEP BY STEP

$$\begin{aligned} E[n^2] &= \int_0^{T_b} \int_0^{T_b} E[w(\tau)w(u)] \psi(\tau)\psi(u) d\tau du \\ &= \int_0^{T_b} \int_0^{T_b} \frac{N_0}{2} \delta(\tau - u) \psi(\tau)\psi(u) d\tau du \\ &= \frac{N_0}{2} \int_0^{T_b} \psi^2(\tau) d\tau = \frac{N_0}{2} \end{aligned}$$

The inner integral sifts $\delta(\tau - u)$ against $\psi(u)$, leaving $\psi(\tau)$. The unit energy of ψ then makes the answer $N_0/2$, independent of the interval length.

2.3 The decision and its error probability

WHAT THE DETECTOR IS GIVEN

$$\sigma_n^2 = \frac{N_0}{2}$$

$$y = s_m + n \sim \mathcal{N}\left(s_m, \frac{N_0}{2}\right)$$

The two-sided convention arrives here unchanged. Reading the density as one-sided makes every error probability in the course 3 dB too optimistic, and nothing in the algebra shows it.

The detector decides s_1 when $y > \lambda$. Write each conditional error as a Gaussian tail, using $s_0 = -\sqrt{E_b}$, $s_1 = +\sqrt{E_b}$ and $\sigma = \sqrt{N_0/2}$.

THE CONDITIONAL ERRORS, WITH THE SUBSTITUTION SHOWN

$$\begin{aligned} P(\text{err} \mid s_0) &= \int_{\lambda}^{\infty} \frac{1}{\sqrt{\pi N_0}} e^{-(y+\sqrt{E_b})^2/N_0} dy \\ &= \int_{(\lambda+\sqrt{E_b})/\sigma}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = Q\left(\frac{\lambda + \sqrt{E_b}}{\sigma}\right) \\ P(\text{err} \mid s_1) &= \int_{-\infty}^{\lambda} f_Y(y \mid s_1) dy = Q\left(\frac{\sqrt{E_b} - \lambda}{\sigma}\right) \end{aligned}$$

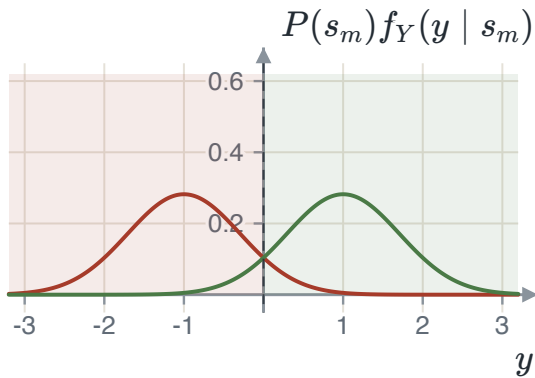
The substitution $z = (y + \sqrt{E_b})/\sigma$ turns the lower limit $y = \lambda$ into $z = (\lambda + \sqrt{E_b})/\sigma$, which is the argument of Q . The second line follows the same way with $z = (\sqrt{E_b} - y)/\sigma$.

The average error $P_e(\lambda)$ weights each conditional error by its prior. Differentiate with respect to λ by the Leibniz rule, set the result to zero, and the two prior-weighted densities must be equal there.

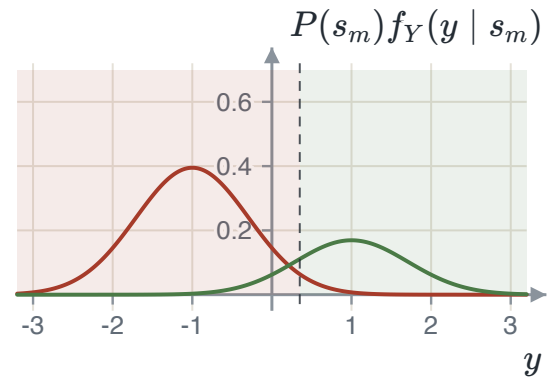
THE OPTIMAL THRESHOLD, DERIVED

$$\begin{aligned} P_e(\lambda) &= P(s_0) P(\text{err} \mid s_0) + P(s_1) P(\text{err} \mid s_1) \\ \frac{dP_e}{d\lambda} &= -P(s_0) f_Y(\lambda \mid s_0) + P(s_1) f_Y(\lambda \mid s_1) = 0 \\ P(s_0) e^{-(\lambda+\sqrt{E_b})^2/N_0} &= P(s_1) e^{-(\lambda-\sqrt{E_b})^2/N_0} \\ 4\lambda\sqrt{E_b} &= N_0 \ln \frac{P(s_0)}{P(s_1)} \\ \lambda_{\text{opt}} &= \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)} \end{aligned}$$

The common factor $1/\sqrt{\pi N_0}$ cancels from the equal-density equation. Taking logarithms and expanding the two squares $(\lambda + \sqrt{E_b})^2$ and $(\lambda - \sqrt{E_b})^2$ leaves only the term $4\lambda\sqrt{E_b}$, since the λ^2 and E_b terms cancel. Equal priors put the threshold at the midpoint. If s_0 is more likely, the threshold moves in the positive direction, enlarging the decision region for s_0 .



Equal priors: the threshold sits midway and the shaded decision regions are symmetric.



$P(s_0) = 0.7$: the more likely symbol's curve is taller and the crossing has moved towards the less likely one.

The Gaussian tail used above is defined by an integral, and it has no closed form. A few of its values are used throughout the course.

THE Q FUNCTION

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-z^2/2} dz = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right)$$

$Q(x)$ is the probability that a Gaussian variable of mean 0 and variance 1 exceeds x . Three values recur: $Q(1) = 0.159$, $Q(2) = 0.0228$, $Q(3) = 1.35 \times 10^{-3}$.

With equal priors the threshold is at the origin, $\lambda = 0$, and both conditional errors become the same Gaussian tail.

THE RESULT OF THE CHAPTER

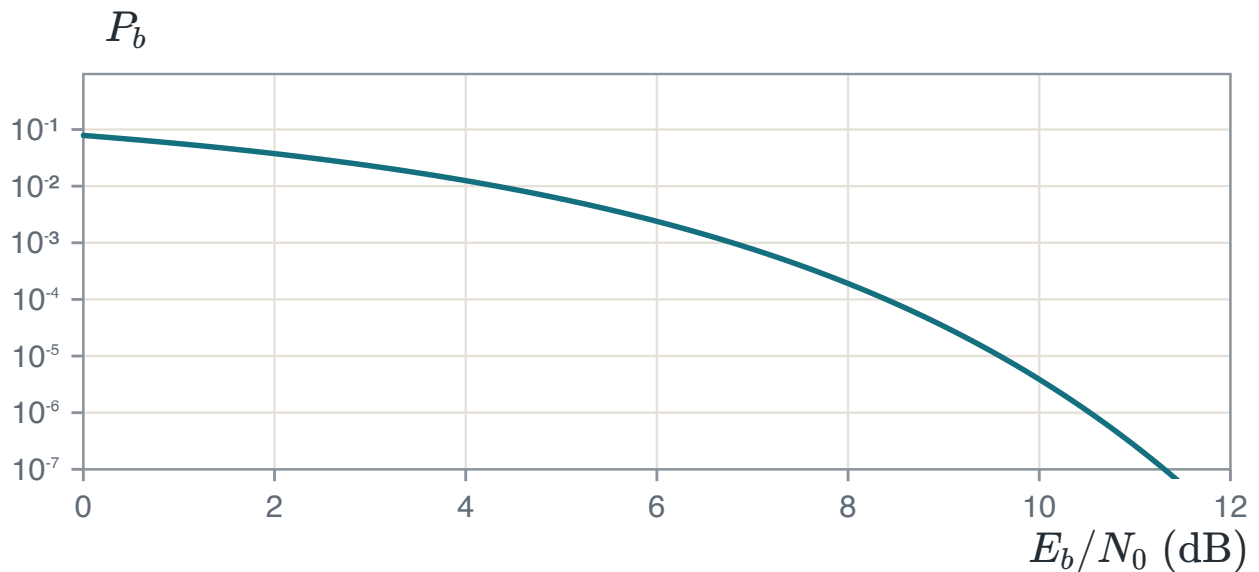
$$\frac{\sqrt{E_b}}{\sigma} = \frac{\sqrt{E_b}}{\sqrt{N_0/2}} = \sqrt{\frac{2E_b}{N_0}}$$

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

It depends on E_b/N_0 and on nothing else — not on the amplitude, not on the bit duration, not on the pulse shape. Doubling the amplitude and quartering the duration change neither the energy per bit nor the answer.

ANTIPODAL AND ORTHOGONAL SIGNALS

Antipodal signaling gives $Q\left(\sqrt{2E_b/N_0}\right)$. On-off or orthogonal signaling gives $Q\left(\sqrt{E_b/N_0}\right)$ and needs 3 dB more energy. For on-off signaling, half the symbols have zero energy. Therefore, average bit energy is half the nonzero-symbol energy.



Bit error probability against E_b/N_0 . Past about 8 dB every extra decibel is worth roughly an order of magnitude in error rate.

EXAMPLE 2.2 – UNEQUAL PRIORS

GIVEN A binary PAM system with correlator output $y = \pm\sqrt{E_b} + n$, where $P(s_1) = 0.3$, $E_b = 1$ and $N_0 = 0.1$.

FIND The optimal threshold and the average error probability there.

METHOD Threshold from the log-ratio of the priors. Each conditional error from a Gaussian tail. Average by weighting.

SOLUTION $\lambda = \frac{0.1}{4} \ln \frac{0.7}{0.3} = 0.0212$ and $\sigma = \sqrt{0.05} = 0.2236$. Then $P(\text{err} | s_0) = Q(4.567) = 2.475 \times 10^{-6}$ and $P(\text{err} | s_1) = Q(4.377) = 6.005 \times 10^{-6}$, so $P_e = 0.7(2.475) + 0.3(6.005)$ in units of 10^{-6} , giving 3.534×10^{-6} .

CHECK Leaving the threshold at zero would give $Q(\sqrt{20}) = 3.872 \times 10^{-6}$. The ratio $3.872/3.534 = 1.096$, so moving the threshold improves the answer by about 10 per cent. A real gain and a small one. This is what a shift of a tenth of a standard deviation buys.

2.4 Intersymbol interference

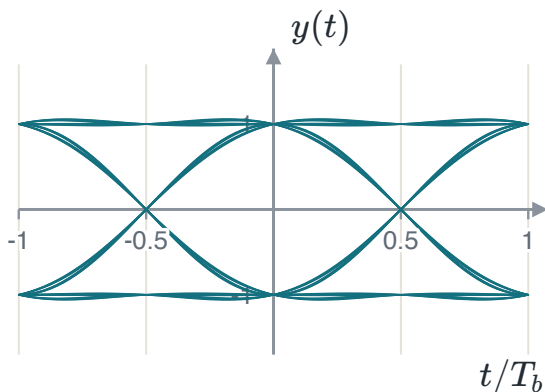
Everything above assumed an ideal channel. A real bandlimited channel spreads each pulse in time. Therefore, one pulse can enter adjacent symbol intervals. Write the received signal as a train of the overall pulse p . Sampling at $t_i = iT_b$ separates the wanted pulse from intersymbol interference.

WHERE THE INTERFERENCE COMES FROM

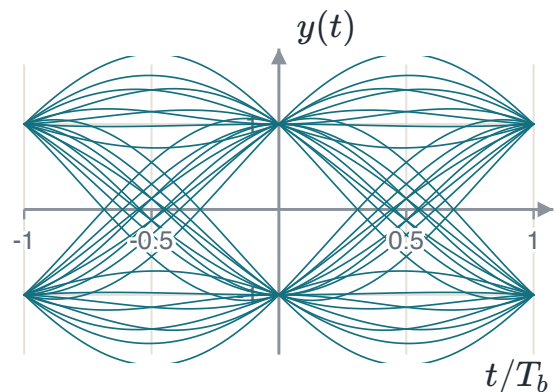
$$y(t_i) = \underbrace{\mu a_i}_{\text{wanted}} + \underbrace{\mu \sum_{k \neq i} a_k p((i-k)T_b)}_{\text{intersymbol interference}} + n(t_i)$$

The middle term is not noise. It is caused by the data itself, and raising the transmit power raises it by the same factor, so more power does not help. At high signal-to-noise ratio it is the only thing limiting the system.

An **eye pattern** shows this interference. Divide the received waveform into bit-length segments. Align the segments with the clock and draw them together. The opening height gives the noise margin. Its width gives the permitted sampling-time error. Crossing slope shows timing sensitivity, and crossing spread shows timing jitter. A closed eye has no sampling time that gives correct decisions for every bit pattern.



A wide-open eye: a pulse that decays quickly. Every trace passes close to ± 1 at the centre.



A slowly decaying pulse. The opening has narrowed and the crossings have scattered, with the noise unchanged.

A first-order RC lowpass channel makes the interference concrete. Its transfer function has one parameter, the 3 dB bandwidth B .

THE CHANNEL AND ITS RESPONSE TO ONE BIT

$$H(f) = \frac{1}{1 + jf/B}, \quad \tau = \frac{1}{2\pi B}$$

$$r(T_b) = 1 - q, \quad q = e^{-2\pi B T_b}$$

τ is the time constant of the channel. A single bit reaches only the fraction $1 - q$ of its own level by the end of the bit. The rest arrives after the bit ends, as a decaying tail.

A bit that ended m bits ago still contributes to the present sample, because its tail has not fully decayed.

ONE EARLIER BIT'S SHARE OF THE PRESENT SAMPLE, AND THE WORST CASE

$$(1 - q) q^m$$

$$\sum_{m=1}^{\infty} (1 - q) q^m = q$$

The sum is a geometric series and adds to q . When every earlier bit opposes the present one, the sample shrinks from $1 - q$ to $1 - 2q$, and the eye opening is $2(1 - 2q)$.

The eye closes when the opening reaches zero. Set $1 - 2q = 0$ and solve for the bandwidth.

EYE CLOSURE

$$1 - 2q = 0 \implies e^{-2\pi BT_b} = \frac{1}{2} \implies BT_b = \frac{\ln 2}{2\pi} = 0.110$$

Below $BT_b = 0.110$ some bit patterns cross zero at the sampling instant, and decisions fail even without noise. At $BT_b = 0.25$, $q = 0.208$, so $1 - q = 0.792$ and $1 - 2q = 0.584$. At $BT_b = 0.3$, $q = 0.152$ and the opening $2(1 - 2q)$ is 1.39.

EXAMPLE 2.3 – A SAMPLE WITH A BIT-PATTERN HISTORY

- GIVEN** Bits 00011101 sent after a line idle at the level of a 0 ($a_k = -1$), through an RC channel with $BT_b = 0.25$, so $q = 0.208$.
- FIND** The sample at the end of the seventh bit, y_7 .
- METHOD** Add the bit's own share $(1 - q)$ or $-(1 - q)$ to the tails of every earlier bit, each weighted by $(1 - q)q^m$.
- SOLUTION** The seventh bit is a 0, contributing $-(1 - q)$. The three 1s before it (bits four to six) contribute $(1 - q)(q + q^2 + q^3)$. Every bit before that, and the idle line, is a 0 and contributes $-(1 - q) \sum_{m \geq 4} q^m$. So $y_7 = -(1 - q) + (1 - q)(q + q^2 + q^3) - (1 - q) \sum_{m \geq 4} q^m = -1 + 2q - 2q^4 = -0.588$.
- CHECK** The sample is negative, so the receiver still decides 0 correctly. The result depends on the whole bit history through the geometric tails, not on the previous bit alone.

COMMON ERROR

Use the full history of bits, not only the previous bit. Each earlier bit still contributes through its own tail $(1 - q)q^m$, however small.

2.5 Nyquist's criterion and the raised cosine

The interference vanishes exactly when the overall pulse is zero at every sampling instant but its own. Sampling p at those instants and transforming turns that requirement into a statement about the spectrum.

NYQUIST'S CRITERION FOR DISTORTIONLESS TRANSMISSION

$$p((i-k)T_b) = \begin{cases} 1, & i = k \\ 0, & i \neq k \end{cases}$$

$$R_b \sum_n P(f - nR_b) = 1, \quad R_b = \frac{1}{T_b}$$

In words: the replicas of the pulse spectrum, spaced by the symbol rate, must add to a constant. The simplest spectrum that does it is a rectangle of width $2W$ with $W = R_b/2$, the **Nyquist bandwidth**, whose pulse is $\text{sinc}(2Wt)$.

THE RATE A BANDWIDTH SUPPORTS

A channel of bandwidth W carries at most $2W$ symbols per second with no interference. This is the counterpart of the sampling theorem, reached from the other end of the same argument.

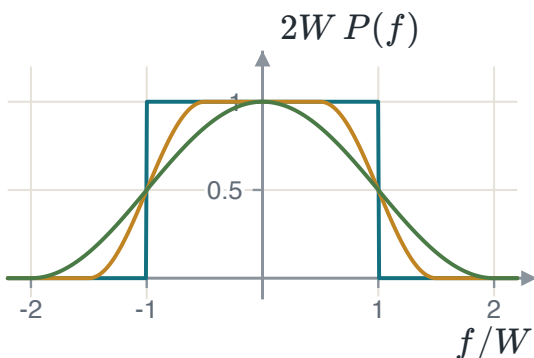
The sinc pulse of the ideal Nyquist channel is impractical. Its terms fall only as $1/(\pi|k|)$, so the sum of their sizes grows like $\sum 1/k$. For example, $\text{sinc}(0.1) = 0.984$ is barely smaller than 1. A small timing error still collects interference from many neighbours at once, because so many terms are involved. The fix is to widen the spectrum and taper its edges while keeping the tiling property.

THE RAISED COSINE

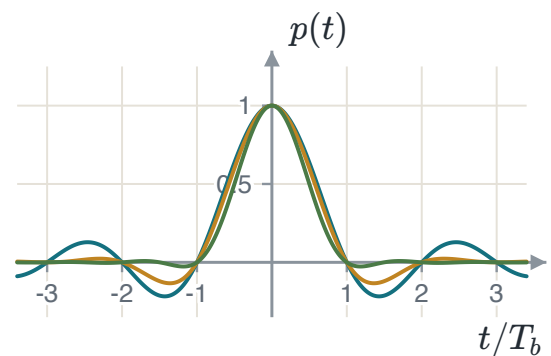
$$P(f) = \begin{cases} \frac{1}{2W}, & |f| < f_1 \\ \frac{1}{4W} \left[1 + \cos \frac{\pi(|f| - f_1)}{2W - 2f_1} \right], & f_1 \leq |f| < 2W - f_1 \\ 0, & |f| \geq 2W - f_1 \end{cases}$$

$$p(t) = \text{sinc}\left(\frac{t}{T_b}\right) \frac{\cos(\pi\alpha t/T_b)}{1 - 4\alpha^2 t^2/T_b^2}, \quad f_1 = W(1 - \alpha)$$

$W = R_b/2$ and α runs from 0 to 1. The sinc factor in $p(t)$ keeps the zero crossings at $t = iT_b$, so the criterion still holds. For $\alpha > 0$ the second factor falls as $1/t^2$, so the pulse tails fall as $1/|t|^3$. The price is bandwidth: $B_T = W(1 + \alpha) = \frac{R_b}{2}(1 + \alpha)$, an excess of αW over the Nyquist bandwidth W .



The spectrum at $\alpha = 0, 0.5$ and 1 . All three tile the axis at spacing $2W$, so all three give zero interference.



Their pulses. All three vanish at every non-zero multiple of T_b . The tails differ by orders of magnitude.

EXAMPLE 2.4 – FITTING A RATE INTO A CHANNEL

GIVEN	A channel of bandwidth 48 kHz is to carry 64 kbit/s with raised-cosine shaping.
FIND	The Nyquist bandwidth and the largest roll-off that fits.
METHOD	$W = R_b/2$, then $(1 + \alpha)W \leq B$.
SOLUTION	$W = 32$ kHz, and $(1 + \alpha)(32) \leq 48$ gives $\alpha \leq 0.5$. At $\alpha = 0.5$ the signal uses the whole 48 kHz, with an excess of 16 kHz over the Nyquist bandwidth.
CHECK	At $\alpha = 0$, the signal needs only 32 kHz and still fits. However, its pulse decays as $1/t$, and its ideal filter cannot be built. Using the available bandwidth gives a pulse that decays as $1/t^3$. The spectral efficiency is $64/48 = 1.33$ bit/s/Hz, below the theoretical maximum of 2.

EXAMPLE 2.5 – THE BANDWIDTH OF A RAISED-COSINE LINK

GIVEN	A link sends $R_b = 20$ kb/s with raised-cosine pulses and $\alpha = 0.5$.
FIND	W , f_1 and B_T .
METHOD	$W = R_b/2$, then $f_1 = W(1 - \alpha)$ and $B_T = W(1 + \alpha)$.
SOLUTION	$W = 20/2 = 10$ kHz. $f_1 = 10(1 - 0.5) = 5$ kHz. $B_T = 10(1 + 0.5) = 15$ kHz.
CHECK	$B_T - W = W - f_1 = 5$ kHz, so the roll-off is symmetric about W .

COMMON ERROR

Use $W = R_b/2 = 10$ kHz, not $W = R_b = 20$ kHz. The wrong W gives $B_T = 30$ kHz, twice the true value.

The raised-cosine filter can be split between the transmitter and the receiver. Each half is a square root of the raised cosine.

THE ROOT-RAISED-COSINE SPLIT

$$H_T(f) H_R(f) = P(f), \quad H_T(f) = H_R(f) = \sqrt{P(f)}$$

For a real, even $\sqrt{P(f)}$, $H_R = H_T^*$, so the receive filter is matched to the transmitted pulse. The link then has both the smallest error probability and zero interference. At $\alpha = 0.5$ the raised cosine is at half height at $f = W$, so $\sqrt{2W P(W)} = \sqrt{0.5} = 0.707$.

2.6 Summary

RESULT	STATEMENT	ANCHOR
Matched filter	$h_{\text{opt}}(t) = g(T - t)$	PS CH8.3.2
What it achieves	$\eta_{\text{max}} = 2E/N_0$, independent of the pulse shape	PS CH8.3.2
Decision statistic	$y = s_m + n$, $n \sim \mathcal{N}(0, N_0/2)$	PS CH8.3.1
Optimal threshold	$\lambda = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_0)}{P(s_1)}$	PS CH8.3.3
Antipodal error	$P_b = Q\left(\sqrt{2E_b/N_0}\right)$	PS CH8.3.3
On-off error	$P_b = Q\left(\sqrt{E_b/N_0}\right)$, three decibels worse	PS CH8.3.3
RC-channel interference	worst case $\sum_m (1 - q)q^m = q$, $q = e^{-2\pi BT_b}$	PS CH10.1.1
Eye closure	opening $2(1 - 2q)$ vanishes at $BT_b = \ln 2/(2\pi) = 0.110$	PS CH10.1.1
Nyquist criterion	$R_b \sum_n P(f - nR_b) = 1$	PS CH10.3.1
Nyquist bandwidth	$W = R_b/2$, the least bandwidth with zero interference	PS CH10.3.1
Raised cosine	$B_T = W(1 + \alpha)$	PS CH10.3.1
Root-raised-cosine split	$H_T(f) = H_R(f) = \sqrt{P(f)}$, matched and free of interference	PS CH10.5.1

Two waveforms have become two points on a line, and the error probability has turned out to depend on the distance between them. Chapter 3 asks what happens when there are more than two waveforms, and finds that the same picture works with more axes.

CHAPTER 3

Geometric representation of signal waveforms

Chapter 2 wrote two opposite waveforms as two points on a line. This chapter writes any set of waveforms as a set of points. Energy becomes a squared length, and the energy of a difference becomes a squared distance. The receiver of Chapter 4 works on these points.

3.1 Signals as vectors

In Chapter 2 the two waveforms were opposites, so one shape $\psi(t)$ carried both. Now take two waveforms on $[0, T]$ that take only the values ± 1 . The first, $s_0(t)$, is $+1$ on the first half and -1 on the second. The second, $s_1(t)$, is $+1$ on the outer quarters and -1 on the middle half.

Neither waveform is a multiple of the other. No single function $\psi(t)$ writes both as $s_m\psi(t)$, so the one-number receiver of Chapter 2 does not apply. The receiver instead computes one number for each waveform.

TWO CORRELATORS

$$y_0 = \int_0^T x(t) s_0(t) dt, \quad y_1 = \int_0^T x(t) s_1(t) dt$$

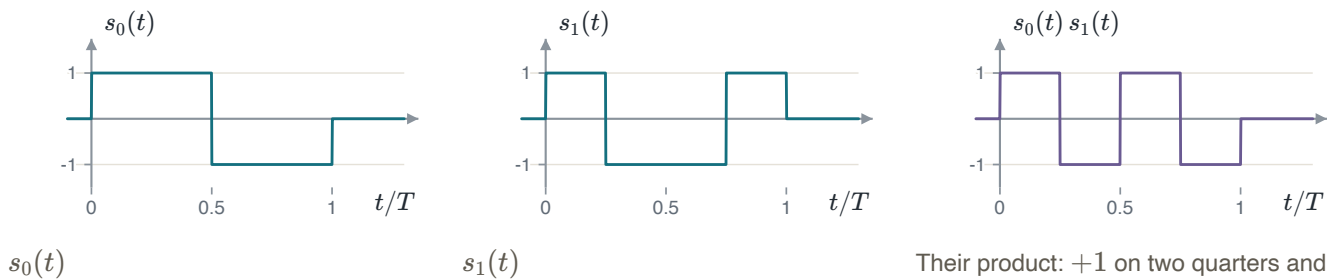
The pair (y_0, y_1) is a point in a plane. The rest of this chapter describes that plane.

Multiply the two waveforms and integrate. The product is $+1$ on two quarters of the interval and -1 on the other two, so the areas cancel.

THE PRODUCT INTEGRATES TO ZERO

$$\int_0^T s_0(t) s_1(t) dt = \frac{T}{2} - \frac{T}{2} = 0$$

Two signals whose product integrates to zero are called **orthogonal**.



Their product: $+1$ on two quarters and -1 on two.

Vectors give the pattern to copy. A vector written against unit axes is the list of its coordinates, $\mathbf{a} = (a_1, \dots, a_N)$. Its length and its angle to another vector come from that list.

INNER PRODUCT AND LENGTH, FOR VECTORS AND FOR SIGNALS

$$\langle \mathbf{a}, \mathbf{b} \rangle = \sum_{k=1}^N a_k b_k, \quad \|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle}$$

$$\langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y(t) dt, \quad \|x\|^2 = \int_{-\infty}^{\infty} x^2(t) dt = E_x$$

For signals the sum over entries becomes an integral over time. The squared length of a signal is its energy.

Axes for signals are functions with unit energy that are orthogonal in pairs. Such a set is called **orthonormal**.

AN ORTHONORMAL SET

$$\int_{-\infty}^{\infty} \psi_j(t) \psi_k(t) dt = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

The first case says each function has unit energy. The second says any two of them are orthogonal.

ORTHOGONAL SIGNALS

Pulses that never overlap are orthogonal. Where one pulse is nonzero, the other is zero. Therefore, their product is zero everywhere.

A pulse $\psi(t) = c$ on $[0, 2]$ has energy $\int_0^2 c^2 dt = 2c^2$. Unit energy needs $2c^2 = 1$, so $c = 1/\sqrt{2} = 0.707$.

With the axes fixed, a waveform is written against them as a vector is. One integral for each axis takes the waveform apart. This step is called **analysis**.

ANALYSIS AND SYNTHESIS

$$s_{ij} = \int_0^T s_i(t) \psi_j(t) dt, \quad j = 1, \dots, N$$
$$s_i(t) = \sum_{j=1}^N s_{ij} \psi_j(t), \quad \mathbf{s}_i = (s_{i1}, \dots, s_{iN})$$

The second line, **synthesis**, builds the waveform back from its N numbers. It is exact when every signal of the set is a combination of the ψ_j . The list $\mathbf{s}_i$ is the **signal vector**.

For example, take $\psi_1 = 1$ on $[0, 1)$ and $\psi_2 = 1$ on $[1, 2)$. Let $s(t) = 3$ on $[0, 1)$ and $s(t) = -1$ on $[1, 2)$. Each coordinate is one integral over the interval where its axis is nonzero.

THE COORDINATES OF ONE WAVEFORM

$$s_1 = \int_0^1 3 \cdot 1 dt = 3$$
$$s_2 = \int_1^2 (-1) \cdot 1 dt = -1$$

So $\mathbf{s} = (3, -1)$, and $s(t) = 3\psi_1(t) - \psi_2(t)$.

The **analyzer** is a bank of N correlators. Correlator j multiplies the waveform by $\psi_j(t)$ and integrates over $[0, T]$. Its output is the coordinate s_{ij} .

The **synthesizer** does the reverse. It scales each $\psi_j(t)$ by s_{ij} and adds the results. So the transmitter can build every waveform of the set from N basis functions.

NUMBER OF CORRELATORS

The analyzer needs one correlator for each basis function, not one for each waveform. A set of $M = 8$ waveforms that spans $N = 2$ dimensions needs 2 correlators.

The translation keeps inner products. The proof takes three moves. Write both signals in the basis. Move the integral inside both sums. Then use orthonormality: each cross term gives 0 and each matching term gives 1.

INNER PRODUCTS ARE PRESERVED

$$\begin{aligned}\int_{-\infty}^{\infty} x(t) y(t) dt &= \int_{-\infty}^{\infty} \left(\sum_{j=1}^N x_j \psi_j(t) \right) \left(\sum_{k=1}^N y_k \psi_k(t) \right) dt \\ &= \sum_{j=1}^N \sum_{k=1}^N x_j y_k \int_{-\infty}^{\infty} \psi_j(t) \psi_k(t) dt \\ &= \sum_{k=1}^N x_k y_k = \langle \mathbf{x}, \mathbf{y} \rangle\end{aligned}$$

An integral over two waveforms has become a sum over N pairs of numbers.

Check it on $x = 2\psi_1 + \psi_2$ and $y = \psi_1 - \psi_2$, with the two unit pulses as basis. The product $x(t)y(t)$ is 2 on $[0, 1)$ and -1 on $[1, 2)$.

THE SIGNED AREA AND THE COORDINATES AGREE

$$\int_0^2 x(t) y(t) dt = 2 - 1 = 1, \quad x_1 y_1 + x_2 y_2 = 2(1) + 1(-1) = 1$$

Two special cases follow. Put $y = x$ to get the energy. Apply the result to the difference $s_i - s_k$, whose vector is $\mathbf{s}_i - \mathbf{s}_k$, to get a distance.

ENERGY AND DISTANCE

$$\begin{aligned}E_i &= \int_0^T s_i^2(t) dt = \sum_{j=1}^N s_{ij}^2 = \|\mathbf{s}_i\|^2 \\ d_{ik}^2 &= \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \int_0^T (s_i(t) - s_k(t))^2 dt\end{aligned}$$

Energy is the squared distance from a point to the origin. The energy of a difference is the squared distance between two points.

Noise moves the received point. Two points far apart are hard to confuse, so the distance d_{ik} decides the error rate. Chapter 4 derives the rule.

For example, $\mathbf{s}_1 = (1, 1)$ and $\mathbf{s}_2 = (1, -1)$ differ by $(0, 2)$. So $d_{12} = \sqrt{0^2 + 2^2} = 2$, and the energy of $\mathbf{s}_1 - \mathbf{s}_2$ is 4.

EXAMPLE 3.1 – A BASIS BY INSPECTION

GIVEN	Four signals on $[0, 2)$. Each is $+1$ or -1 on $[0, 1)$ and $+1$ or -1 on $[1, 2)$, giving all four sign pairs.
FIND	An orthonormal basis, the four signal vectors and the number of basis functions.
METHOD	Every signal is constant on the same two intervals. So read the basis off those pieces, then check unit energy and orthogonality.
SOLUTION	Take $\psi_1 = 1$ on $[0, 1)$ and $\psi_2 = 1$ on $[1, 2)$. Each has energy 1, and $\int \psi_1 \psi_2 dt = 0$ because the pulses never overlap. Each coordinate is the height on that half times its width 1. So $\mathbf{s}_1 = (1, 1)$, $\mathbf{s}_2 = (1, -1)$, $\mathbf{s}_3 = (-1, 1)$, $\mathbf{s}_4 = (-1, -1)$, and $N = 2$.
CHECK	From the vectors, each energy is $1^2 + 1^2 = 2$. From the waveforms, $\int_0^2 (\pm 1)^2 dt = 2$ as well.

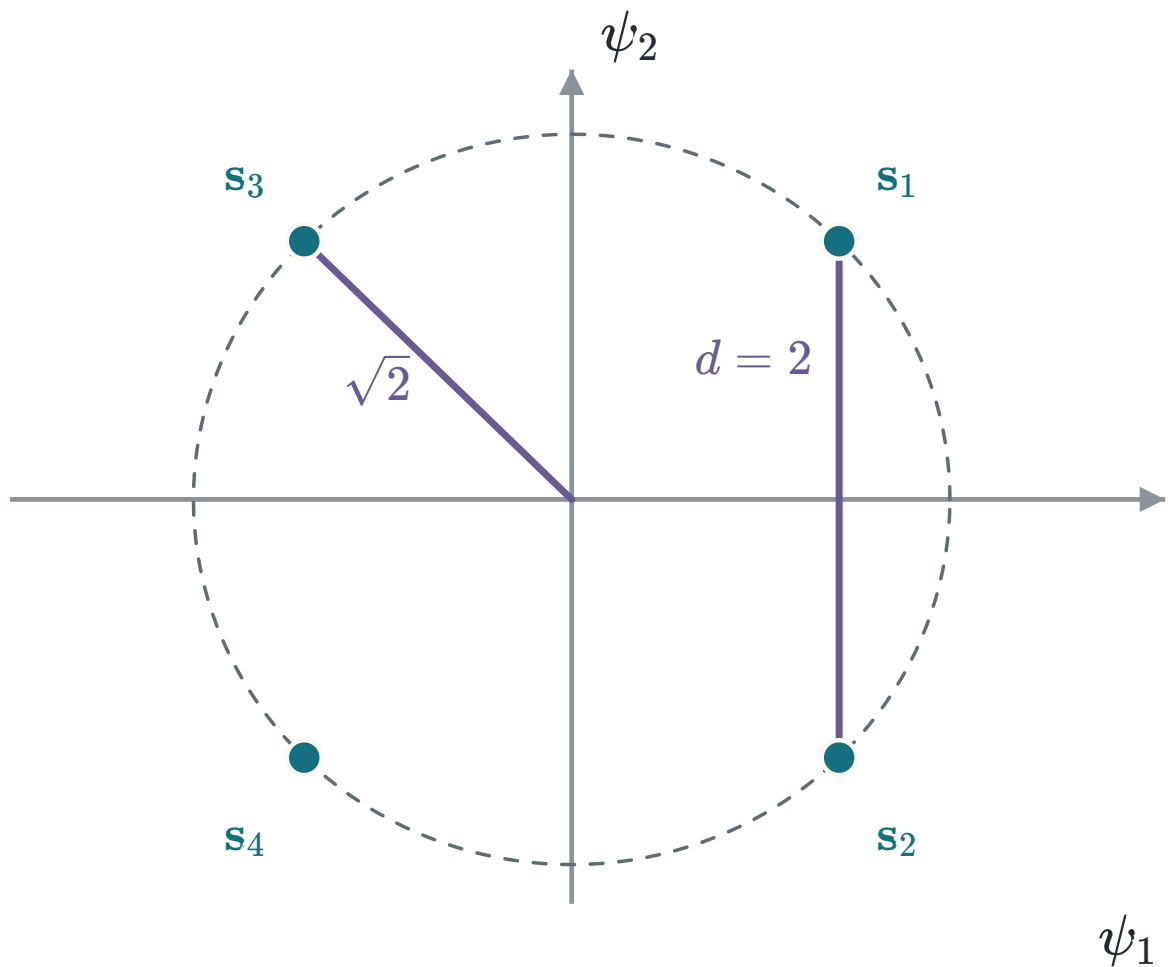
COMMON ERROR

Use a unit-energy function as an axis, not s_1 itself. s_1 has energy 2, so every coordinate on it comes out a factor $\sqrt{2}$ wrong.

3.2 Constellations

The **constellation diagram** draws the signal vectors in the space of their basis. It has one point for each waveform and one axis for each basis function.

Three things are read off it. The distance from the origin to a point is $\sqrt{E_i}$. The distance between two points is the root of the energy of their difference. The number of axes is the number of correlators the receiver needs.



The four signals of Example 3.1 as a constellation. Every point lies on the circle of radius $\sqrt{E} = \sqrt{2}$, and neighbours are $d = 2$ apart.

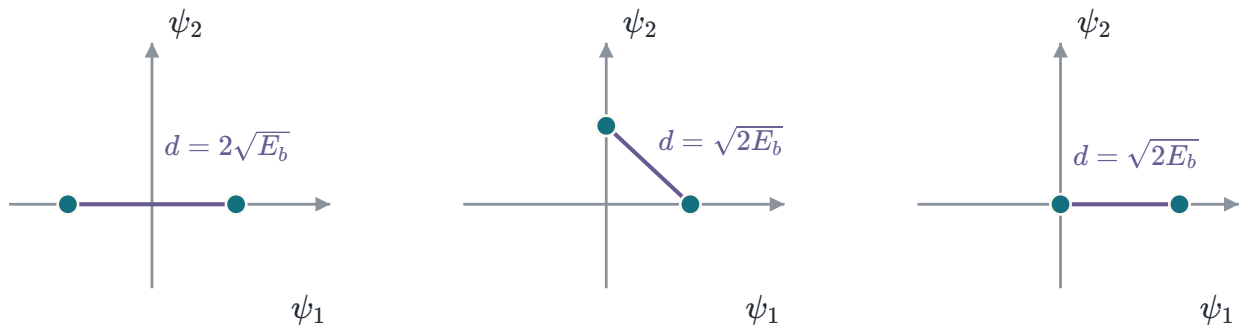
In this square the smallest distance is $d_{\min} = 2$, between neighbours such as $(1, 1)$ and $(1, -1)$. The diagonal pairs are $2\sqrt{2} = 2.83$ apart.

Three binary signal sets can now be compared by the distance between their two points. Each set has the same average energy per bit, E_b .

Antipodal signals satisfy $s_2(t) = -s_1(t)$. They need one axis, and the points $\pm\sqrt{E_b}$ are $d = 2\sqrt{E_b}$ apart. Polar NRZ and BPSK are antipodal.

Orthogonal signals satisfy $\langle s_1, s_2 \rangle = 0$. They need two axes, and the points $(\sqrt{E_b}, 0)$ and $(0, \sqrt{E_b})$ are $d = \sqrt{2E_b}$ apart. PPM and FSK are orthogonal.

On-off signals send 0 or $s(t)$ with equal priors. The average energy is E_b , so $s(t)$ has energy $2E_b$. The points are 0 and $\sqrt{2E_b}$, again $d = \sqrt{2E_b}$ apart.



Antipodal: one axis. Drawn in units of $\sqrt{E_b}$.

Orthogonal: two axes. Drawn in units of $\sqrt{E_b}$.

On-off: one point at the origin. Drawn in units of $\sqrt{E_b}$.

Find the energy each set needs to reach the same distance d . Square the two distance formulas and set them equal.

ENERGY FOR EQUAL DISTANCE

$$4E_b^{\text{anti}} = d^2 = 2E_b^{\text{orth}}$$

$$\frac{E_b^{\text{orth}}}{E_b^{\text{anti}}} = 2 = 3.01 \text{ dB}$$

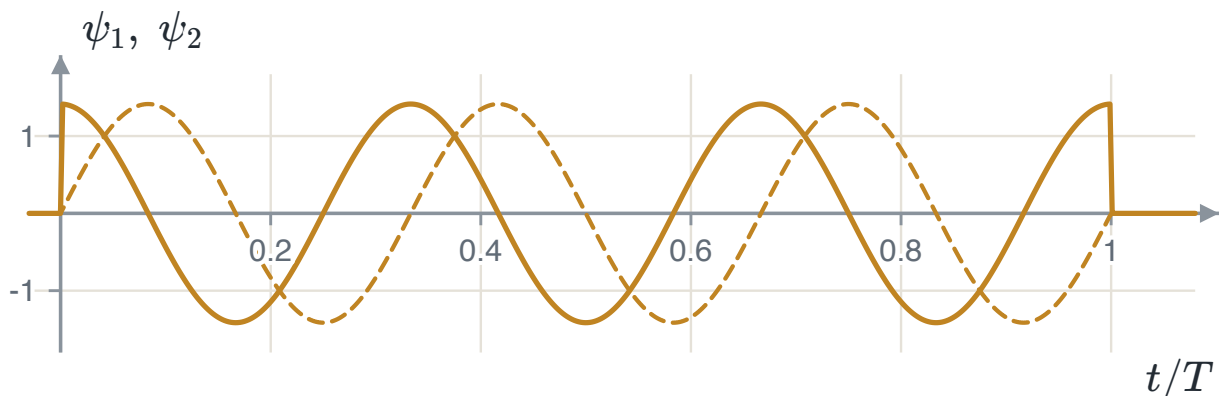
Orthogonal and on-off signalling need twice the energy of antipodal signalling for the same distance. This is the 3 dB gap between the error probabilities $Q\left(\sqrt{2E_b/N_0}\right)$ and $Q\left(\sqrt{E_b/N_0}\right)$ of Chapter 2, where Q is the Gaussian tail.

A carrier gives two axes. A cosine and a sine at the same frequency f_c are orthonormal over $[0, T]$ when each holds whole cycles there.

THE TWO CARRIER AXES

$$\psi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_c t), \quad \psi_2(t) = \sqrt{\frac{2}{T}} \sin(2\pi f_c t), \quad 0 \leq t \leq T$$

$f_c T$ is a whole number, so each function holds whole cycles in $[0, T]$.



The two axes with $T = 1$ and $f_c T = 3$: ψ_1 solid and ψ_2 dashed. Each holds whole cycles in $[0, T]$.

Check unit energy first. Use $2 \cos^2 a = 1 + \cos 2a$, then integrate term by term.

UNIT ENERGY

$$\begin{aligned}\int_0^T \psi_1^2 dt &= \frac{2}{T} \int_0^T \cos^2(2\pi f_c t) dt \\ &= \frac{1}{T} \int_0^T [1 + \cos(4\pi f_c t)] dt \\ &= 1 + \frac{\sin(4\pi f_c T)}{4\pi f_c T} = 1\end{aligned}$$

The sine is zero because $4\pi f_c T$ is a whole multiple of 2π . The same steps with $2\sin^2 a = 1 - \cos 2a$ give unit energy for ψ_2 .

Then check orthogonality. Use $2\cos a \sin a = \sin 2a$.

THE COSINE AND THE SINE ARE ORTHOGONAL

$$\begin{aligned}\int_0^T \psi_1 \psi_2 dt &= \frac{2}{T} \int_0^T \cos(2\pi f_c t) \sin(2\pi f_c t) dt \\ &= \frac{1}{T} \int_0^T \sin(4\pi f_c t) dt \\ &= \frac{1 - \cos(4\pi f_c T)}{4\pi f_c T} = 0\end{aligned}$$

The product holds $2f_c T$ whole cycles, and $\cos(4\pi f_c T) = 1$.

A carrier of energy E and phase θ is one point in this plane. Expand the cosine of a difference to find its coordinates.

A CARRIER WITH PHASE θ

$$\begin{aligned}s(t) &= \sqrt{\frac{2E}{T}} \cos(2\pi f_c t - \theta) \\ &= \sqrt{\frac{2E}{T}} [\cos \theta \cos(2\pi f_c t) + \sin \theta \sin(2\pi f_c t)] \\ &= \sqrt{E} \cos \theta \psi_1(t) + \sqrt{E} \sin \theta \psi_2(t)\end{aligned}$$

The point is $(\sqrt{E} \cos \theta, \sqrt{E} \sin \theta)$, on the circle of radius $\sqrt{E}$. At $\theta = 90^\circ$ it is $(0, \sqrt{E})$, because $\cos(x - 90^\circ) = \sin x$.

ORTHOGONAL CARRIERS

Two sinusoids are orthogonal over T when each holds a whole number of cycles in T and the numbers differ. Wi-Fi spaces its subcarriers $\Delta f = 312.5$ kHz apart. So each holds whole cycles in one symbol of $T = 3.2 \mu\text{s}$.

EXAMPLE 3.2 – FOUR CARRIER WAVEFORMS

GIVEN	For $0 \leq t \leq T$ and whole $f_c T$: $s_{1,2}(t) = \mp \sqrt{2/T} \cos(2\pi f_c t) + \sqrt{2/T} \sin(2\pi f_c t)$ and $s_{3,4}(t) = \mp \sqrt{2/T} \cos(2\pi f_c t) - \sqrt{2/T} \sin(2\pi f_c t)$.
FIND	A basis, the four signal vectors, their energies and $d_{\min}$.
METHOD	Every waveform is already a sum of ψ_1 and ψ_2 above. So the basis is found by inspection, and the coefficients are the coordinates.
SOLUTION	Each waveform is $s_i(t) = s_{i1}\psi_1(t) + s_{i2}\psi_2(t)$ with $s_{ij} = \pm 1$. So $\mathbf{s}_1 = (-1, 1)$, $\mathbf{s}_2 = (1, 1)$, $\mathbf{s}_3 = (-1, -1)$ and $\mathbf{s}_4 = (1, -1)$. Each energy is $1 + 1 = 2$. Neighbours differ by 2 in one coordinate, so $d_{\min} = 2$.
CHECK	The four points form the square of Example 3.1. The waveforms differ, and the constellation is the same.

COMMON ERROR

Use ± 1 as the coordinate, not $\pm \sqrt{2/T}$. The factor $\sqrt{2/T}$ belongs to the basis function ψ_j , so it is already inside the axis.

The pulse set of Example 3.1 and the carrier set of Example 3.2 share one constellation. Sets like these are called **geometrically equivalent**.

Geometrically equivalent sets need the same receiver. They also have the same error probability in white Gaussian noise, because both follow from the points alone.

BANDWIDTH

The constellation does not fix the bandwidth. The pulse set occupies frequencies near $f = 0$, and the carrier set occupies frequencies near f_c . Bandwidth follows from the waveform shapes.

M carriers of equal energy at equally spaced phases form **M-ary phase shift keying**, or M-PSK. Every point lies on one circle.

THE M-PSK SIGNAL SET

$$s_i(t) = A \cos\left(2\pi f_c t - \frac{2\pi i}{M}\right), \quad i = 0, \dots, M-1$$
$$\mathbf{s}_i = \sqrt{E} \left(\cos \frac{2\pi i}{M}, \sin \frac{2\pi i}{M} \right), \quad E = \frac{A^2 T}{2}$$

The second line is the carrier result above with $\theta = 2\pi i/M$ and $A = \sqrt{2E/T}$.

Neighbouring points are $2\pi/M$ apart in angle. Take the difference of $\mathbf{s}_0$ and $\mathbf{s}_1$, square its length, and use $1 - \cos 2a = 2 \sin^2 a$.

THE DISTANCE BETWEEN NEIGHBOURS

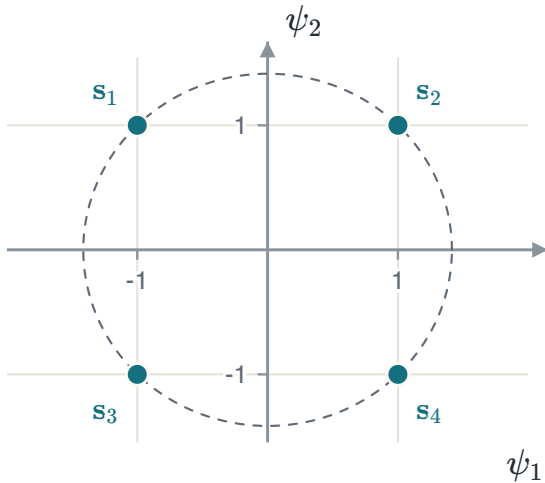
$$\begin{aligned} d_{\min}^2 &= E \left[\left(1 - \cos \frac{2\pi}{M}\right)^2 + \sin^2 \frac{2\pi}{M} \right] \\ &= E \left[2 - 2 \cos \frac{2\pi}{M} \right] \\ &= 4E \sin^2 \frac{\pi}{M} \end{aligned}$$

$$d_{\min} = 2\sqrt{E} \sin \frac{\pi}{M}$$

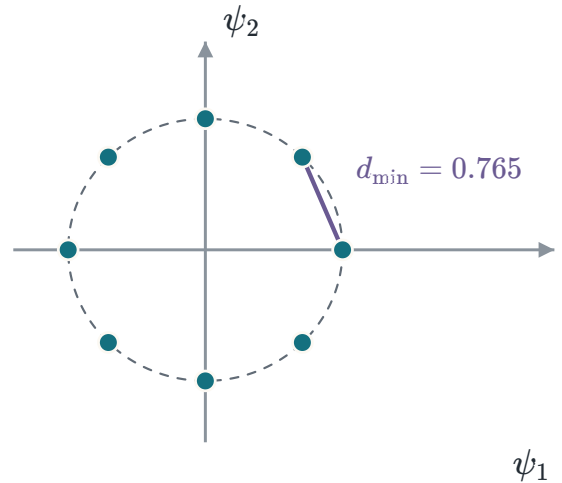
The middle step uses $\cos^2 a + \sin^2 a = 1$. Each doubling of M adds one bit a symbol and brings neighbours closer.

For 8-PSK at $E = 1$, $d_{\min} = 2 \sin(\pi/8) = 2(0.383) = 0.765$. QPSK at $E = 2$ gives $d_{\min} = 2\sqrt{2} \sin(\pi/4) = 2\sqrt{2}(0.707) = 2$, the square of Example 3.2.

Square grids of points, called QAM, carry more bits a symbol. At average energy 1, $d_{\min}$ is 2 for BPSK, 0.632 for 16-QAM and 0.309 for 64-QAM.



Example 3.2 as QPSK: four points on the circle of radius $\sqrt{2}$, with $d_{\min} = 2$.



8-PSK at $E = 1$: eight points on the unit circle. Neighbours are 0.765 apart.

3.3 The Gram–Schmidt procedure

So far each basis was found by inspection. The **Gram–Schmidt procedure** finds one for any set of M waveforms. It takes the signals in turn and keeps only what is new in each.

Start with the first signal and scale it to unit energy.

STEP 1

$$E_1 = \int s_1^2(t) dt, \quad \psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}}$$

The first axis is the first signal scaled to unit energy. The coordinate of s_1 on it is $s_{11} = \sqrt{E_1}$.

For each later signal s_k , find its coordinates on the axes found so far. Subtract those parts. Then scale the nonzero remainder to unit energy.

STEP k

$$s_{ki} = \int s_k(t) \psi_i(t) dt, \quad g_k(t) = s_k(t) - \sum_{i=1}^{k-1} s_{ki} \psi_i(t)$$

$$\psi_k(t) = \frac{g_k(t)}{\sqrt{E_{g_k}}}, \quad E_{g_k} = \int g_k^2(t) dt$$

The remainder $g_k(t)$ is orthogonal to every earlier axis, so ψ_k is a new axis.

STOP RULE

If $g_k(t) = 0$, then s_k is a combination of earlier signals and adds no axis. So $N \leq M$, with $N = M$ only when no signal is a combination of the others.

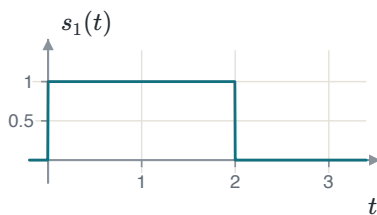
The energy of a remainder follows from the energy result. With one earlier axis, g_2 is orthogonal to ψ_1 . So the energy of s_2 splits into two squared lengths.

THE ENERGY OF THE REMAINDER

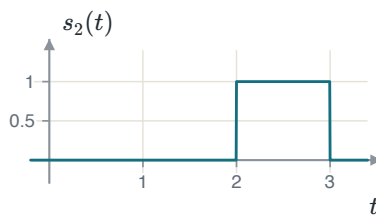
$$E_2 = s_{21}^2 + E_{g_2} \implies E_{g_2} = E_2 - s_{21}^2$$

For example, $E_2 = 25$ and $s_{21} = 3$ give $E_{g_2} = 25 - 3^2 = 16$.

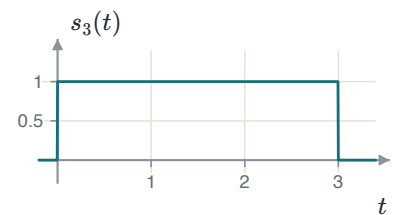
Example 3.3 runs the procedure on three rectangular pulses. The third pulse is the first one followed by the second.



s_1



s_2



$s_3 = s_1 + s_2$

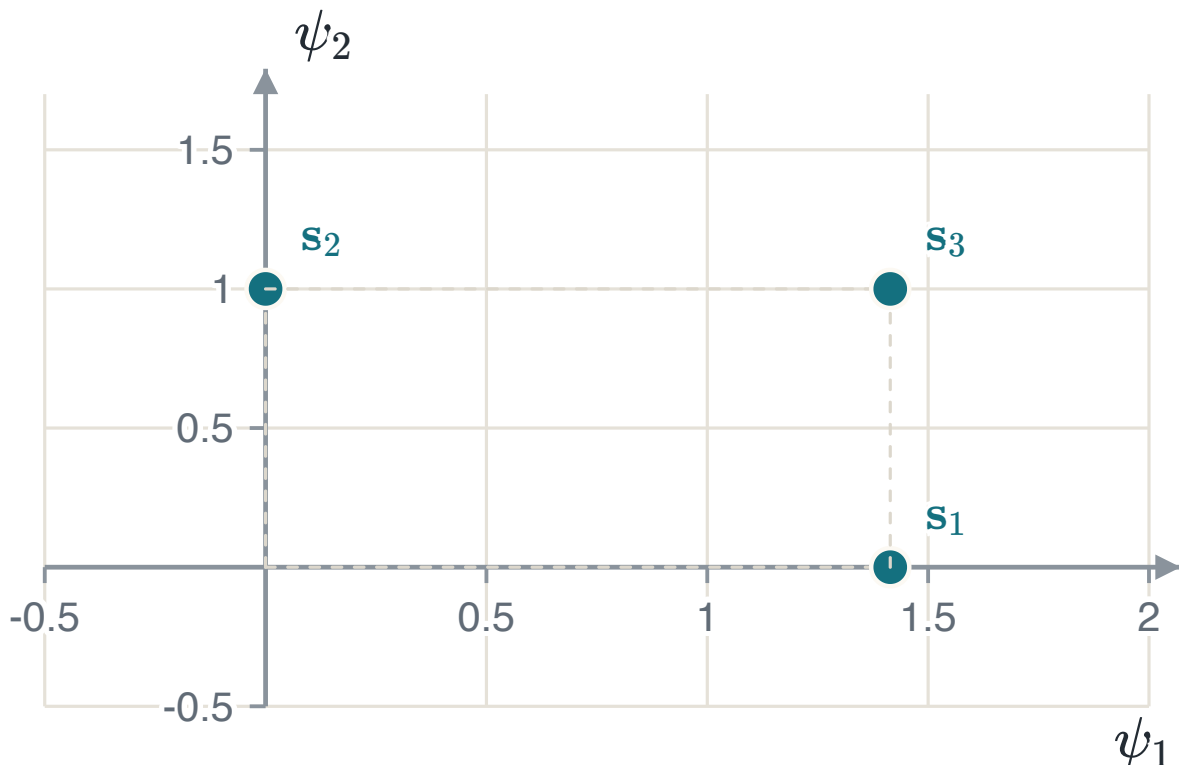
EXAMPLE 3.3 – THREE PULSES, TWO AXES

- GIVEN** $s_1(t) = 1$ on $[0, 2]$, $s_2(t) = 1$ on $[2, 3]$ and $s_3(t) = 1$ on $[0, 3]$. Each is zero elsewhere.
- FIND** An orthonormal basis, the three signal vectors and the constellation.
- METHOD** Gram–Schmidt needs no guess about the basis. Normalise the first signal. For each later signal, subtract its parts along the axes found so far and normalise the remainder.
- STEP 1** $E_1 = \int_0^2 1^2 dt = 2$, so $\psi_1(t) = 1/\sqrt{2}$ on $[0, 2]$ and $s_{11} = \sqrt{2}$.
- STEP 2** $s_{21} = \int s_2 \psi_1 dt = 0$, because the two never overlap. So $g_2 = s_2$, its energy is 1, and $\psi_2(t) = s_2(t)$.
- STEP 3** $s_{31} = \int_0^2 1 \cdot \frac{1}{\sqrt{2}} dt = \sqrt{2}$ and $s_{32} = \int_2^3 1 \cdot 1 dt = 1$. Then $g_3 = s_3 - \sqrt{2} \psi_1 - \psi_2 = 0$, so no third axis is added.
- SOLUTION** $\mathbf{s}_1 = (\sqrt{2}, 0)$, $\mathbf{s}_2 = (0, 1)$ and $\mathbf{s}_3 = (\sqrt{2}, 1)$, with $N = 2$.
- CHECK** Energies from the vectors are 2, 1 and 3. From the waveforms, height 1 over lengths 2, 1 and 3 gives the same values. Three waveforms need only two axes, because $s_3 = s_1 + s_2$.

COMMON ERROR

Use $\psi_1 = 0$ on $[2, 3]$, not $\psi_1 = 1/\sqrt{2}$ there. The integral for s_{31} then runs over $[0, 2]$ only, and $s_{31} = \sqrt{2}$.

The constellation of Example 3.3 has two axes and three points.



Two axes carry three signals. The point furthest from the origin is $\mathbf{s}_3$, whose energy is 3.

Example 3.4 needs three axes for four signals. Its pieces overlap, so no basis can be read off at once.

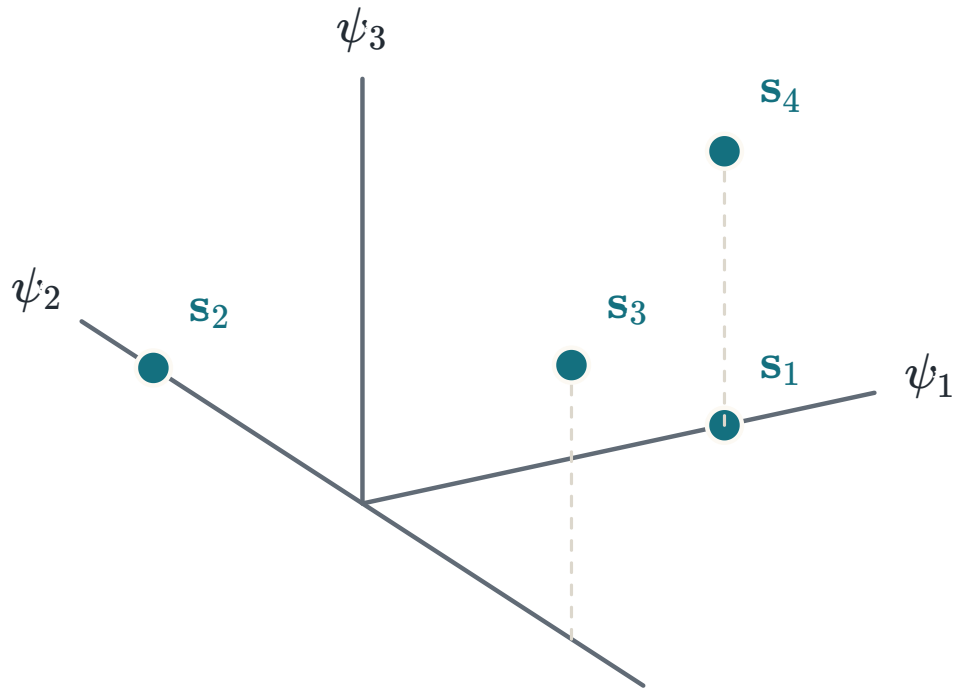
EXAMPLE 3.4 – FOUR SIGNALS, THREE AXES

GIVEN	$s_1 = 1$ on $[0, 2)$. $s_2 = 1$ on $[0, 1)$ and -1 on $[1, 2)$. $s_3 = -1$ on $[0, 1)$ and 1 on $[1, 3)$. $s_4 = 1$ on $[0, 3)$. Each is zero elsewhere.
FIND	The basis, the four signal vectors and the number of dimensions N .
METHOD	Run Gram–Schmidt in the order s_1, s_2, s_3, s_4 . Each coordinate is a sum of piece heights times piece widths.
STEP 1	$E_1 = \int_0^2 1^2 dt = 2$, so $\psi_1 = s_1/\sqrt{2}$.
STEP 2	$s_{21} = \frac{1}{\sqrt{2}}(1 - 1) = 0$. So $g_2 = s_2$, with energy 2, and $\psi_2 = s_2/\sqrt{2}$.
STEP 3	$s_{31} = \frac{1}{\sqrt{2}}(-1 + 1) = 0$ and $s_{32} = \frac{1}{\sqrt{2}}(-1 - 1) = -\sqrt{2}$. Then $g_3 = s_3 + \sqrt{2}\psi_2 = s_3 + s_2$, which is 1 on $[2, 3)$ and zero elsewhere. Its energy is 1, so $\psi_3 = g_3$.
STEP 4	$s_{41} = \frac{1}{\sqrt{2}}(1 + 1) = \sqrt{2}$, $s_{42} = \frac{1}{\sqrt{2}}(1 - 1) = 0$ and $s_{43} = \int_2^3 1 dt = 1$. Then $g_4 = s_4 - \sqrt{2}\psi_1 - \psi_3 = 0$, so s_4 adds no axis.
SOLUTION	$\mathbf{s}_1 = (\sqrt{2}, 0, 0)$, $\mathbf{s}_2 = (0, \sqrt{2}, 0)$, $\mathbf{s}_3 = (0, -\sqrt{2}, 1)$ and $\mathbf{s}_4 = (\sqrt{2}, 0, 1)$, with $N = 3$.
CHECK	Energies from the vectors are 2, 2, 3 and 3, equal to $\int s_i^2 dt$. Also $\mathbf{s}_1 + \mathbf{s}_2 + \mathbf{s}_3 = \mathbf{s}_4$, matching $s_4 = s_1 + s_2 + s_3$.

COMMON ERROR

Count one axis for each nonzero remainder, not one for each signal. Here $g_4 = 0$, so $N = 3$, not 4.

The four vectors of Example 3.4 are drawn in three dimensions below, seen from an angle.



Four signals need three axes. s_4 sits one unit above s_1 , along ψ_3 .

The basis is not unique. Taking the signals in another order in Gram–Schmidt gives another orthonormal basis of the same space. The new axes are the old ones turned or reflected.

The coordinates change. The number of axes, the energies, the inner products and the distances do not, because each is an integral of the waveforms.

UNCHANGED BY A CHANGE OF BASIS

$$N, \quad \|s_i\|^2 = E_i, \quad \langle s_i, s_k \rangle, \quad \|s_i - s_k\|$$

For example, turn the axes of Example 3.1 by 45° . The new first axis is $\psi'_1 = (\psi_1 + \psi_2)/\sqrt{2}$, and the new second axis is $\psi'_2 = (\psi_2 - \psi_1)/\sqrt{2}$.

THE NEW COORDINATES OF $s_1 = (1, 1)$

$$\langle s_1, \psi'_1 \rangle = \frac{1+1}{\sqrt{2}} = \sqrt{2}$$

$$\langle s_1, \psi'_2 \rangle = \frac{-1+1}{\sqrt{2}} = 0$$

All the length $\sqrt{2}$ of s_1 now lies on ψ'_1 . Its energy is still 2.

3.4 Summary

RESULT	STATEMENT	ANCHOR
Orthonormal set	$\int \psi_j \psi_k dt = 1$ if $j = k$, else 0	PS CH8.1
Analysis	$s_{ij} = \int_0^T s_i \psi_j dt$, one correlator for each basis function	PS CH8.1
Synthesis	$s_i(t) = \sum_j s_{ij} \psi_j(t)$	PS CH8.1
Inner products preserved	$\int xy dt = \sum_k x_k y_k$	PS CH8.1
Energy	$E_i = \ \mathbf{s}_i\ ^2$	PS CH8.1
Distance	$d_{ik}^2 = \ \mathbf{s}_i - \mathbf{s}_k\ ^2 = \int (s_i - s_k)^2 dt$	PS CH8.1
Constellation	one point for each waveform, one axis for each basis function	PS CH8.1
Binary sets	antipodal $d = 2\sqrt{E_b}$, orthogonal and on-off $d = \sqrt{2E_b}$, a 3 dB gap	PS CH8.2
Carrier axes	$\sqrt{2/T} \cos(2\pi f_c t)$ and $\sqrt{2/T} \sin(2\pi f_c t)$, orthonormal for whole $f_c T$	PS CH8.6.1
QPSK	$(\pm 1, \pm 1)$, energy 2, $d_{\min} = 2$	PS CH8.6.1
M-PSK	$d_{\min} = 2\sqrt{E} \sin(\pi/M)$	PS CH8.6.1
Geometric equivalence	same constellation, same receiver and error probability, bandwidth not fixed	PS CH8.1
Gram–Schmidt	$g_k = s_k - \sum_{i < k} s_{ki} \psi_i$, and $\psi_k = g_k / \sqrt{E_{g_k}}$	PS CH8.1
Dimension	$N \leq M$, and a zero remainder adds no axis	PS CH8.1
Change of basis	N , energies and distances unchanged	PS CH8.1

Every waveform set is now a set of points, with energy as squared length and the energy of a difference as squared distance. Chapter 4 builds the receiver on these points. It correlates to find the received point and picks the nearest signal point. Its error probability depends only on the distances between the points.

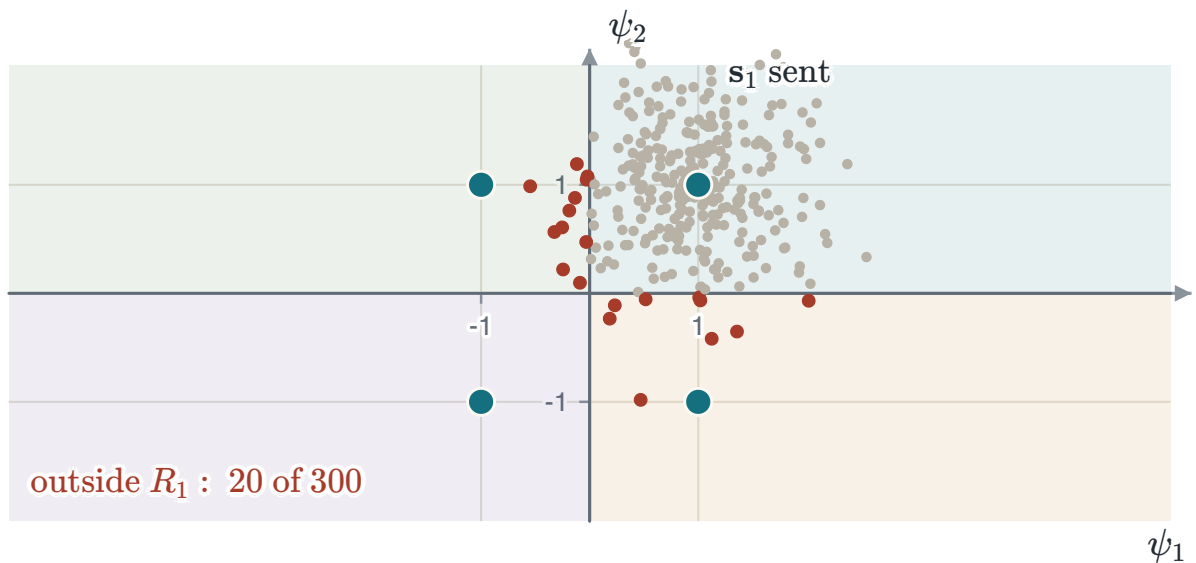
CHAPTER 4

The optimal receiver in additive white Gaussian noise

One of M signals is sent and the channel adds noise. The receiver turns what arrives into a point and picks the nearest signal point.

A bank of correlators turns $r(t)$ into a point $\mathbf{r}$. The best rule picks the nearest signal point, with a handicap for each prior. How often the receiver is wrong depends only on the distances between the points. The union bound turns those distances into a number.

The figure below sends one point many times and marks the received points that land in the wrong region.



Four signal points and their decision regions. The grey dots are received points when s_1 is sent, and the red ones fall outside its region R_1 .

4.1 The observation

Bits into symbols

With M waveforms, each symbol carries $k = \log_2 M$ bits. The bit stream is read in blocks of k bits, and each block picks one waveform.

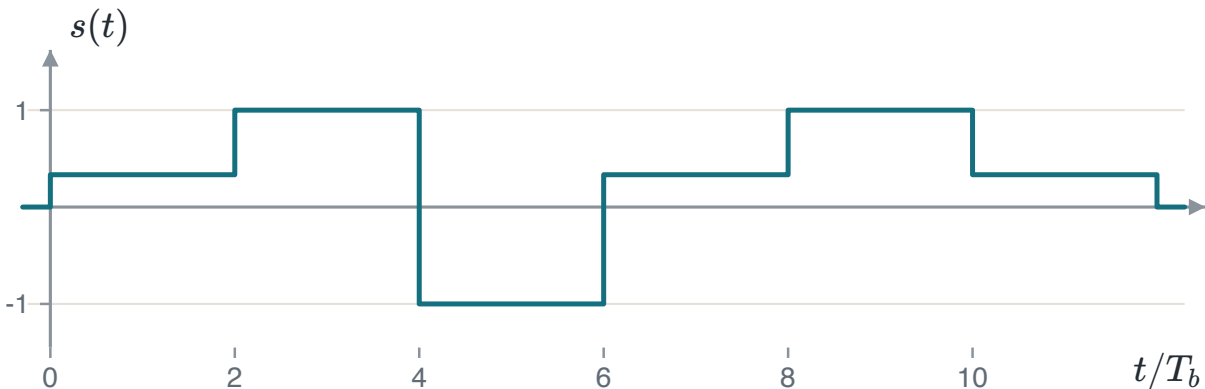
BITS AND SYMBOLS

$$k = \log_2 M, \quad R_s = \frac{1}{T}, \quad R_b = kR_s, \quad T_b = \frac{T}{k}$$

Each symbol carries k bits and lasts T . The symbol rate R_s counts waveforms a second. The bit rate R_b counts bits.

Each T seconds the transmitter sends one of the M waveforms $s_1(t), \dots, s_M(t)$. The receiver must name it from a noisy $r(t)$.

The figure below reads twelve bits in blocks of two. Each block picks one of four levels and holds it for $T = 2T_b$.



Twelve bits read in blocks of $k = 2$, so $M = 2^k = 4$ and $T = kT_b = 2T_b$. The number under each block is the level it picks, from 0 to 3.

For example, a modem sends 2400 symbols a second with $M = 16$. Then $k = \log_2 16 = 4$ bits a symbol, and $R_b = 4 \times 2400 = 9600$ b/s.

The observation vector

Over one symbol the receiver sees $r(t) = s_i(t) + n(t)$. Here $n(t)$ is white Gaussian noise of two-sided power spectral density $N_0/2$.

The receiver correlates $r(t)$ with each of the N orthonormal basis functions $\psi_1(t), \dots, \psi_N(t)$ of the signal set. Each correlator multiplies by one basis function and integrates over one symbol.

CORRELATOR OUTPUTS

$$\begin{aligned} r_j &= \int_0^T r(t) \psi_j(t) dt \\ &= \int_0^T s_i(t) \psi_j(t) dt + \int_0^T n(t) \psi_j(t) dt \\ &= s_{ij} + n_j, \quad j = 1, \dots, N \end{aligned}$$

The integral is linear, so it splits over the sum. The first integral is the coordinate s_{ij} of the sent signal. The second is a noise coordinate n_j .

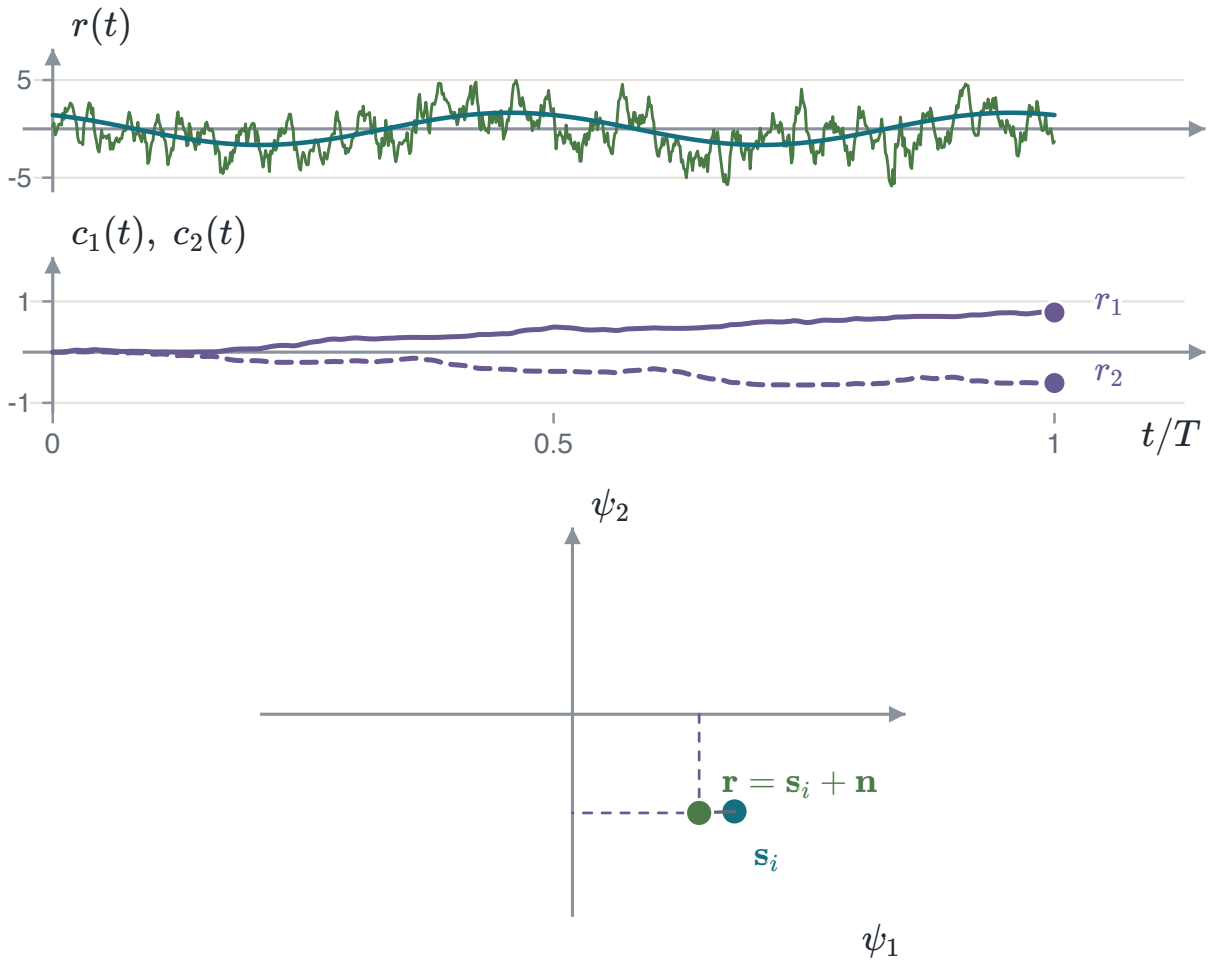
Stack the N outputs into one vector. The result is the observation the receiver decides from.

THE OBSERVATION VECTOR

$$\mathbf{r} = \mathbf{s}_i + \mathbf{n}$$

The receiver now works with one point $\mathbf{r}$. The noise moves it away from the sent point $\mathbf{s}_j$.

The figure below follows one received waveform, with $T = 1$, through two correlators.



Top: $r(t)$, green, is $s_i(t)$, cyan, plus noise. Middle: the running integrals $c_1(t)$, solid, and $c_2(t)$, dashed, of $r(t) \psi_j(t)$. Bottom: their end values r_1 and r_2 are the coordinates of $\mathbf{r}$.

For example, let $\mathbf{s}_i = (2, -1)$ be sent with noise coordinates $n_1 = 0.3$ and $n_2 = -0.4$. Then $\mathbf{r} = \mathbf{s}_i + \mathbf{n} = (2 + 0.3, -1 - 0.4) = (2.3, -1.4)$.

The matched-filter bank

A bank of matched filters gives the same numbers as the correlators. The filter for axis j is its basis function turned around in time.

MATCHED FILTER

$$h_j(t) = \psi_j(T - t)$$

The filter output is the convolution of $r(t)$ with $h_j(t)$. Write the convolution, then put in $h_j(t - \tau) = \psi_j(T - t + \tau)$.

$$\begin{aligned} y_j(t) &= \int_0^T r(\tau) h_j(t - \tau) d\tau \\ &= \int_0^T r(\tau) \psi_j(T - t + \tau) d\tau \end{aligned}$$

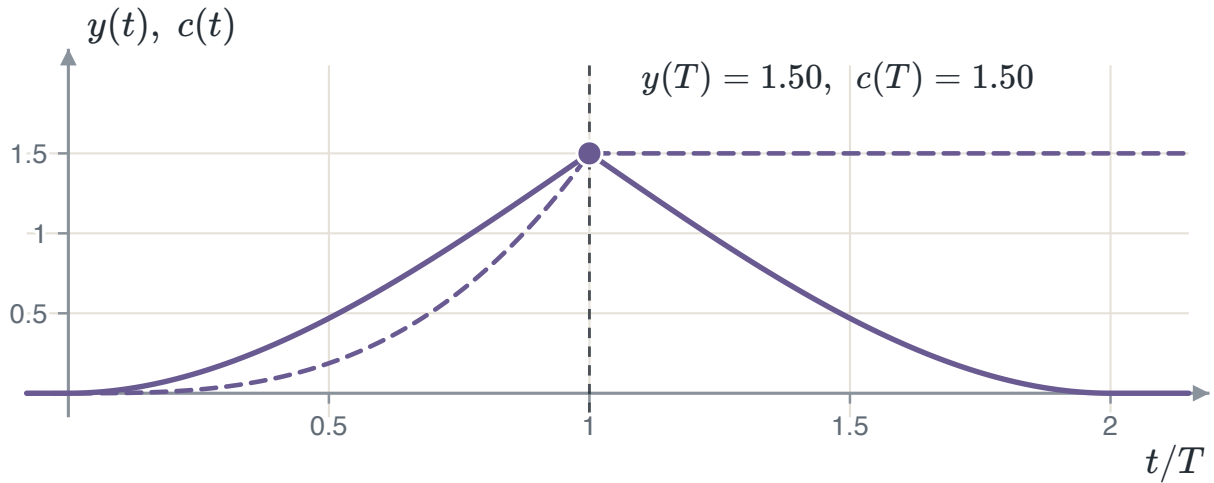
Now sample the output at $t = T$. The shift $T - t$ becomes zero, and the integral is the correlator.

SAMPLE AT T

$$y_j(T) = \int_0^T r(\tau) \psi_j(\tau) d\tau = r_j$$

At any other time the two outputs differ.

The figure below uses $\psi(t) = \sqrt{3}t$ on $[0, 1]$ and $r(t) = 1.5\psi(t)$, with no noise. The two curves meet only at $t = T = 1$.



Solid: the matched filter output $y(t)$. Dashed: the correlator output $c(t)$. At the sampling time $t_0 = T$ both equal 1.5.

The value at T follows from the energy of ψ . The correlator gives $y(1) = 1.5 \int_0^1 \psi^2(t) dt = 1.5 \int_0^1 3t^2 dt = 1.5$.

The noise the receiver ignores

Noise has parts in every direction, not only in the signal space. The part outside the span of $\psi_1, \dots, \psi_N$ is written $n'(t)$. It does not depend on the signal sent.

Let $\tilde{\mathbf{r}}$ be the received waveform with nothing dropped. It is $\mathbf{r}$ plus n' , and n' is at right angles to the signal space. So Pythagoras splits each squared distance into two parts.

$$\tilde{\mathbf{r}} - \mathbf{s}_j = (\mathbf{r} - \mathbf{s}_j) + n'$$

THE NOISE OUTSIDE THE SIGNAL SPACE

$$\|\tilde{\mathbf{r}} - \mathbf{s}_j\|^2 = \|\mathbf{r} - \mathbf{s}_j\|^2 + \|n'\|^2$$

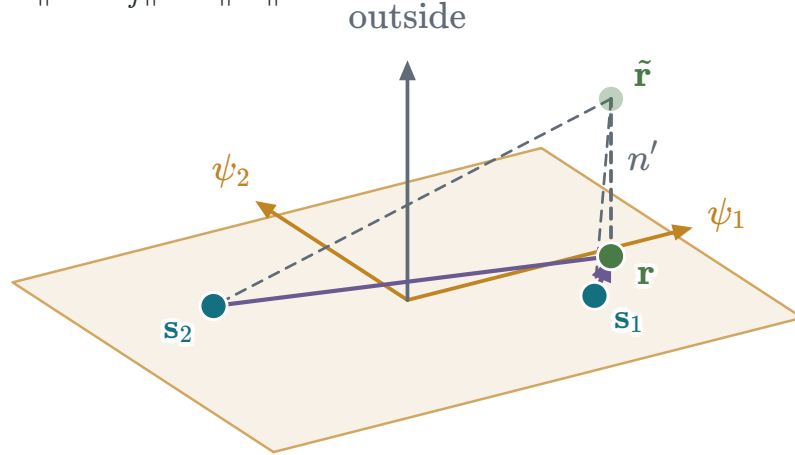
The second part is the same for every j . It cannot change which signal is nearest, so the receiver can drop it.

SUFFICIENT STATISTICS

The N correlator outputs $r_1, \dots, r_N$ hold everything the decision needs. Nothing is lost by dropping $n'(t)$. Outputs with this property are called **sufficient statistics**.

The figure below draws the signal space as a sheet in three dimensions.

$$\|\tilde{\mathbf{r}} - \mathbf{s}_j\|^2 = \|\mathbf{r} - \mathbf{s}_j\|^2 + \|n'\|^2$$



The amber sheet is the signal space, spanned by ψ_1 and ψ_2 . The noise splits into a part in the sheet and n' outside it. Only the part in the sheet moves $\mathbf{r}$.

For example, let the squared distances in the sheet be $\|\mathbf{r} - \mathbf{s}_1\|^2 = 0.5$ and $\|\mathbf{r} - \mathbf{s}_2\|^2 = 2.0$, with $\|n'\|^2 = 0.8$. The full squared distances are $0.5 + 0.8 = 1.3$ and $2.0 + 0.8 = 2.8$. The gap stays 1.5, so $\mathbf{s}_1$ still wins.

The noise components

Each noise coordinate is the noise correlated with one basis function. It is a weighted integral of a Gaussian process, so it is Gaussian.

ONE COMPONENT

$$n_j = \int_0^T n(t) \psi_j(t) dt, \quad E[n_j] = 0$$

The noise has mean zero, and the integral keeps the mean at zero.

The variance of one component, and the correlation of two, follow from the autocorrelation of white noise, $E[n(t)n(u)] = \frac{N_0}{2} \delta(t - u)$.

$$\begin{aligned} E[n_j n_k] &= \int_0^T \int_0^T E[n(t)n(u)] \psi_j(t) \psi_k(u) dt du \\ &= \int_0^T \int_0^T \frac{N_0}{2} \delta(t - u) \psi_j(t) \psi_k(u) dt du \\ &= \frac{N_0}{2} \int_0^T \psi_j(t) \psi_k(t) dt \end{aligned}$$

The delta function collapses the integral over u . Orthonormality then gives 1 for the last integral when $j = k$, and 0 otherwise.

TWO COMPONENTS

$$E[n_j^2] = \frac{N_0}{2}, \quad E[n_j n_k] = 0, \quad j \neq k$$

Gaussian variables that are uncorrelated are independent. So the N components are independent, each with variance $N_0/2$.

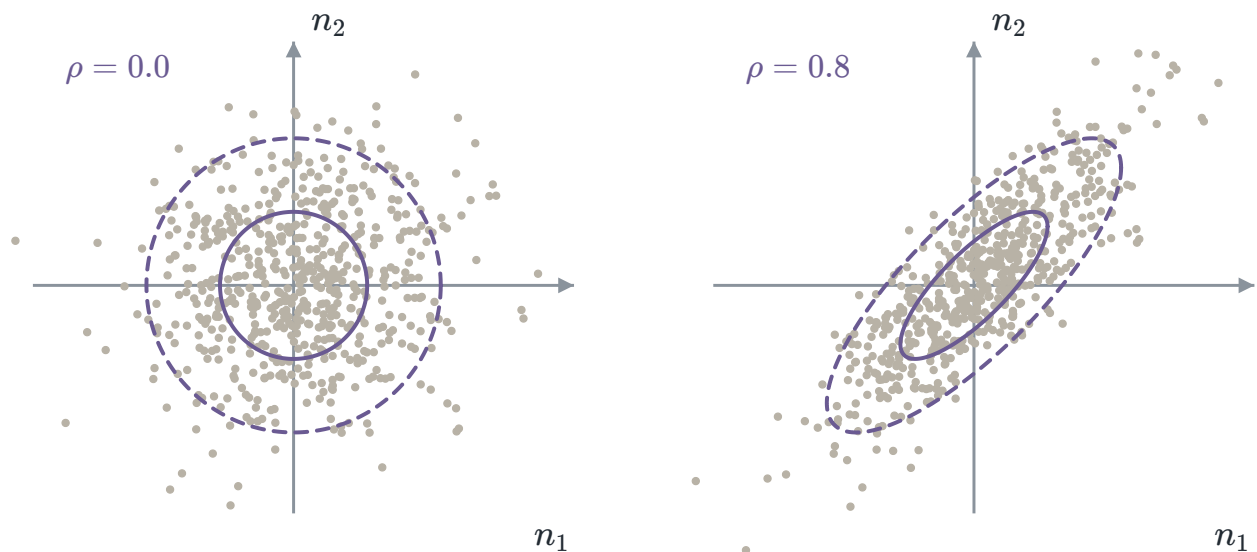
Every axis has the same variance, and the axes are independent. So the cloud of received points around a signal point is round.

AN ELLIPSE

Coloured noise gives correlated components, with correlation coefficient $\rho \neq 0$. The cloud tilts into an ellipse, and the nearest point is no longer the best guess.

For example, noise with $\rho = 0.8$ reaches the two correlators. A positive ρ makes n_2 follow n_1 , so the cloud stretches along the line $n_1 = n_2$.

The figure below draws two components of variance 1 for two values of ρ .



$\rho = 0$: independent components and a round cloud. The solid and dashed curves hold about 39% and 86% of the points.

$\rho = 0.8$: n_2 follows n_1 , and the cloud stretches along the line $n_1 = n_2$.

EXAMPLE 4.1 – FOUR-LEVEL PAM

GIVEN Four-level PAM, $s_m \in \{-1.5, -0.5, 0.5, 1.5\}$ on a unit-energy pulse, with $N_0 = 0.1$.

FIND The conditional density $f(r | s_m)$ of the correlator output for each level, and the height of each bell.

METHOD One basis function means one correlator and one number, $r = s_m + n$. The noise n is Gaussian with mean 0 and variance $\sigma^2 = N_0/2$. So r is Gaussian and centred on s_m .

SOLUTION Put $\sigma^2 = N_0/2$ into the Gaussian density.

$$f(r | s_m) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(r-s_m)^2/2\sigma^2} = \frac{1}{\sqrt{\pi N_0}} e^{-(r-s_m)^2/N_0}$$

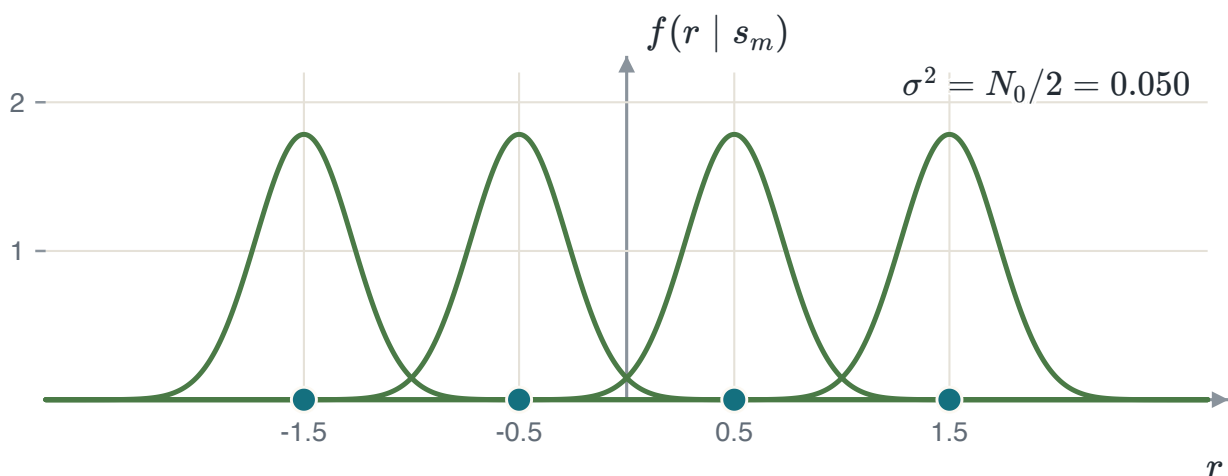
Here $\sigma^2 = 0.05$. The peak is $1/\sqrt{\pi N_0} = 1/\sqrt{0.1\pi} = 1.78$. The four bells have one shape, each centred on its level.

CHECK Each bell has $\sigma = \sqrt{0.05} = 0.22$, well inside half the spacing between levels, 0.5. At $N_0 = 0.6$, $\sigma = 0.55$ and neighbouring bells overlap.

COMMON ERROR

Use $N_0/2$, not N_0 , as the variance. N_0 makes each bell $\sqrt{2}$ times too wide.

The figure below draws the four densities of Example 4.1.



The four conditional densities of Example 4.1, drawn for $N_0 = 0.1$. Each bell sits on one level, and its width grows with $\sqrt{N_0}$.

EXAMPLE 4.2 – FOUR ORTHOGONAL SIGNALS

GIVEN Four orthogonal signals of energy $E = 4$, with $N_0 = 1$. $\mathbf{s}_1 = (2, 0, 0, 0)$ is sent.

FIND The density of each correlator output r_k , and the mean of r_1 .

METHOD Each signal lies along its own axis, at distance $\sqrt{E}$ from the origin. So only the first correlator sees the signal.

$$r_1 = \sqrt{E} + n_1, \quad r_k = n_k, \quad k = 2, 3, 4$$

SOLUTION Each n_k is Gaussian with variance $N_0/2 = 0.5$, so $2\sigma^2 = 1$.

$$f(r_1 | \mathbf{s}_1) = \frac{e^{-(r_1-2)^2}}{\sqrt{\pi}}, \quad f(r_k | \mathbf{s}_1) = \frac{e^{-r_k^2}}{\sqrt{\pi}}$$

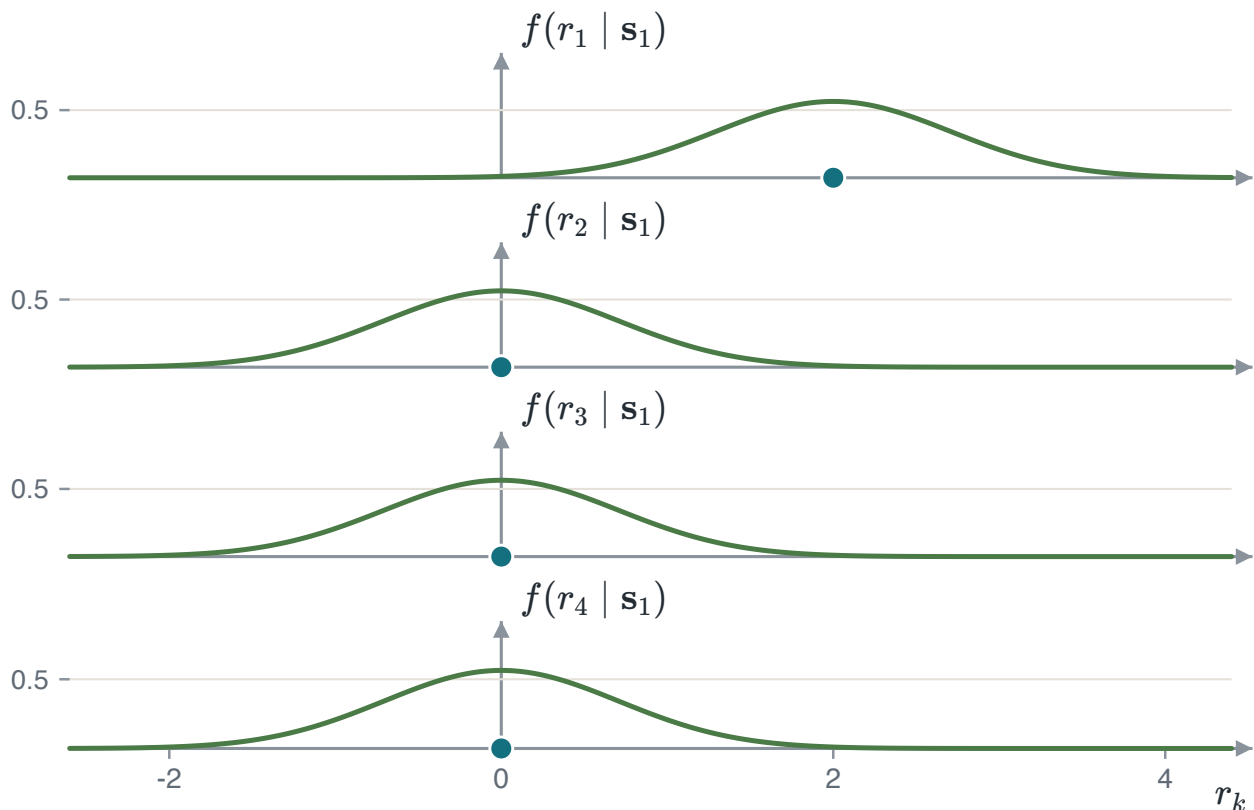
The mean of r_1 is $\sqrt{E} = 2$. Each density has peak $1/\sqrt{\pi} = 0.56$, and the four outputs are independent.

CHECK The means $(2, 0, 0, 0)$ have squared length $2^2 = 4 = E$, as they must.

COMMON ERROR

Use $\sqrt{E} = 2$, not $E = 4$, as the mean of r_1 . The coordinate of $\mathbf{s}_1$ is its length, and E is its squared length.

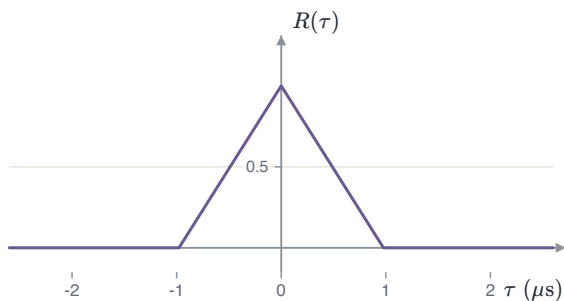
The figure below draws the four densities of Example 4.2.



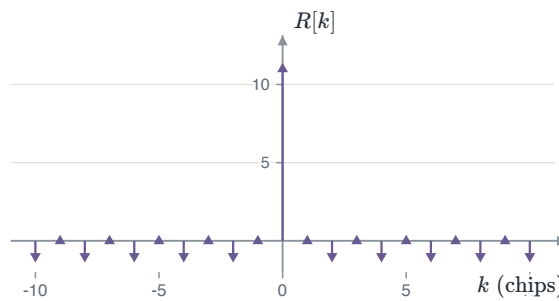
The four correlator outputs of Example 4.2 when $\mathbf{s}_1$ is sent. Only r_1 is centred away from zero.

Correlators around us

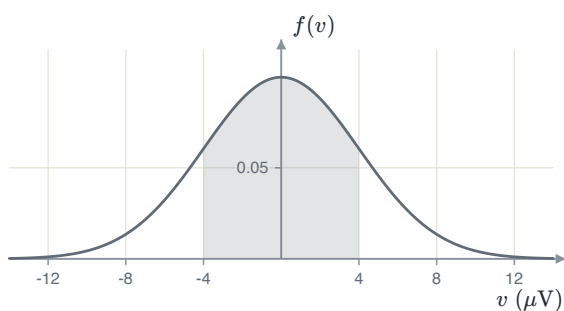
Four everyday receivers correlate what arrives with a known waveform.



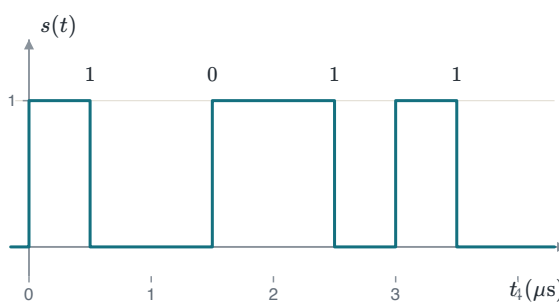
A GPS receiver correlates the C/A code with a local copy.
 $R(\tau) = 1 - |\tau|/T_c$, with $T_c = 0.978 \mu\text{s}$, peaks when the two align.



802.11b Wi-Fi spreads each bit with the 11-chip Barker code. Its correlation $R[k]$ is 11 at $k = 0$ and 0 or -1 elsewhere.



A 50Ω resistor at 290 K, over 20 MHz, gives noise with $\sigma = \sqrt{4kTRB} = 4.0 \mu\text{V}$. Every correlator output carries some.



Aircraft ADS-B sends a 1 as a $0.5 \mu\text{s}$ pulse in the first half of the bit, a 0 in the second. Two correlators give (r_1, r_2) .

CORRELATE, THEN DECIDE

Each receiver multiplies what arrives by a known waveform and integrates. The number it gets is one coordinate of $\mathbf{r}$.

NOISE IN EVERY OUTPUT

Thermal noise reaches every correlator. Each output is the signal coordinate plus a Gaussian sample.

THE WINDOW

The integral runs over one symbol. A GPS copy that is late by T_c gives $R = 0$ and loses the whole signal.

4.2 The decision rule

MAP and ML

The receiver should name the signal that is most probable once $\mathbf{r}$ has been seen. This is the maximum a posteriori, or **MAP**, rule.

Bayes' rule writes that probability with the prior $P(s_i)$ and the likelihood $f(\mathbf{r} | s_i)$.

$$P(s_i | \mathbf{r}) = \frac{P(s_i) f(\mathbf{r} | s_i)}{f(\mathbf{r})}$$

The denominator $f(\mathbf{r})$ is the same for every i , so it cannot change which i wins.

MAP

$$\hat{s} = \arg \max_i P(s_i | \mathbf{r}) = \arg \max_i P(s_i) f(\mathbf{r} | s_i)$$

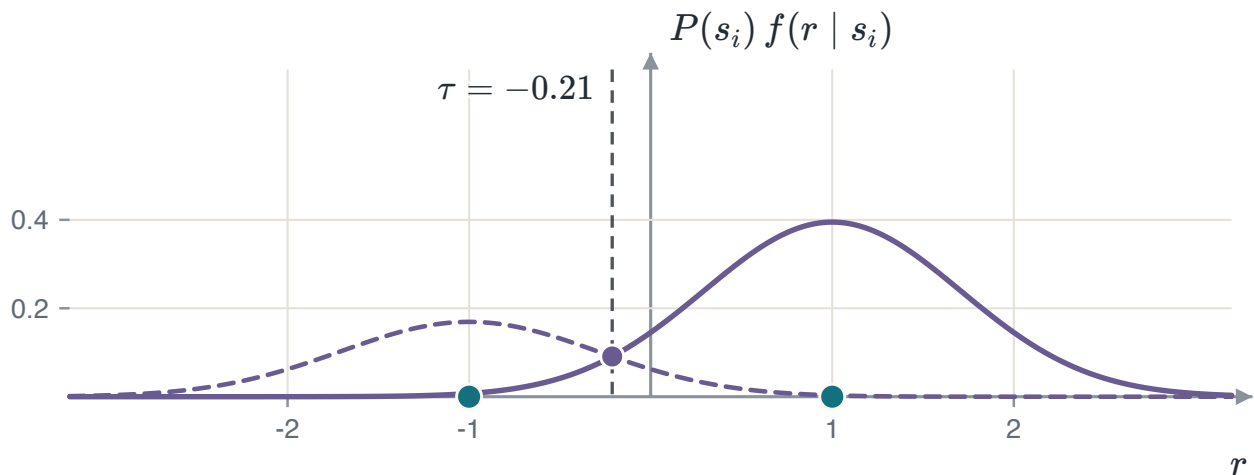
With equal priors the factor $P(s_i)$ is common to every i and drops too. MAP then becomes the maximum-likelihood, or **ML**, rule.

ML

$$\hat{s} = \arg \max_i f(\mathbf{r} | s_i)$$

For example, let $P(s_1) = 0.8$ and $P(s_2) = 0.2$. At some r the likelihoods are $f(r | s_1) = 0.2$ and $f(r | s_2) = 0.5$. MAP compares $0.8(0.2) = 0.16$ with $0.2(0.5) = 0.10$ and picks s_1 . ML picks s_2 , because $0.5 > 0.2$.

In one dimension the MAP rule picks the taller of the two weighted densities $P(s_i) f(r | s_i)$. The figure below shows that their crossing τ is the threshold.



$s_1 = +1$ and $s_2 = -1$, with $N_0 = 1$, drawn for $P(s_1) = 0.7$. Solid: $P(s_1) f(r | s_1)$. Dashed: $P(s_2) f(r | s_2)$. The crossing τ sits on the side of the less likely s_2 .

Minimum distance

In Gaussian noise the likelihood has a simple form. The N noise components are independent, so the density of $\mathbf{r}$ is a product of N bells.

$$f(\mathbf{r} | \mathbf{s}_i) = \prod_{j=1}^N \frac{1}{\sqrt{\pi N_0}} e^{-(r_j - s_{ij})^2 / N_0}$$

$$= \frac{1}{(\pi N_0)^{N/2}} e^{-\|\mathbf{r} - \mathbf{s}_i\|^2 / N_0}$$

The sum of the squares in the exponent is the squared distance $\|\mathbf{r} - \mathbf{s}_i\|^2$. Now take the logarithm of the MAP product. The logarithm is increasing, so the best signal does not change.

$$\ln(P(\mathbf{s}_i) f(\mathbf{r} | \mathbf{s}_i)) = \ln P(\mathbf{s}_i) - \frac{N}{2} \ln(\pi N_0) - \frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0}$$

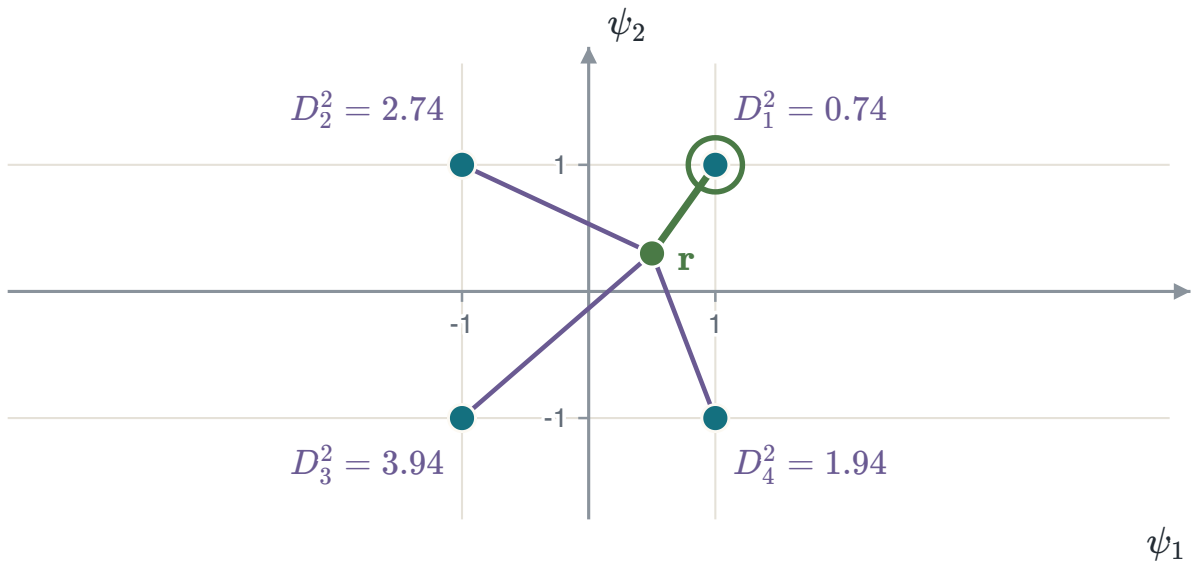
The middle term is the same for every i and drops. Multiply what is left by $-N_0$, which turns the largest into the smallest.

MINIMUM DISTANCE

$$\hat{s} = \arg \min_i \left(\|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right)$$

With equal priors the MAP receiver picks the signal point nearest to $\mathbf{r}$. Unequal priors subtract $N_0 \ln P(\mathbf{s}_i)$ from each squared distance, a handicap in favour of the likely signals.

The figure below takes $\mathbf{r} = (0.5, 0.3)$ and four equally likely points $(\pm 1, \pm 1)$. It marks the squared distance D_i^2 to each point.



The received point $\mathbf{r} = (0.5, 0.3)$ and the points $\mathbf{s}_1 = (1, 1)$, $\mathbf{s}_2 = (-1, 1)$, $\mathbf{s}_3 = (-1, -1)$ and $\mathbf{s}_4 = (1, -1)$. The ring marks the nearest one, $\mathbf{s}_1$.

For $\mathbf{s}_1$ the squared distance is $D_1^2 = (0.5 - 1)^2 + (0.3 - 1)^2 = 0.25 + 0.49 = 0.74$. The others are 2.74, 3.94 and 1.94. With equal priors the receiver picks $\mathbf{s}_1$.

Now let $N_0 = 1$, with $P(\mathbf{s}_4) = 0.7$ and 0.1 for each of the others. Each score adds $-\ln P(\mathbf{s}_i)$ to the squared distance.

$$\begin{aligned}
\mathbf{s}_1 : \quad & 0.74 - \ln 0.1 = 0.74 + 2.30 = 3.04 \\
\mathbf{s}_2 : \quad & 2.74 + 2.30 = 5.04 \\
\mathbf{s}_3 : \quad & 3.94 + 2.30 = 6.24 \\
\mathbf{s}_4 : \quad & 1.94 - \ln 0.7 = 1.94 + 0.36 = 2.30
\end{aligned}$$

The smallest score is 2.30, so MAP picks $\mathbf{s}_4$.

The correlation metric

Expanding the squared distance gives the form in which a receiver is built.

$$\begin{aligned}
\|\mathbf{r} - \mathbf{s}_i\|^2 &= \sum_{j=1}^N (r_j - s_{ij})^2 \\
&= \sum_{j=1}^N r_j^2 - 2 \sum_{j=1}^N r_j s_{ij} + \sum_{j=1}^N s_{ij}^2 \\
&= \|\mathbf{r}\|^2 - 2 \mathbf{r} \cdot \mathbf{s}_i + E_i
\end{aligned}$$

Here $E_i = \|\mathbf{s}_i\|^2$ is the energy of $s_i(t)$. The term $\|\mathbf{r}\|^2$ is the same for every i and drops. Halve the rest and change its sign, so the smallest becomes the largest.

CORRELATION METRIC

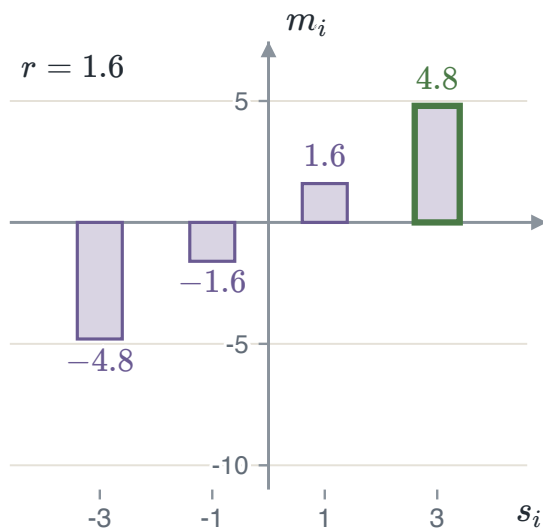
$$\hat{s} = \arg \max_i \left(\mathbf{r} \cdot \mathbf{s}_i - \frac{E_i}{2} + \frac{N_0}{2} \ln P(\mathbf{s}_i) \right)$$

This is the same rule as minimum distance. With equal energies and equal priors only $\mathbf{r} \cdot \mathbf{s}_i$ is left: pick the largest correlation.

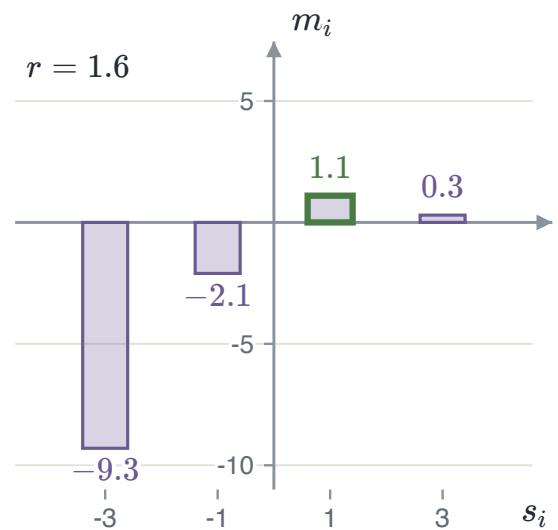
With unequal energies the term $-E_i/2$ must stay. Take 4-PAM at $s_i = -3, -1, 1, 3$ with $r = 1.6$ and equal priors. Each metric is $r s_i - s_i^2/2$.

$$\begin{aligned}
s = -3 : \quad & -4.8 - 4.5 = -9.3 \\
s = -1 : \quad & -1.6 - 0.5 = -2.1 \\
s = 1 : \quad & 1.6 - 0.5 = 1.1 \\
s = 3 : \quad & 4.8 - 4.5 = 0.3
\end{aligned}$$

The largest is 1.1, so the receiver picks $s = 1$, the nearest level. The correlation alone is largest for $s = 3$, at 4.8. The figure below draws both sets of bars.



The correlation $r s_i$ alone. The largest bar, outlined in green, is $s = 3$.



The metric $r s_i - s_i^2/2$. The energy term gives the win to $s = 1$.

Two receiver structures

There are two ways to build the receiver. Both compute the correlation metric for every signal and keep the largest.

Method I computes $\mathbf{r} = (r_1, \dots, r_N)$ with N correlators. Then it forms $\mathbf{r} \cdot \mathbf{s}_i + a_i$ for each i and keeps the largest.

THE BIAS

$$a_i = \frac{N_0}{2} \ln P(\mathbf{s}_i) - \frac{E_i}{2}$$

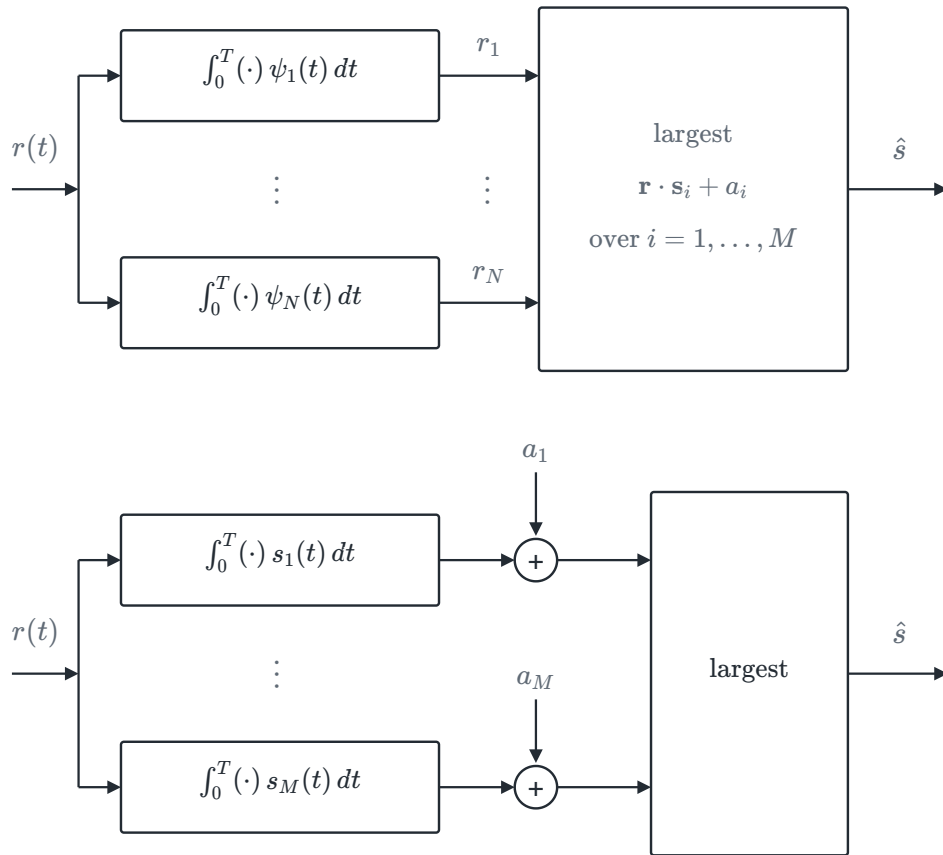
There is one constant for each signal. The priors and the energies fix it before any signal arrives.

Method II correlates $r(t)$ with each waveform $s_i(t)$ directly. Inner products are preserved, so each of these correlators gives $\mathbf{r} \cdot \mathbf{s}_i$ at once.

$$\int_0^T r(t) s_i(t) dt = \mathbf{r} \cdot \mathbf{s}_i$$

Method I needs N correlators and Method II needs M . So Method I is cheaper whenever $N < M$. For example, $M = 16$ signals in $N = 2$ dimensions need 16 correlators in Method II and 2 in Method I.

The figure below draws both structures as block diagrams.



Top: Method I correlates with the N basis functions, then forms M metrics. Bottom: Method II correlates with the M waveforms and adds each bias a_i . Both give the same $\hat{s}$.

Optimality of the MAP rule

No other rule has a smaller error probability than MAP. In one dimension the reason is a picture, because the error probability is an area.

Take two signals and a threshold τ , with s_1 decided above τ . Each way to be wrong is a weighted density on the wrong side of τ .

ERROR AS AN AREA

$$P_e(\tau) = P(s_1) \int_{-\infty}^{\tau} f(r | s_1) dr + P(s_2) \int_{\tau}^{\infty} f(r | s_2) dr$$

Move τ right by a small Δ . The first area gains $P(s_1)f(\tau | s_1)\Delta$, and the second loses $P(s_2)f(\tau | s_2)\Delta$. So the slope of P_e is the difference.

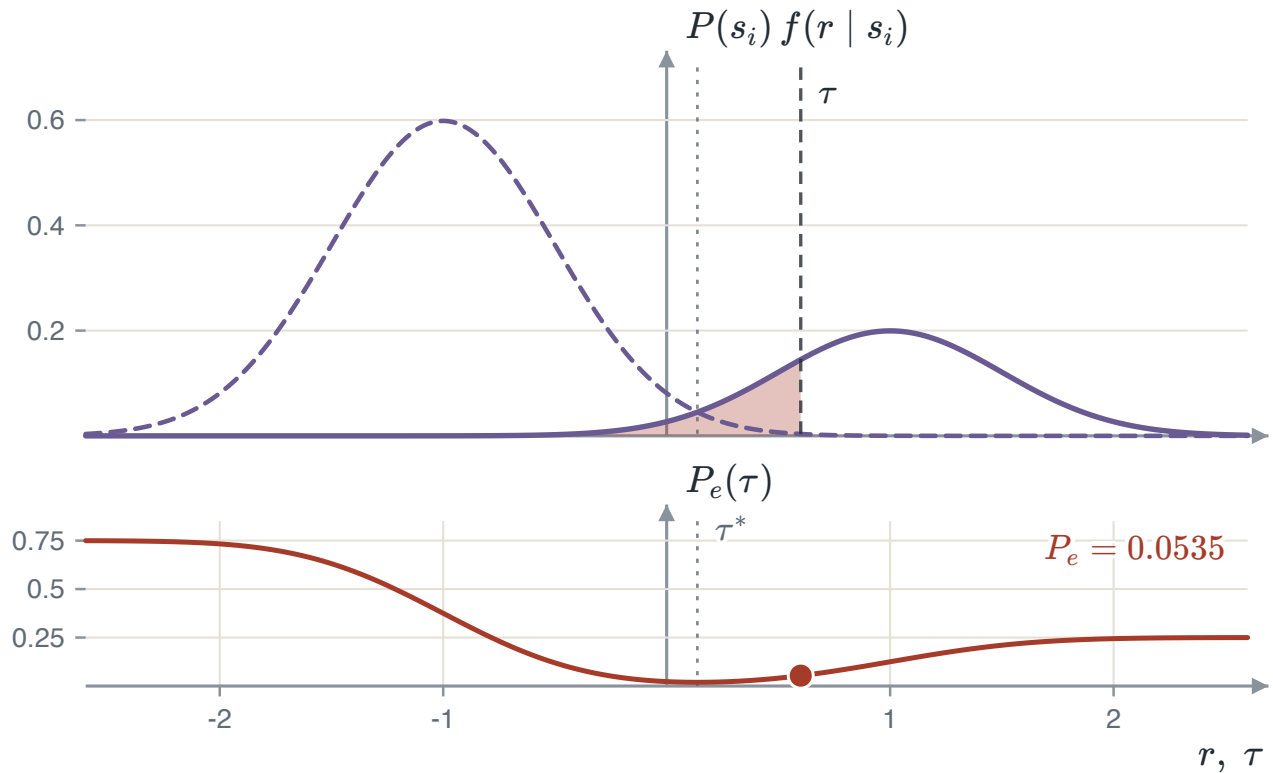
$$\frac{dP_e}{d\tau} = P(s_1) f(\tau | s_1) - P(s_2) f(\tau | s_2)$$

The slope is zero where the two weighted densities are equal. That point is the MAP threshold, and no move of τ helps there.

IN ANY DIMENSION

Give each $\mathbf{r}$ to the signal whose weighted density $P(s_i)f(\mathbf{r} | s_i)$ is largest there. Any other choice adds area, so MAP has the smallest P_e .

The figure below uses ± 1 with $P(s_1) = 0.25$ at $+1$ and $N_0 = 0.5$. Its lower panel is $P_e(\tau)$ as the threshold moves.



Drawn for $\tau = 0.6$. Top: the red areas are the weighted densities on the wrong side of τ . Bottom: their sum $P_e(\tau)$, smallest at $\tau^* = 0.137$, where the weighted densities cross.

The minimum sits at $\tau^* = 0.137$, with $P_e = 0.0192$. At $\tau = 0$ it is 0.0228.

EXAMPLE 4.3 – A MAP THRESHOLD

- GIVEN** Antipodal signals ± 1 , so $E_b = 1$, with $P(s_1) = 0.25$ at $+1$, $P(s_2) = 0.75$ at -1 , and $N_0 = 0.5$.
- FIND** The MAP threshold τ , the direction it moves, and the error probability.
- METHOD** The MAP threshold is where the two weighted densities are equal. Set them equal at $r = \tau$ and take logarithms. The common factor $1/\sqrt{\pi N_0}$ cancels.

$$\begin{aligned}\ln P(s_1) - \frac{(\tau - \sqrt{E_b})^2}{N_0} &= \ln P(s_2) - \frac{(\tau + \sqrt{E_b})^2}{N_0} \\ \frac{(\tau + \sqrt{E_b})^2 - (\tau - \sqrt{E_b})^2}{N_0} &= \ln \frac{P(s_2)}{P(s_1)} \\ \frac{4\sqrt{E_b}\tau}{N_0} &= \ln \frac{P(s_2)}{P(s_1)}\end{aligned}$$

The squares τ^2 and E_b cancel in the second line. So $\tau = \frac{N_0}{4\sqrt{E_b}} \ln \frac{P(s_2)}{P(s_1)}$.

- SOLUTION** $\tau = \frac{0.5}{4} \ln \frac{0.75}{0.25} = 0.125 \ln 3 = 0.137$. It is positive, so it moves toward $+1$, the less likely signal. Each Q argument is the distance from a point to τ , over $\sigma = \sqrt{N_0/2} = 0.5$.

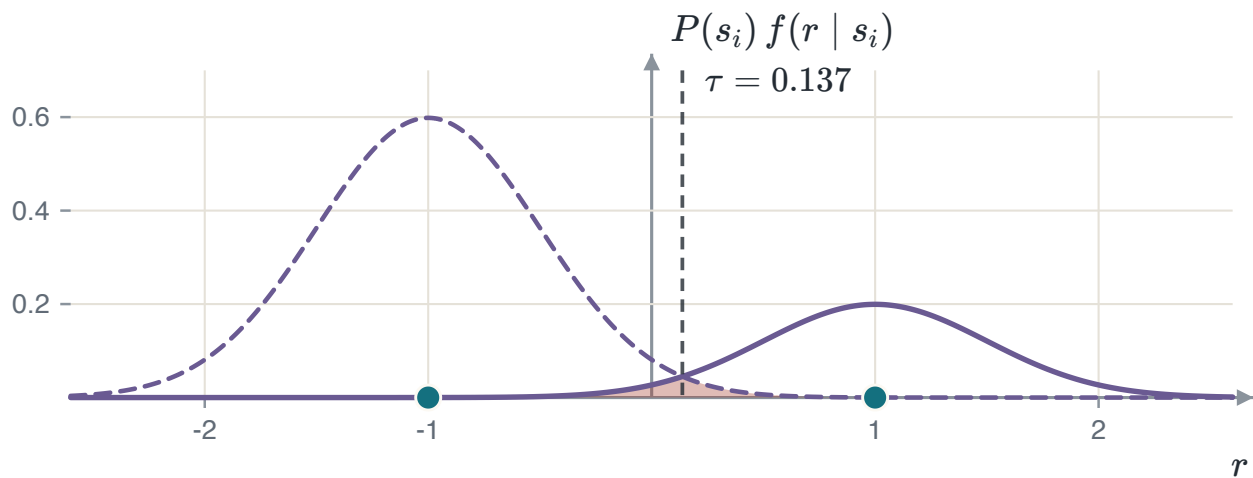
$$\begin{aligned}P_e &= 0.25 Q\left(\frac{1 - 0.137}{0.5}\right) + 0.75 Q\left(\frac{1 + 0.137}{0.5}\right) \\ &= 0.25 Q(1.73) + 0.75 Q(2.27) \\ &= 0.25(0.0422) + 0.75(0.0115) = 0.0192\end{aligned}$$

- CHECK** ML keeps $\tau = 0$ and gets $Q(1/0.5) = Q(2) = 0.0228$. MAP is about 16% better.

COMMON ERROR

Use $\ln(P(s_2)/P(s_1))$, not its inverse. The inverse gives $\tau = -0.137$.

The figure below marks the threshold and the two error areas of Example 4.3.



Example 4.3. Solid: $0.25 f(r | s_1)$ at $+1$. Dashed: $0.75 f(r | s_2)$ at -1 . The two red areas are the two ways to be wrong, and their sum is $P_e = 0.0192$.

EXAMPLE 4.4 – A RECEIVER FOR FOUR WAVEFORMS

GIVEN Four equally likely waveforms on $[0, 2)$. $s_1(t) = 2$ on $[0, 1)$ and $s_2(t) = 1$ on $[0, 2)$. $s_3(t)$ is -1 on $[0, 1)$ and 1 on $[1, 2)$, and $s_4(t) = -1$ on $[0, 2)$. $\mathbf{r} = (1.2, 0.3)$ arrives.

FIND The receiver and its decision for $\mathbf{r}$.

METHOD Every waveform is constant on each half, so two unit pulses form an orthonormal basis: $\psi_1 = 1$ on $[0, 1)$ and $\psi_2 = 1$ on $[1, 2)$. Read each point from the heights on the two halves.

$$\mathbf{s}_1 = (2, 0), \quad \mathbf{s}_2 = (1, 1), \quad \mathbf{s}_3 = (-1, 1), \quad \mathbf{s}_4 = (-1, -1)$$

The energies $E_i = \|\mathbf{s}_i\|^2$ are 4, 2, 2, 2. They differ, so the metric keeps $-E_i/2$.

SOLUTION With equal priors the metric is $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

$$\mathbf{s}_1 : 1.2(2) + 0.3(0) - 2 = 0.4$$

$$\mathbf{s}_2 : 1.2(1) + 0.3(1) - 1 = 0.5$$

$$\mathbf{s}_3 : 1.2(-1) + 0.3(1) - 1 = -1.9$$

$$\mathbf{s}_4 : 1.2(-1) + 0.3(-1) - 1 = -2.5$$

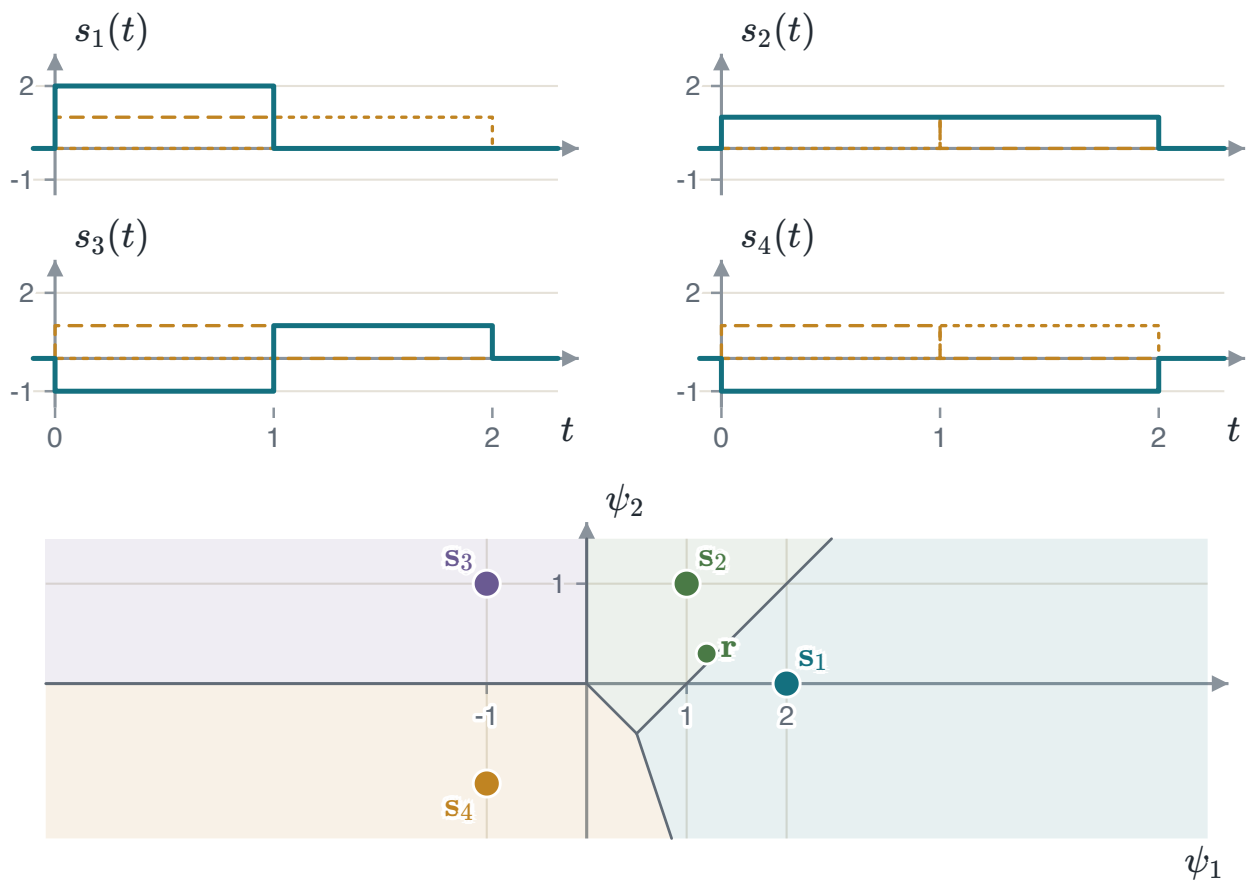
The largest is 0.5, so $\hat{s} = s_2$.

CHECK $\|\mathbf{r} - \mathbf{s}_1\|^2 = 0.8^2 + 0.3^2 = 0.73$ and $\|\mathbf{r} - \mathbf{s}_2\|^2 = 0.2^2 + 0.7^2 = 0.53$. So $\mathbf{s}_2$ is also the nearest point, and $\mathbf{r}$ lies in its region.

COMMON ERROR

Use $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$, not $\mathbf{r} \cdot \mathbf{s}_i$, if the energies differ. Correlation alone picks s_1 .

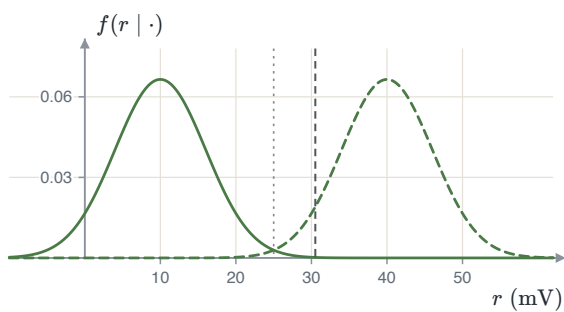
The figure below draws the waveforms, the basis and the regions of Example 4.4.



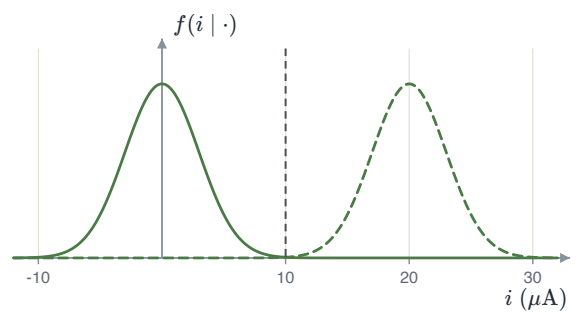
Example 4.4. Top: the four waveforms, with the two unit pulses dashed in amber. Bottom: the four points, the regions of the minimum-distance receiver, and $\mathbf{r} = (1.2, 0.3)$ in the region of s_2 .

Decision rules around us

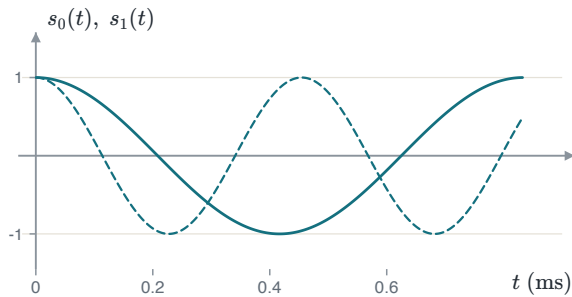
Four everyday receivers use the MAP threshold, the energy term or the correlation receiver.



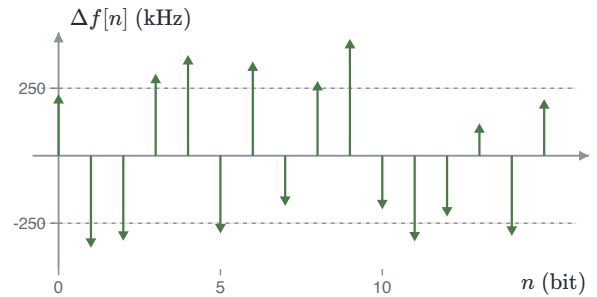
A smoke alarm reads 10 mV in clean air and 40 mV in smoke, with $\sigma = 6$ mV. Smoke is rare, $P = 0.01$, so the threshold moves from 25 to 30.5 mV.



An optical receiver sees 0 or 20 μA . The energy term puts the threshold at 10 μA . With $\sigma = 3$ μA , $P_e = Q(10/3) = 4.3 \times 10^{-4}$.



Caller ID sends 1200 Hz for a 1 and 2200 Hz for a 0. The receiver correlates with both tones over $T = 0.833$ ms: Method II.



Bluetooth Low Energy shifts the carrier by about ± 250 kHz. With equal priors the receiver decides by the sign of each sample.

PRIORS MOVE THE THRESHOLD

A rare event gets a smaller region. The threshold moves toward the less likely signal.

THE ENERGY TERM

With unequal energies the metric keeps $-E_i/2$. For on-off keying that puts the threshold at half the on level.

UNKNOWN PRIORS

When the priors are not known, the receiver uses ML. It is optimal only when the priors are equal.

4.3 Decision regions

Regions with equal priors

Group the observations that give the same decision. The space splits into M **decision regions**, one for each signal.

Start with two points. The points $\mathbf{r}$ equally far from $\mathbf{s}_i$ and $\mathbf{s}_j$ form a line, the **perpendicular bisector** of the segment between them. Expanding both squared distances shows why.

$$\begin{aligned}\|\mathbf{r} - \mathbf{s}_i\|^2 &= \|\mathbf{r} - \mathbf{s}_j\|^2 \\ -2\mathbf{r} \cdot \mathbf{s}_i + E_i &= -2\mathbf{r} \cdot \mathbf{s}_j + E_j \\ 2\mathbf{r} \cdot (\mathbf{s}_j - \mathbf{s}_i) &= E_j - E_i\end{aligned}$$

The $\|\mathbf{r}\|^2$ terms cancel, and the last line is linear in $\mathbf{r}$. So the boundary is a straight line, at right angles to $\mathbf{s}_j - \mathbf{s}_i$.

A REGION

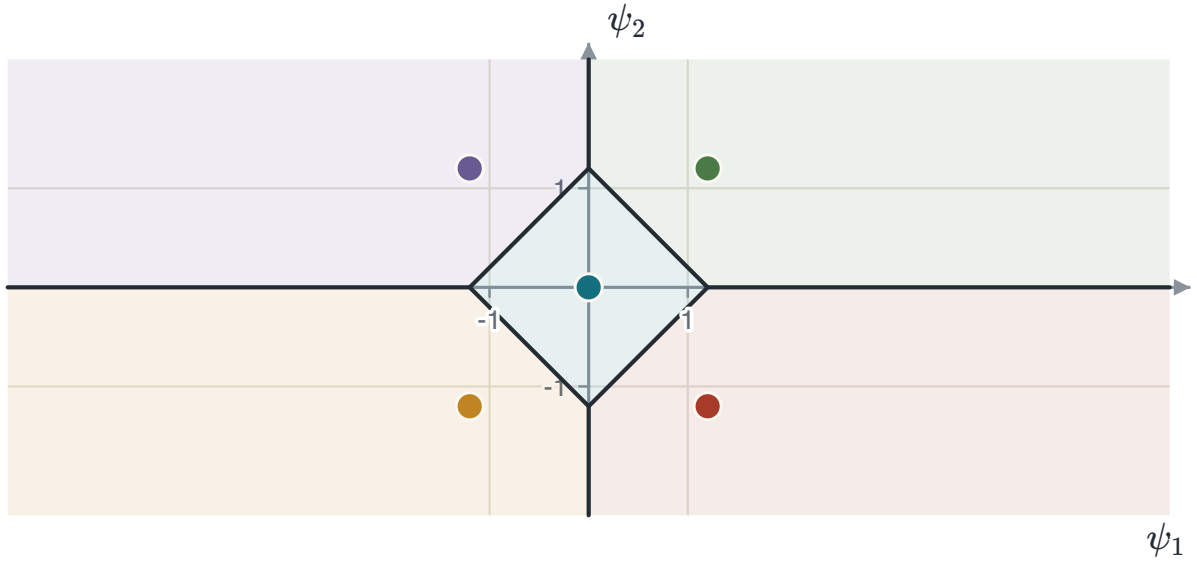
$$R_i = \{\mathbf{r} : \|\mathbf{r} - \mathbf{s}_i\| < \|\mathbf{r} - \mathbf{s}_j\| \text{ for all } j \neq i\}$$

Each region is cut out by bisectors, so it is a convex polygon.

UNBOUNDED REGIONS

Outer points have regions that run off to infinity. Noise that pushes an outer point outward causes no error.

The figure below takes five equally likely points: the corners $(\pm 1.2, \pm 1.2)$ and the centre.



Five equally likely points and their regions. The centre region is the square $|r_1| + |r_2| < 1.2$, turned by 45° . The four corner regions are unbounded.

The centre region has four straight faces. The bisector between $(0, 0)$ and $(1.2, 1.2)$ is $r_1 + r_2 = 1.2$. The other three corners give the other three faces.

Regions with unequal priors

With unequal priors the boundary between $\mathbf{s}_i$ and $\mathbf{s}_j$ is where the two MAP scores are equal.

A BOUNDARY

$$\|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) = \|\mathbf{r} - \mathbf{s}_j\|^2 - N_0 \ln P(\mathbf{s}_j)$$

The $\|\mathbf{r}\|^2$ terms still cancel. The boundary is still a straight line, parallel to the bisector.

To find where the line sits, measure along the segment from $\mathbf{s}_i$ to $\mathbf{s}_j$, which is d long. Put the boundary a distance x from $\mathbf{s}_i$.

$$\begin{aligned} x^2 - N_0 \ln P(\mathbf{s}_i) &= (d - x)^2 - N_0 \ln P(\mathbf{s}_j) \\ 2dx - d^2 &= N_0 \ln \frac{P(\mathbf{s}_i)}{P(\mathbf{s}_j)} \end{aligned}$$

Solve for x and call the result μ_i .

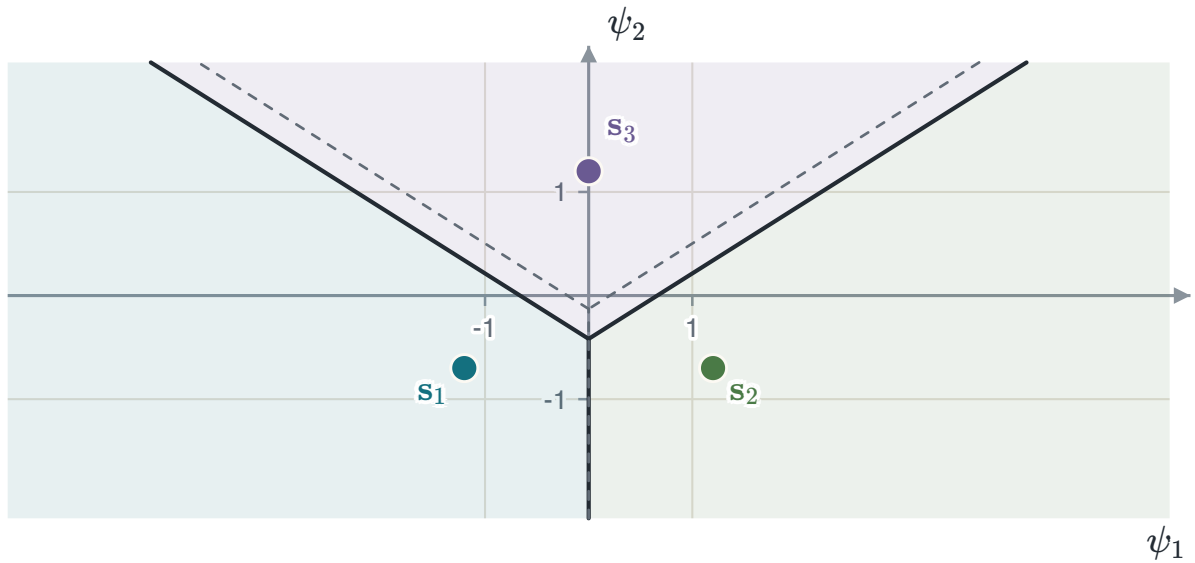
THE SHIFT

$$\mu_i = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(\mathbf{s}_i)}{P(\mathbf{s}_j)}$$

μ_i is the distance from $\mathbf{s}_i$ to its boundary with $\mathbf{s}_j$. A likely point pushes the line away from itself, toward the less likely point.

For example, take $d = 2$, $N_0 = 1$, $P(\mathbf{s}_i) = 0.8$ and $P(\mathbf{s}_j) = 0.2$. Then $\mu_i = 1 + \frac{1}{4} \ln 4 = 1 + 0.35 = 1.35$. The likely point keeps the larger share.

The figure below gives each of three points a prior, with $N_0 = 1$.



Three points with $N_0 = 1$, drawn for $P(\mathbf{s}_3) = 0.6$ and $P(\mathbf{s}_1) = P(\mathbf{s}_2) = 0.2$. Solid lines are the MAP boundaries, dashed lines the equal-prior ones. The likely $\mathbf{s}_3$ takes more of the plane.

Binary error probability

Two equally likely points a distance d apart give the simplest error probability. The boundary is the midpoint, $d/2$ from each point.

Only the noise along the line through the two points can cause an error. That component has variance $N_0/2$, and an error needs more than $d/2$ of it.

ALONG THE LINE

$$\begin{aligned} P_e &= P\left(n > \frac{d}{2}\right) = Q\left(\frac{d/2}{\sqrt{N_0/2}}\right) \\ &= Q\left(\sqrt{\frac{d^2}{2N_0}}\right) \end{aligned}$$

Here $Q(x) = \frac{1}{2} \operatorname{erfc}(x/\sqrt{2})$ is the Gaussian tail. The last step squares the argument: $(d/2)^2/(N_0/2) = d^2/2N_0$.

Put the distance of each special case into $d^2/2N_0$.

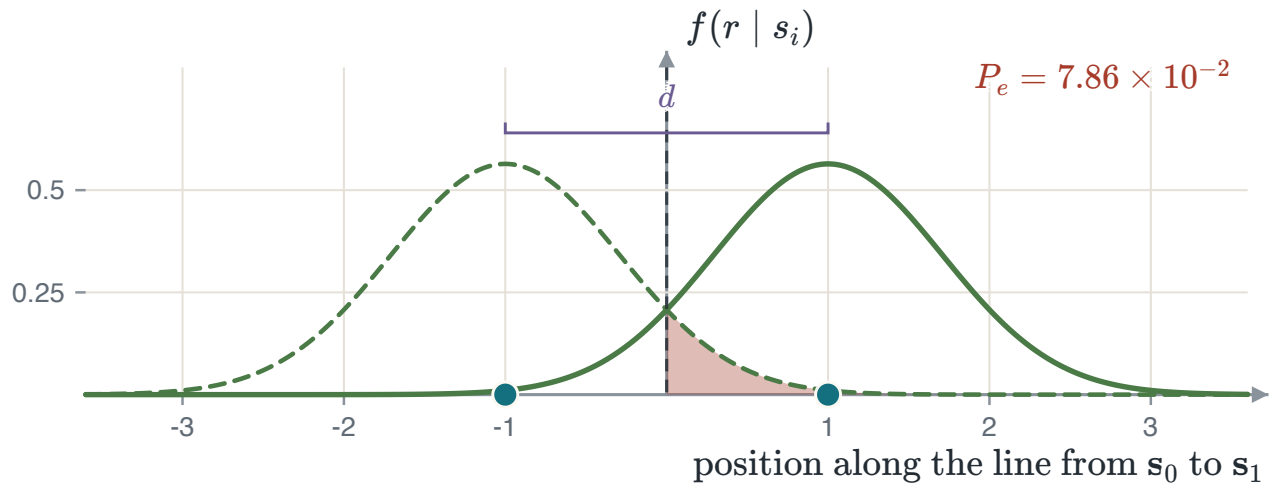
TWO SPECIAL CASES

$$\text{antipodal, } d = 2\sqrt{E_b} : \quad \frac{d^2}{2N_0} = \frac{2E_b}{N_0}, \quad P_e = Q(\sqrt{2E_b/N_0})$$

$$\text{orthogonal or on-off, } d = \sqrt{2E_b} : \quad \frac{d^2}{2N_0} = \frac{E_b}{N_0}, \quad P_e = Q(\sqrt{E_b/N_0})$$

For example, $d = 2$ and $N_0 = 0.5$ give $d^2/2N_0 = 4/1 = 4$, so $P_e = Q(2) = 0.0228$.

The figure below draws the two densities along the line for $d = 2$ and $N_0 = 1$.



Two equally likely points $d = 2$ apart, with $N_0 = 1$. The red tail of the left density past the midpoint is $P_e = Q(\sqrt{2}) = 7.86 \times 10^{-2}$.

Binary error with unequal priors

With unequal priors the MAP boundary moves off the midpoint. Put s_0 and s_1 a distance d apart, with the boundary a distance μ from s_1 .

THE BOUNDARY

$$\mu = \frac{d}{2} + \frac{N_0}{2d} \ln \frac{P(s_1)}{P(s_0)}$$

This is the shift μ_i above, with s_1 in the place of s_i .

If s_0 is sent, an error needs noise past $d - \mu$. If s_1 is sent, it needs noise past μ . Weight each tail by its prior.

TWO TAILS

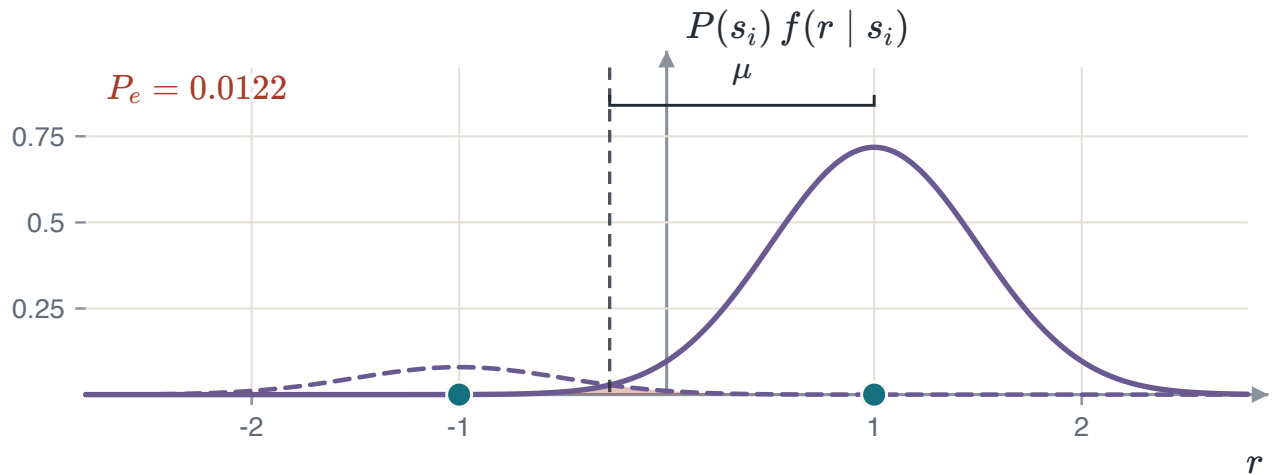
$$P_e = P(s_0) Q\left(\frac{d - \mu}{\sqrt{N_0/2}}\right) + P(s_1) Q\left(\frac{\mu}{\sqrt{N_0/2}}\right)$$

Equal priors give $\mu = d/2$ and the formula for two equally likely points.

For example, take $d = 2$, $N_0 = 0.5$ and $P(s_1) = 0.9$. The boundary and the error follow in two steps.

$$\begin{aligned}\mu &= 1 + \frac{0.5}{4} \ln \frac{0.9}{0.1} = 1 + 0.125(2.197) = 1.27 \\ P_e &= 0.1 Q\left(\frac{2 - 1.27}{0.5}\right) + 0.9 Q\left(\frac{1.27}{0.5}\right) \\ &= 0.1 Q(1.45) + 0.9 Q(2.55) = 0.0122\end{aligned}$$

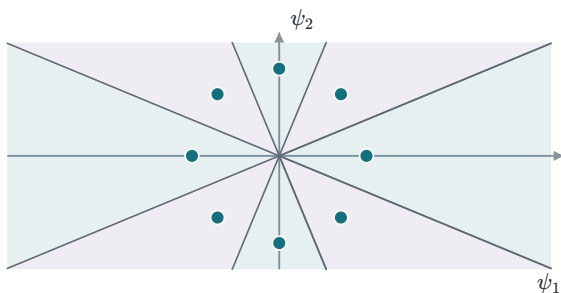
The ML receiver keeps the midpoint and gets $Q(2) = 0.0228$. So the MAP receiver has the smaller P_e .



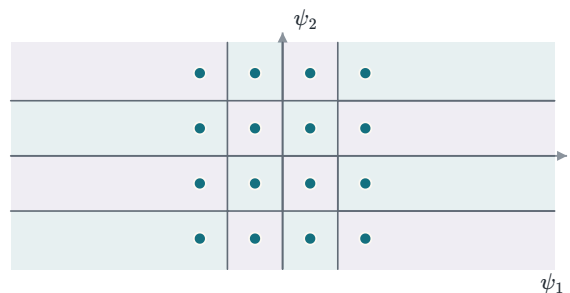
$s_0 = -1$ and $s_1 = +1$, so $d = 2$, with $N_0 = 0.5$, drawn for $P(s_1) = 0.9$. The boundary sits $\mu = 1.27$ from s_1 , and the two red tails add up to $P_e = 0.0122$.

Decision regions around us

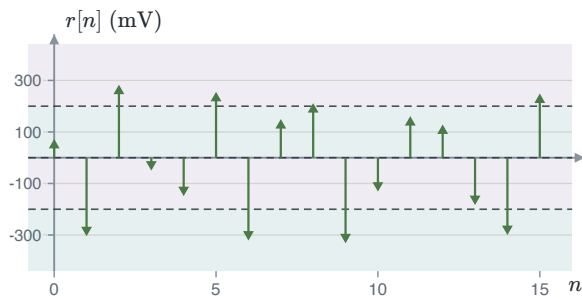
Four everyday systems decide in regions cut by bisectors.



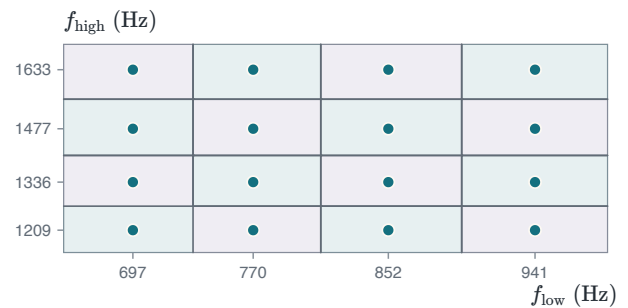
Satellite television can use 8PSK. Its regions are eight 45° sectors, so the receiver decides by the phase of $\mathbf{r}$ alone.



Wi-Fi 16-QAM at unit average energy has square regions of side $d_{\min} = 2/\sqrt{10} = 0.632$. The 12 outer ones are unbounded.



PAM4 in data-centre links sends ± 100 and ± 300 mV. The thresholds 0 and ± 200 mV cut the axis into four intervals.



A phone key sends one low and one high tone. The receiver decides in rectangles with edges halfway between neighbouring tones.

NEAREST POINT

In each system the receiver picks the signal point nearest to $\mathbf{r}$. The regions are cut by perpendicular bisectors.

OUTER REGIONS

Points on the edge of a constellation have unbounded regions. Noise can push them outward without an error.

SCALE

The regions hold only if the receiver knows the scale of the points. A gain error moves every point and every boundary.

4.4 Error probability and the union bound

The exact error probability

The exact error probability adds, over the signals, the part of each Gaussian bell that falls outside its region.

EXACT ERROR

$$P_e = \sum_{i=1}^M P(\mathbf{s}_i) \int_{\mathbf{r} \notin R_i} f(\mathbf{r} | \mathbf{s}_i) d\mathbf{r}$$

Each term is the part of a Gaussian bell that falls outside a polygon.

NO CLOSED FORM

For most polygons this integral has no formula. It needs numerical integration or a simulation.

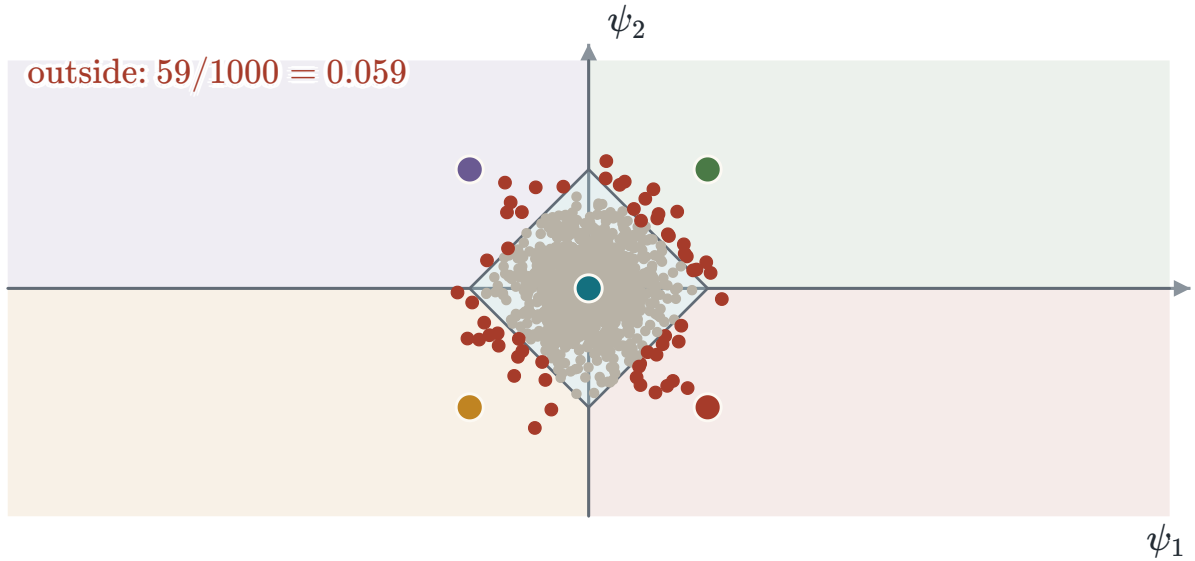
A **simulation** sends many symbols, adds noise and counts the wrong decisions. About 100 errors give a usable estimate. At $P_e = 10^{-4}$ that takes about $100/10^{-4} = 10^6$ symbols.

Sometimes the integral is easy. In the five-point set of Section 4.3, send the centre with $\sigma = 0.4$. Its region is the square $|r_1| + |r_2| < 1.2$, turned by 45° .

Turn the axes with the square. Each new noise component still has variance σ^2 . The region becomes $|u_1| < a$ and $|u_2| < a$, with half-width $a = 1.2/\sqrt{2} = 0.849$.

$$\begin{aligned} P(\text{error} \mid \text{centre}) &= 1 - \left(1 - 2Q(a/\sigma)\right)^2 \\ &= 1 - \left(1 - 2Q(2.12)\right)^2 = 0.067 \end{aligned}$$

The figure below counts the draws that land outside the centre region.



The centre point of the five-point set is sent with $\sigma = 0.4$, drawn for 1000 draws. The red draws fall outside its region, and their fraction settles near the exact 0.067.

The union bound

The union bound avoids the integral. Suppose $\mathbf{s}_i$ was sent, and let A_{ij} be the event that $\mathbf{r}$ is nearer $\mathbf{s}_j$ than $\mathbf{s}_i$.

An error happens when at least one A_{ij} happens, so the error is the union of these events. The probability of a union is at most the sum of the probabilities.

PAIRWISE EVENTS

$$P(\text{error} \mid \mathbf{s}_i) \leq \sum_{j \neq i} P(A_{ij})$$

$$P(A_{ij}) = Q\left(\sqrt{\frac{d_{ij}^2}{2N_0}}\right)$$

Each A_{ij} is a binary error between two points at distance d_{ij} . So its probability is the binary result of Section 4.3.

OVERLAPS COUNTED TWICE

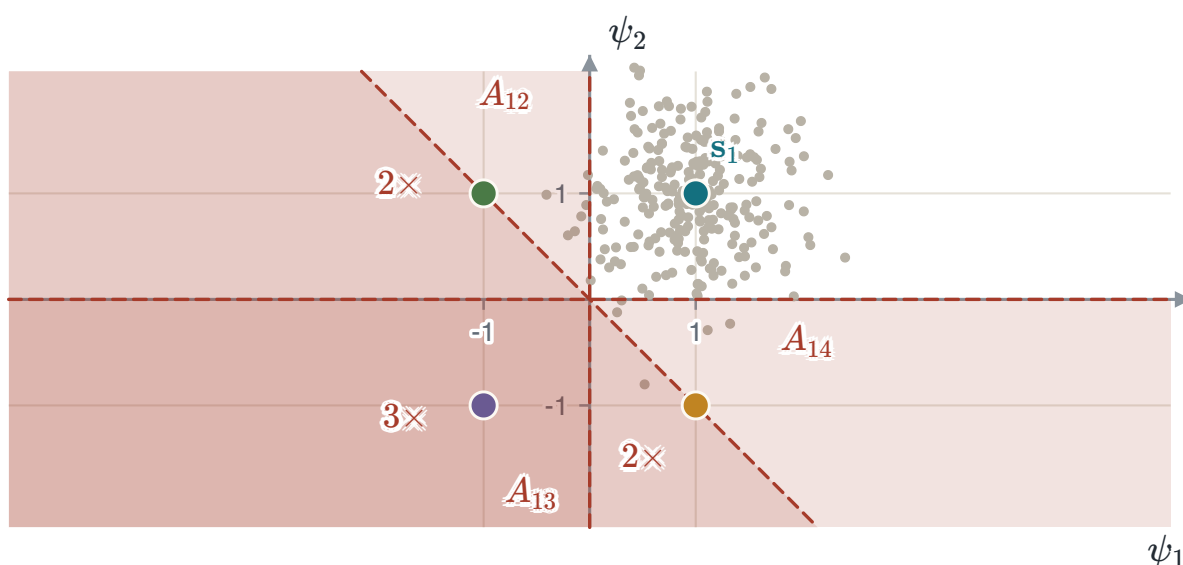
The events overlap. The sum counts each overlap more than once, so it can only be larger than the true probability.

Average over the signals with equal priors $1/M$. This gives the union bound, which needs only the distances between the points.

UNION BOUND

$$P_e \leq \frac{1}{M} \sum_{i=1}^M \sum_{j \neq i} Q \left(\sqrt{\frac{d_{ij}^2}{2N_0}} \right)$$

The figure below draws the three half-planes for $\mathbf{s}_1 = (1, 1)$ of the square.



$\mathbf{s}_1 = (1, 1)$ is sent, and the grey dots are received points. Each red half-plane A_{1j} is where $\mathbf{r}$ is nearer $\mathbf{s}_j$. The labels mark the parts the sum counts two or three times.

For $\mathbf{s}_1$ the bound has $M - 1 = 3$ terms. Two are for the points at $d = 2$, and one is for the diagonal point at $2\sqrt{2}$.

The minimum-distance bound

Every d_{ij} is at least the smallest distance $d_{\min}$, and Q falls as its argument grows. So each term is at most the term at $d_{\min}$.

MINIMUM-DISTANCE BOUND

$$P_e \leq (M - 1) Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

Each of the M signals has $M - 1$ terms, each at most $Q(\sqrt{d_{\min}^2/2N_0})$. The average over the M signals leaves $M - 1$ of them.

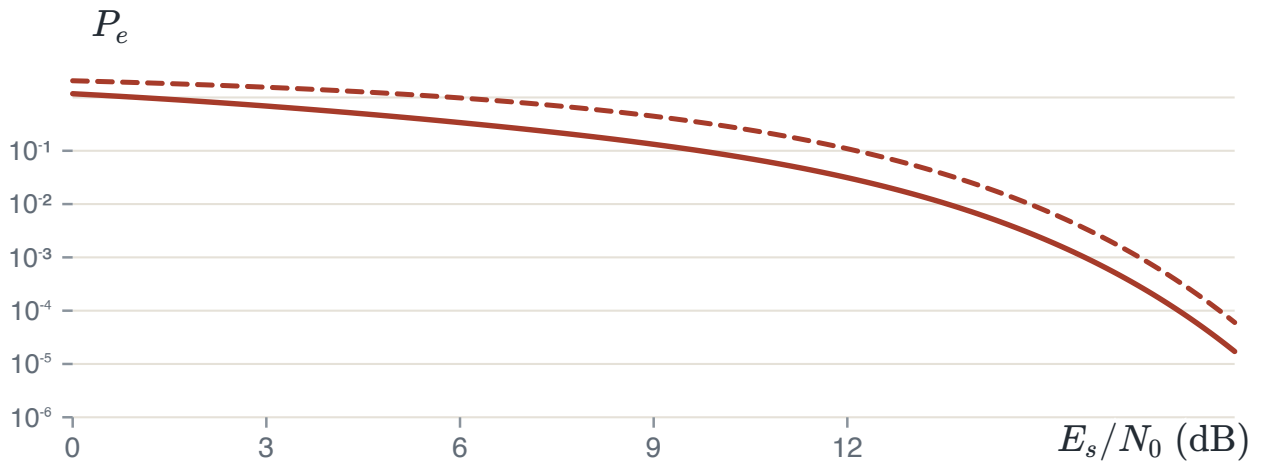
ONE NUMBER

The bound needs only M and the smallest distance. It is quick, and loose when few pairs sit at $d_{\min}$.

8-PSK puts $M = 8$ points on a circle of radius $\sqrt{E_s}$. The point j steps away sits at $d_j = 2\sqrt{E_s} \sin(\pi j/8)$. With $E_s = 1$, $d_{\min} = 2 \sin(\pi/8) = 2(0.383) = 0.77$.

Only two of the seven other points sit at $d_{\min}$. So the bound, which puts all seven there, is loose.

The figure below compares it with the union bound for 8-PSK. Each term uses $d_j^2/2N_0 = (2E_s/N_0) \sin^2(\pi j/8)$.



8-PSK. Solid: the union bound, the sum of all seven pairwise terms. Dashed: the minimum-distance bound, with every distance replaced by $d_{\min}$. The dashed curve stays above.

An exponential bound

The function $Q(x)$ has no closed form. A plain exponential sits above it for every $x \geq 0$.

EXPONENTIAL BOUND

$$Q(x) \leq \frac{1}{2} e^{-x^2/2}, \quad x \geq 0$$

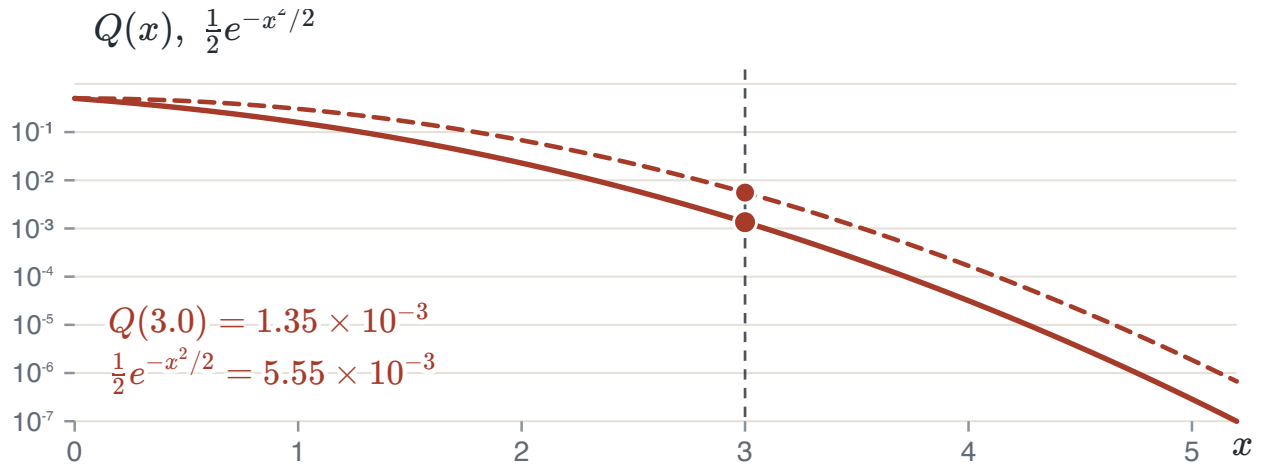
Put $x^2 = d_{\min}^2/2N_0$ into the minimum-distance bound. Then $x^2/2 = d_{\min}^2/4N_0$.

APPLIED TO THE BOUND

$$P_e \leq \frac{M-1}{2} e^{-d_{\min}^2/4N_0}$$

The error falls exponentially in $d_{\min}^2/N_0$.

For example, at $x = 3$ the bound is $\frac{1}{2}e^{-4.5} = 5.6 \times 10^{-3}$ against $Q(3) = 1.35 \times 10^{-3}$. It is about 4.1 times larger. The figure below draws both on a logarithmic axis.



Solid: $Q(x)$. Dashed: $\frac{1}{2}e^{-x^2/2}$, marked at $x = 3$. The bound stays above $Q(x)$, and both fall with the same exponent.

The intelligent union bound

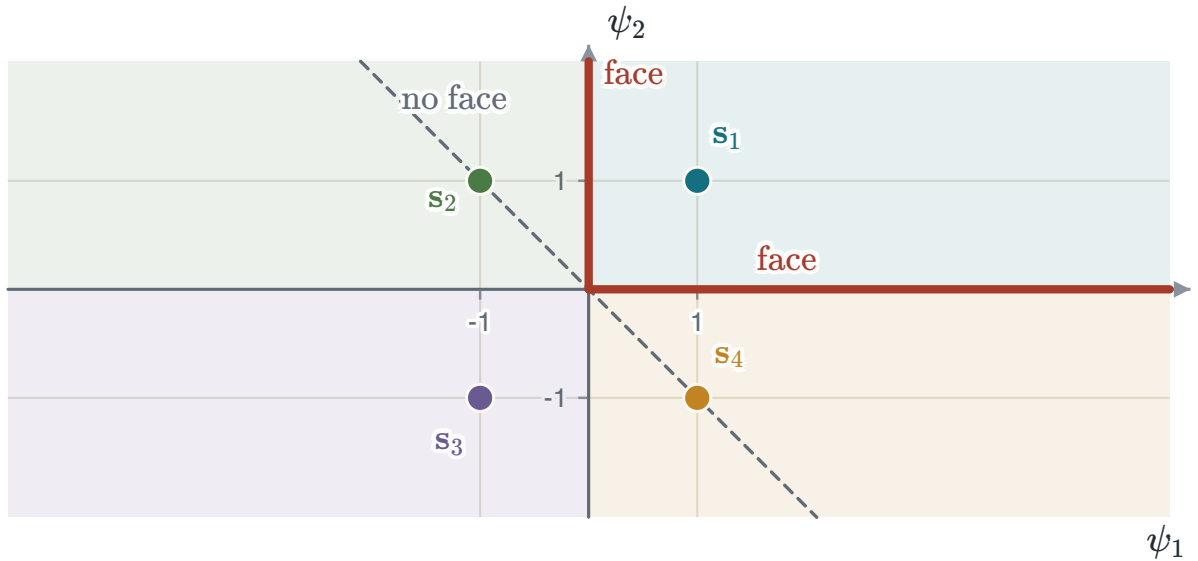
To leave R_i , the point $\mathbf{r}$ must cross one of the faces of R_i . So keep the terms for the points that share a face with R_i , a set written F_i . Drop the others.

INTELLIGENT UNION BOUND

$$P(\text{error} \mid \mathbf{s}_i) \leq \sum_{j \in F_i} Q\left(\sqrt{\frac{d_{ij}^2}{2N_0}}\right)$$

The bound is tighter than the general union bound, and still a bound.

In the square, the region of $\mathbf{s}_1$ is the first quadrant. The figure below marks its two faces, shared with $\mathbf{s}_2$ and $\mathbf{s}_4$. So the bound keeps two terms and drops the term for $\mathbf{s}_3$.



The region of s_1 is the first quadrant. Its two faces, in red, are shared with s_2 and s_4 . The bisector with s_3 , dashed, only touches the corner.

Nearest neighbours

At high SNR the terms at $d_{\min}$ are far larger than the others. Keeping only those terms gives the nearest-neighbour form.

NEAREST NEIGHBOURS

$$P_e \approx \bar{N}_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

$\bar{N}_{\min}$ is the average number of neighbours at $d_{\min}$, taken over the M points.

AN APPROXIMATION

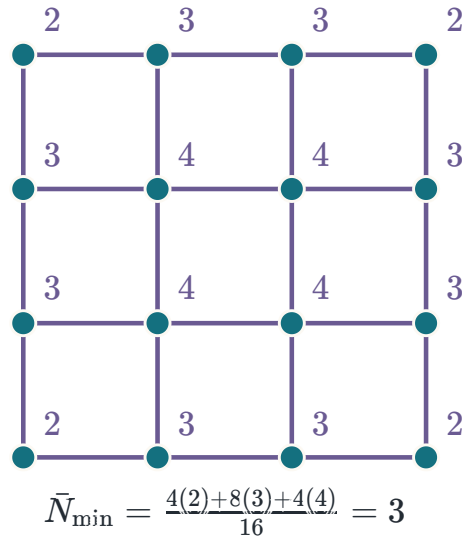
It keeps only part of the union bound, so it is not a bound. It can sit below the exact P_e .

In QPSK on the square $(\pm 1, \pm 1)$, each point has two neighbours at $d_{\min} = 2$. The diagonal point is at $2\sqrt{2}$, so $\bar{N}_{\min} = 2$.

In 16-QAM on the grid $\pm 1, \pm 3$, $d_{\min} = 2$. The four corners have two neighbours at $d_{\min}$, the eight edge points three, and the four inner points four.

$$\bar{N}_{\min} = \frac{4(2) + 8(3) + 4(4)}{16} = \frac{48}{16} = 3$$

The figure below joins every pair of neighbours in 16-QAM.



16-QAM on the grid $\pm 1, \pm 3$. Each line joins two neighbours at $d_{\min} = 2$, and each number counts the neighbours of its point.

EXAMPLE 4.5 – FOUR FORMS ON QPSK

GIVEN QPSK at $(\pm 1, \pm 1)$, equally likely, with $N_0 = 2/9$. So $d_{\min}^2/2N_0 = 4/(4/9) = 9$.

FIND The general union bound, the intelligent bound, the nearest-neighbour form and the minimum-distance bound. Which is largest?

METHOD From $\mathbf{s}_1 = (1, 1)$, two points are at $d = 2$, which gives $x = \sqrt{9} = 3$. One is at $2\sqrt{2}$, which gives $x = \sqrt{8/(4/9)} = \sqrt{18} = 4.24$. The region of $\mathbf{s}_1$ is a quadrant with two faces, and all four points are alike.

$$Q(3) = 1.35 \times 10^{-3}, \quad Q(4.24) = 1.1 \times 10^{-5}$$

SOLUTION

$$\begin{aligned} \text{general} &= 2Q(3) + Q(4.24) = 2.71 \times 10^{-3} \\ \text{intelligent} &= \text{nearest} = 2Q(3) = 2.70 \times 10^{-3} \\ d_{\min} \text{ bound} &= 3Q(3) = 4.05 \times 10^{-3} \end{aligned}$$

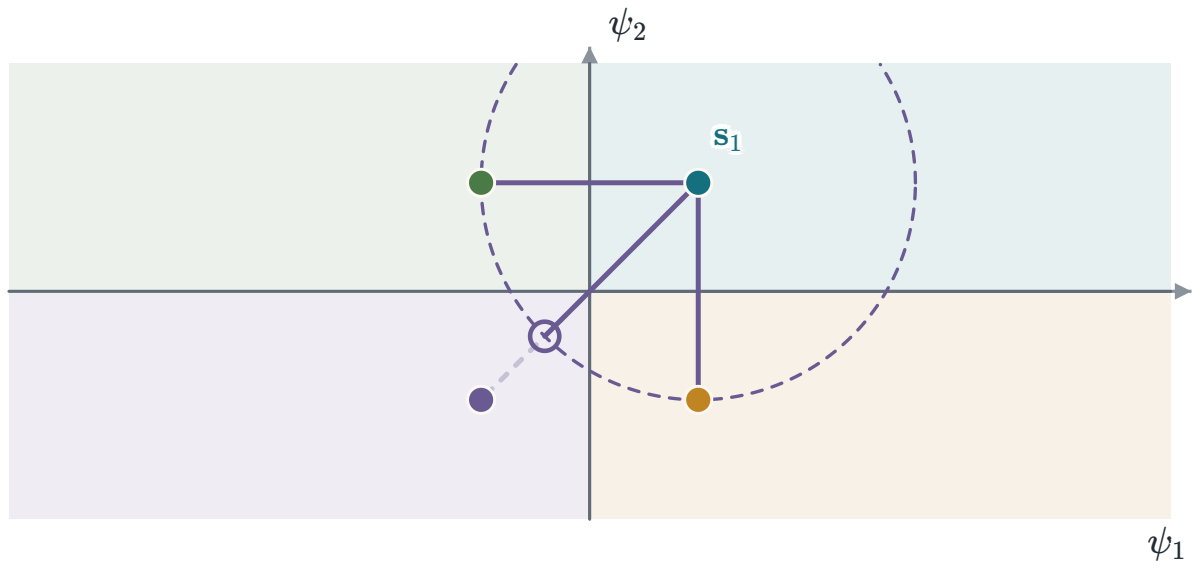
The minimum-distance bound is the largest. It counts three terms at $Q(3)$, and the others count two plus a tiny diagonal term.

CHECK The regions are quadrants, so each noise component acts on its own. The exact value is $1 - (1 - Q(3))^2 = 2.70 \times 10^{-3}$. The diagonal term adds only 0.4%.

COMMON ERROR

Use $\bar{N}_{\min} = 2$, not 3. The diagonal point is at $2\sqrt{2}$, not at $d_{\min} = 2$.

The figure below shows the nearest neighbours of $\mathbf{s}_1$ in Example 4.5.



Example 4.5 from s_1 . The dashed circle of radius $d_{\min}$ passes through the two nearest neighbours. The minimum-distance bound moves the diagonal point in to $d_{\min}$, ringed.

EXAMPLE 4.6 – A RECTANGLE

GIVEN Four equally likely points $(\pm 1.5, \pm 1)$ with $N_0 = 0.4$.

FIND The intelligent bound and the nearest-neighbour form. Which point shares a face with $s_1 = (1.5, 1)$ but is not a nearest neighbour?

METHOD From s_1 the others are at 3, 2 and $\sqrt{13}$. The region of s_1 is the first quadrant, and its faces are shared with the points at 3 and 2. Each Q argument is $d/\sqrt{2N_0} = d/\sqrt{0.8}$.

$$d = 2 : Q(2.24) = 1.27 \times 10^{-2}, \quad d = 3 : Q(3.35) = 4.0 \times 10^{-4}$$

SOLUTION

$$\text{intelligent} = Q(2.24) + Q(3.35) = 1.31 \times 10^{-2}$$

$$\text{nearest} = \bar{N}_{\min} Q(2.24) = 1.27 \times 10^{-2}$$

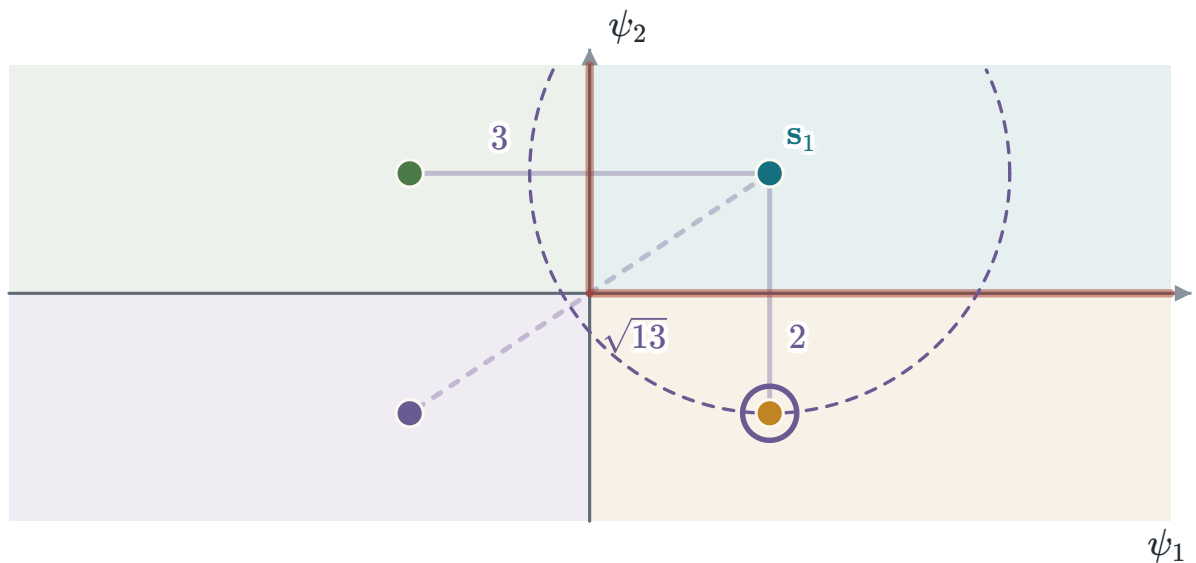
Each point has one neighbour at $d_{\min} = 2$, so $\bar{N}_{\min} = 1$. The point $(-1.5, 1)$ shares the face $r_1 = 0$ but sits at 3, more than $d_{\min}$.

CHECK The regions are quadrants, so the exact value is $1 - (1 - Q(3.35))(1 - Q(2.24)) = 1.31 \times 10^{-2}$. The nearest-neighbour form is 3% below it, so it is not a bound.

COMMON ERROR

Use $Q(2.24) + Q(3.35)$ for both faces, not only $Q(2.24)$ for the nearest neighbour.

The figure below marks the faces and the nearest neighbour of s_1 in Example 4.6.



Example 4.6. The two red faces of R_1 are shared with the points at 3 and 2. The dashed circle of radius $d_{\min} = 2$ passes through one nearest neighbour, ringed.

Tightness of the bounds

The bounds differ most at low SNR. Many terms are then large, and they overlap.

LOW SNR

The union bound can pass 1 and then tells nothing. For 16-QAM with $E_s = 10$ at $E_s/N_0 = 0$ dB, it adds 15 heavily overlapping terms and gives 2.79.

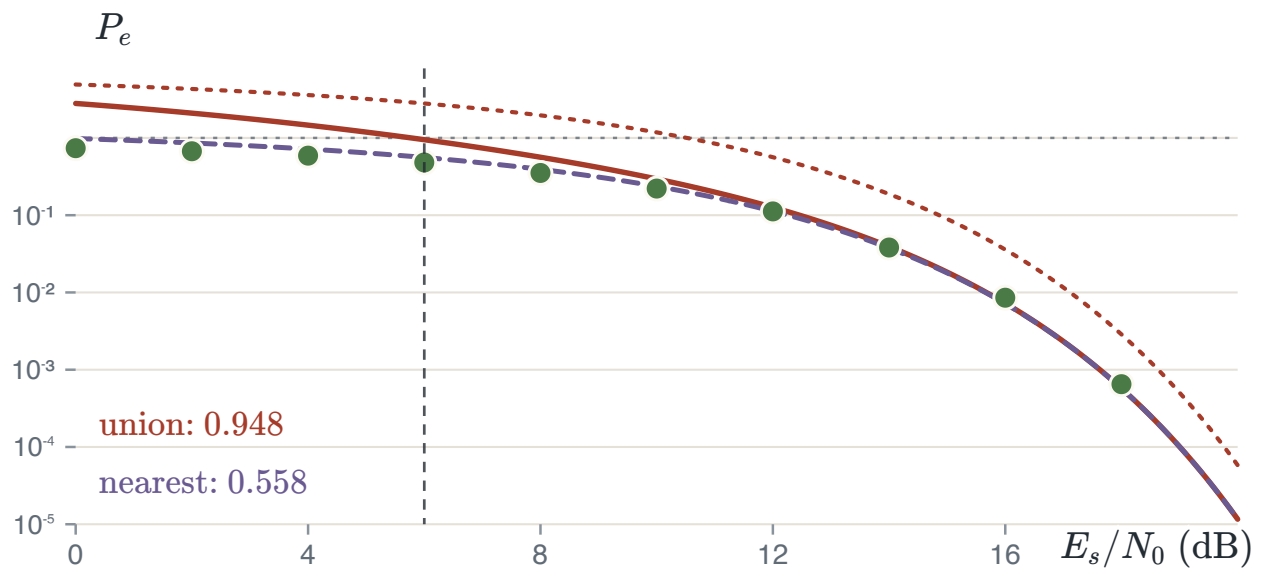
HIGH SNR

The terms at $d_{\min}$ dominate. The union bound, the nearest-neighbour form and the simulation meet.

THE MINIMUM-DISTANCE BOUND

It counts all 15 other points at $d_{\min}$, five times the $\bar{N}_{\min} = 3$ of 16-QAM. So it stays five times too high.

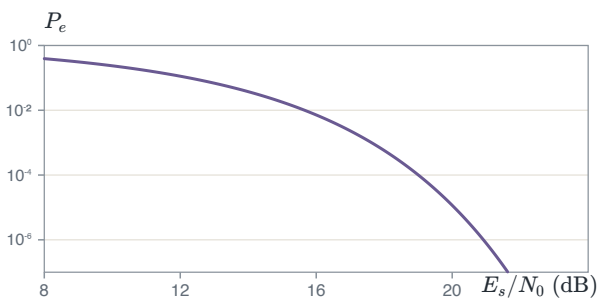
The figure below compares the three forms with a simulation for 16-QAM.



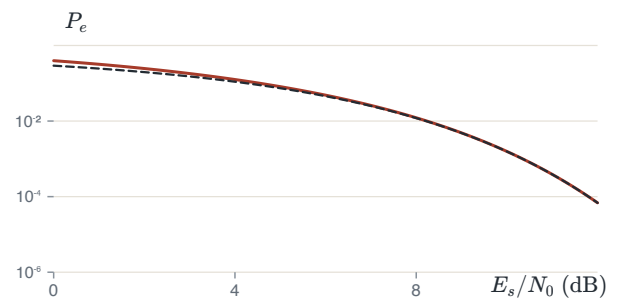
16-QAM with $E_s = 10$, marked at $E_s/N_0 = 6$ dB. Solid red: the union bound. Dotted red: the minimum-distance bound. Dashed violet: the nearest-neighbour form. Green dots: a simulation of 20 000 symbols. The dashed line at the top is $P_e = 1$.

Error probability around us

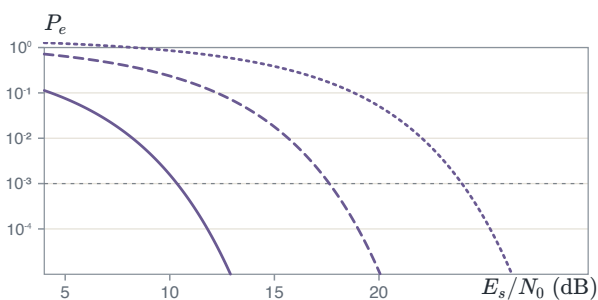
Wi-Fi, LTE and 5G use the nearest-neighbour form and the union bound, and a tester measures the error rate.



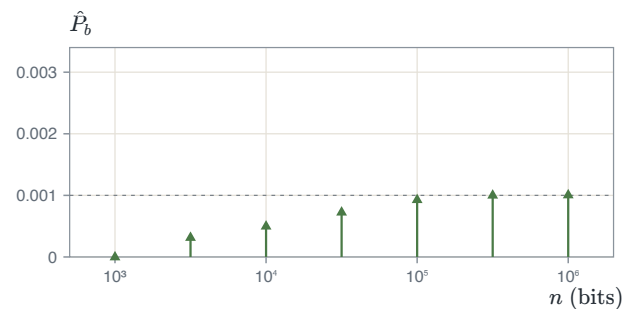
Wi-Fi 16-QAM has $d_{\min}^2 = 0.4E_s$ and $\bar{N}_{\min} = 3$. The nearest-neighbour form is $P_e \approx 3Q(\sqrt{E_s/5N_0})$.



LTE control channels use QPSK. Solid: the union bound. Dashed: the exact P_e . They differ by less than 1% above 10 dB.



A 5G phone picks its modulation from the channel. For $P_e = 10^{-3}$, QPSK, 16-QAM and 64-QAM need about 10.3, 17.6 and 24.0 dB.



A bit-error-rate tester divides the errors by the bits sent. At $P_b = 10^{-3}$ the estimate settles once about 100 errors are in.

NEAREST NEIGHBOURS FIRST

At useful error rates the terms at $d_{\min}$ dominate. Designers quote $\bar{N}_{\min} Q\left(\sqrt{d_{\min}^2/2N_0}\right)$.

A TARGET ERROR RATE

A link fixes a target such as 10^{-3} . Each constellation then needs its own E_s/N_0 .

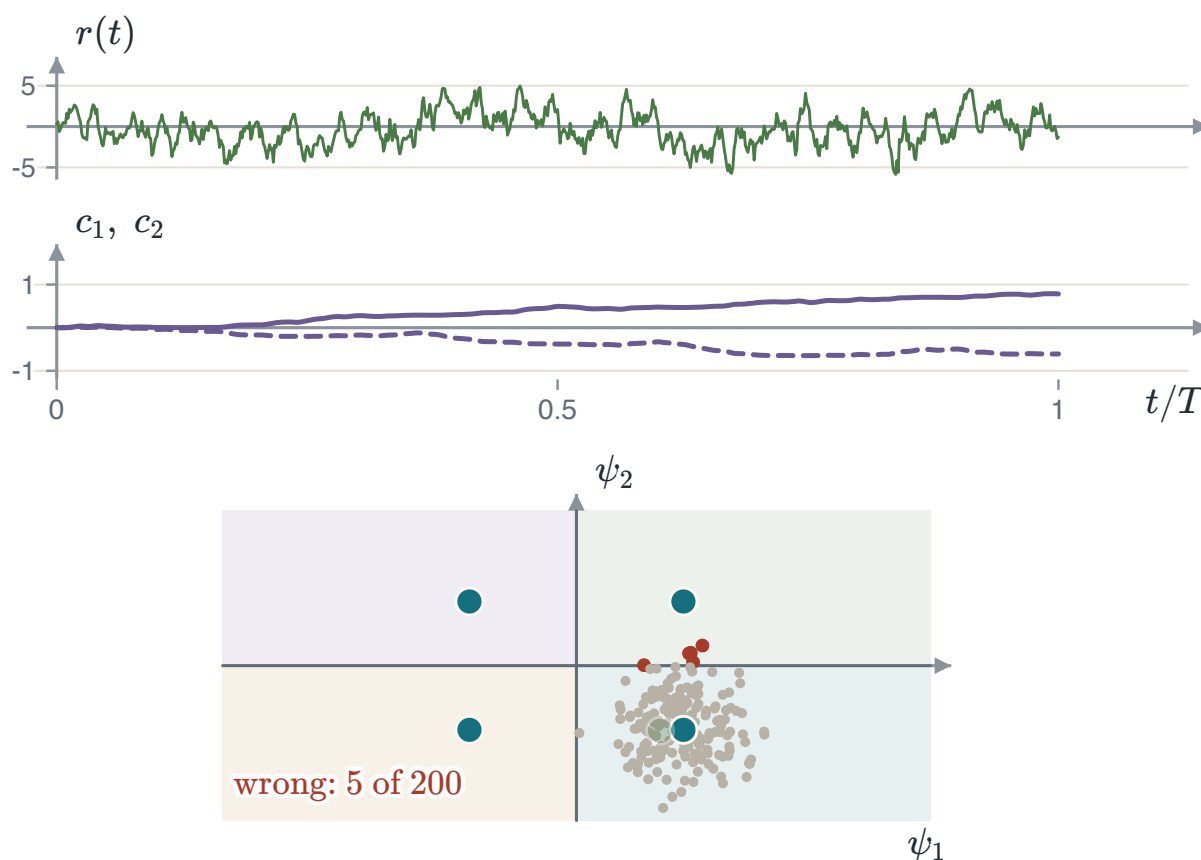
FEW ERRORS

An error rate measured from a few errors is noisy. A tester waits for about 100 errors.

4.5 Summary

From waveform to error rate

The figure below follows one symbol through the receiver and then repeats it.



One symbol through the receiver: $r(t)$, the correlator outputs c_1 and c_2 , and the regions of four points $(\pm 1, \pm 0.6)$. The 200 repeats with $\sigma = 0.3$ estimate P_e , and the wrong ones are red.

The receiver of this chapter makes four moves.

1. Correlate $r(t)$ with the basis to get $\mathbf{r}$.
2. Pick the best metric: the nearest point for equal priors.
3. Read the regions from the bisectors.

4. Bound P_e from the distances.

Each modulation of Chapter 5 is a constellation. Its error rate follows from its distances and this receiver.

For example, take QPSK with $d_{\min}^2/2N_0 = 16$. Here $\bar{N}_{\min} = 2$ and $Q(4) = 3.17 \times 10^{-5}$, so the nearest-neighbour form gives $P_e \approx 6.3 \times 10^{-5}$.

Quick check

1 A set of $M = 32$ signals. How many bits does each symbol carry?

Answer: 5 bits, since $\log_2 32 = 5$.

2 $N_0 = 0.4$. What is the noise variance of each correlator output?

Answer: $N_0/2 = 0.2$.

3 $\mathbf{r} = (0.2, 0.9)$, with $(0, 1)$ and $(1, 0)$ equally likely. Which point does the receiver pick?

Answer: $(0, 1)$. The squared distances are 0.05 and 1.45, and the smaller wins.

4 $P(\mathbf{s}_1)$ rises. Which way does the boundary between $\mathbf{s}_1$ and $\mathbf{s}_2$ move?

Answer: Away from $\mathbf{s}_1$. A likely point gets a larger region, so the line moves toward $\mathbf{s}_2$.

5 Two equally likely points, $d = 2$ and $N_0 = 2$. What is P_e ?

Answer: $Q(1)$, since $d^2/2N_0 = 4/4 = 1$.

6 A set of $M = 8$ points. For one sent point, how many terms does the general union bound have?

Answer: 7 terms, one for each other point, $M - 1 = 7$.

Summary of results

RESULT	STATEMENT	ANCHOR
Bits and symbols	$k = \log_2 M$, so $R_b = kR_s$ and $T_b = T/k$	PS CH8.4
Correlator bank	$\mathbf{r} = \mathbf{s}_i + \mathbf{n}$, with $r_j = \int_0^T r(t) \psi_j(t) dt$	PS CH8.4.1
Matched filter	the same number at the sampling time $t = T$, with impulse response $\psi_j(T - t)$	PS CH8.4.1
Noise outside the signal space	does not depend on the signal and adds the same $\ n'\ ^2$ to every distance	PS CH8.4.1
Noise components	independent Gaussians with mean 0 and variance $N_0/2$. The cloud is round	PS CH8.4.1
MAP and ML	MAP maximises $P(\mathbf{s}_i) f(\mathbf{r} \mathbf{s}_i)$. With equal priors it is ML	PS CH8.4.1
Minimum distance	ML in Gaussian noise. MAP subtracts $N_0 \ln P(\mathbf{s}_i)$ from each squared distance	PS CH8.4.1
Correlation metric	$\mathbf{r} \cdot \mathbf{s}_i - E_i/2 + \frac{N_0}{2} \ln P(\mathbf{s}_i)$. Keep the largest	PS CH8.4.1
Decision regions	convex polygons cut by bisectors. Unequal priors shift each line toward the less likely point	PS CH8.4.1
Binary error	$Q(\sqrt{d^2/2N_0})$ for equal priors	PS CH8.3.3, 8.4.2
Union bound	$P_e \leq \frac{1}{M} \sum_i \sum_{j \neq i} Q(\sqrt{d_{ij}^2/2N_0})$. The intelligent bound keeps only faces	PS CH8.4.2
Nearest neighbours	$\bar{N}_{\min} Q(\sqrt{d_{\min}^2/2N_0})$. Close at high SNR, but not a bound	

Correlate to get $\mathbf{r}$, pick the best metric, and bound P_e from the distances. Chapter 5 uses this receiver for every modulation.

Projects to try

Four optional projects use the chapter on simulated signals. Each gives an aim, what it practises, a few steps and what to look for.

A CORRELATION RECEIVER FROM SAMPLES

Build the whole receiver for QPSK from sampled waveforms and measure how often it is wrong.

Practises: correlators computed as sums over samples. Scaling noise to a chosen N_0 . Counting errors to estimate P_e .

1. Build two carrier basis functions over one symbol at 100 samples.
2. Send random QPSK symbols and add Gaussian noise of variance $N_0/2$ to each sample, scaled for the sample spacing.
3. Correlate, pick the nearest point and count the errors.
4. Repeat at several N_0 and plot P_e on a log axis.

Look for: the measured points follow $2Q(\sqrt{2E_b/N_0})$ once enough errors are counted.

MAP AGAINST ML

See how much a receiver gains by knowing the priors.

Practises: the MAP threshold for a binary link. Why the gain grows as the priors become uneven. What happens when the assumed priors are wrong.

1. Send antipodal bits with $P(s_1) = p$ for several p from 0.5 to 0.95.
2. Decide each with the midpoint and with the MAP threshold.
3. Plot both error rates against p .
4. Decide with a threshold built for the wrong p .

Look for: the two receivers agree at $p = 0.5$. The MAP receiver pulls ahead as p grows, and a wrong prior can do worse than ML.

DECISION REGIONS ON A GRID

Draw the regions of any constellation by deciding every point of a fine grid.

Practises: minimum distance as a rule you can run anywhere. How priors move the boundaries. Why the regions are convex polygons.

1. Pick a constellation of five to eight points.
2. Decide every point of a 400×400 grid and colour it by the answer.
3. Give one point a larger prior and redraw.
4. Compare the boundaries with the perpendicular bisectors.

Look for: every boundary is a straight line. A larger prior pushes its point's boundaries outward, parallel to themselves.

Compare the union bound and its variants with a simulation across SNR.

Practises: the general and intelligent union bounds. The nearest-neighbour form. Where each is close to the truth.

1. Take 8-PSK and 16-QAM at unit average energy.
2. Compute every bound from the pairwise distances.
3. Simulate at SNR from 0 to 20 dB.
4. Plot all on one log axis and read the gap at each SNR.

Look for: at low SNR the general bound can pass 1. At high SNR all the curves close in on the simulation.

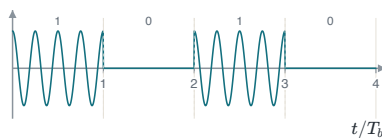
CHAPTER 5

Digital modulation methods

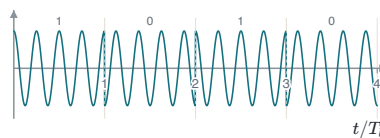
A baseband pulse cannot be sent over a radio channel, so the bits have to ride on a carrier. There are only three things about a sine wave that can be changed: its amplitude, its phase and its frequency. Every scheme in this chapter changes one of them, or two at once. Chapter 4 already gave the receiver and the error probability. All that is left is to place the points and measure the distance between them.

5.1 The three binary schemes

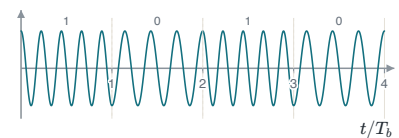
Take one bit at a time and two waveforms. Amplitude-shift keying sends the carrier for a one and nothing for a zero. Phase-shift keying sends the carrier either way and flips its sign. Frequency-shift keying sends one of two frequencies.



Amplitude-shift keying: the carrier is switched on and off.



Phase-shift keying: the carrier is always on and its sign flips.



Frequency-shift keying: one frequency for a one, another for a zero.

To compare them, put each on the axes of chapter 3 and measure. BPSK needs one basis function and its two points are $\pm\sqrt{E_b}$ on it. BFSK needs two, because the two frequencies are orthogonal, and its points sit one on each axis. BASK needs one, and its points are 0 and $\sqrt{2E_b}$ once the average over equally likely bits is taken.

THE THREE DISTANCES AND THE THREE ERROR PROBABILITIES

$$\text{BPSK: } d_{\min}^2 = 4E_b, \quad P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$\text{BFSK and BASK: } d_{\min}^2 = 2E_b, \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

The two answers differ by a factor of two inside the square root. This is 3.01 dB. That gap does not depend on the noise level and cannot be closed by spending more power. It is a fact about where the points are.

THREE-DECIBEL DIFFERENCE

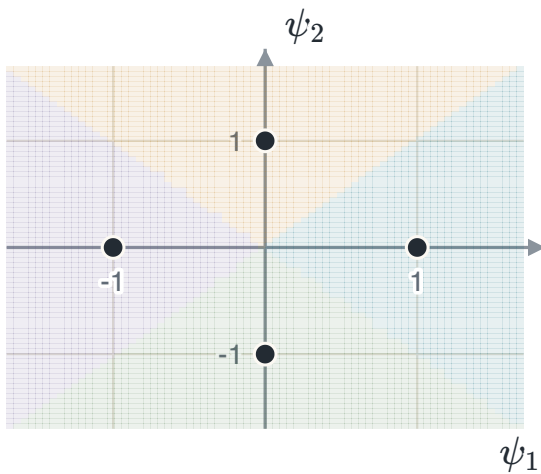
Antipodal points are $2\sqrt{E_b}$ apart. Orthogonal points are $\sqrt{2E_b}$ apart, which is smaller by $\sqrt{2}$. Squaring this ratio gives a factor of two. Therefore, the required energy differs by $10 \log_{10} 2 = 3.01$ dB.

BASK AND BFSK REACH THE SAME ANSWER BY DIFFERENT ROUTES

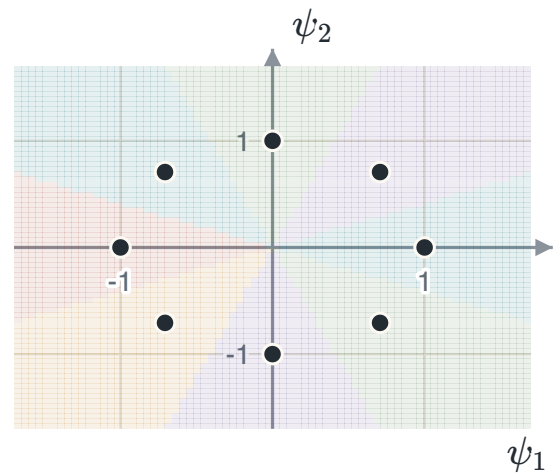
BASK keeps its points on one line but half its symbols cost nothing, so the average energy is half the peak. BFSK keeps the full energy in every symbol but separates the points by a right angle instead of a straight line. Both end at $d^2 = 2E_b$. The mathematics does not prefer either. The transmitter does, and BASK asks it to switch between zero power and full power.

5.2 M-ary phase-shift keying

Sending one bit at a time wastes the two dimensions a carrier offers. Use both, keep the energy fixed, and the constellation becomes M points spaced evenly around a circle of radius $\sqrt{E_s}$. Each symbol now carries $\log_2 M$ bits.



$M = 4$, two bits a symbol. The four regions are the four quadrants.



$M = 8$, three bits. The regions are wedges, and they are narrowing.

Two neighbouring points subtend an angle $2\pi/M$ at the centre. Dropping a perpendicular from the centre onto the line joining them cuts that angle in half and gives the distance directly.

$$d_{\min} = 2\sqrt{E_s} \sin\left(\frac{\pi}{M}\right), \quad N_{\min} = 2$$

$$P_e \approx 2Q\left(\sqrt{\frac{2E_s}{N_0}} \sin \frac{\pi}{M}\right)$$

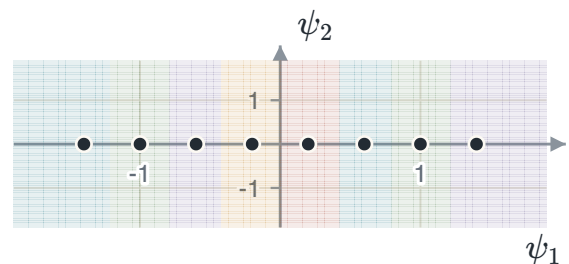
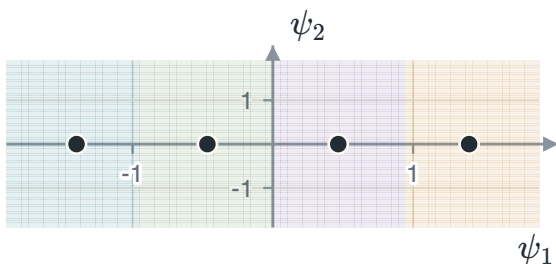
$N_{\min} = 2$ for every M above two, because a point on a circle has one neighbour clockwise and one anticlockwise. At $M = 2$ the formula gives $d_{\min} = 2\sqrt{E_s}$, which is BPSK — a general result that reproduces the case already known.

EXAMPLE 5.1 – WHAT THE STEP FROM FOUR POINTS TO EIGHT COSTS

- GIVEN** QPSK and 8-PSK, at whatever energy each needs to reach the same symbol error probability.
- FIND** The extra energy 8-PSK needs, per symbol and per bit.
- METHOD** A fixed error probability means a fixed Q argument, so $(E_s/N_0) \sin^2(\pi/M)$ must be held constant. Take the ratio.
- SOLUTION** $\frac{\sin^2(\pi/4)}{\sin^2(\pi/8)} = \frac{0.5}{0.1464} = 3.414$, which is 5.33 dB per symbol. 8-PSK carries three bits where QPSK carries two, so per bit the figure is $3.414 \times \frac{2}{3} = 2.276$, or 3.57 dB.
- CHECK** Both numbers are positive, as they must be: at the same radius, eight points are closer together than four. The per-bit figure is the smaller of the two, because the extra bit pays part of its own way.

5.3 M-ary amplitude-shift keying

Change the amplitude among M levels instead. One basis function carries them all, so the constellation is M points on a line at $\pm A, \pm 3A, \dots, \pm(M-1)A$, and neighbouring points are $2A$ apart.



$M = 4$ on one axis. The two end regions run off to infinity.

$M = 8$: the same line, twice as crowded.

$$E_{s,\text{avg}} = \frac{A^2(M^2 - 1)}{3} \implies d_{\min}^2 = \frac{12E_{s,\text{avg}}}{M^2 - 1}$$

$$N_{\min} = \frac{2(M - 1)}{M}, \quad P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right)$$

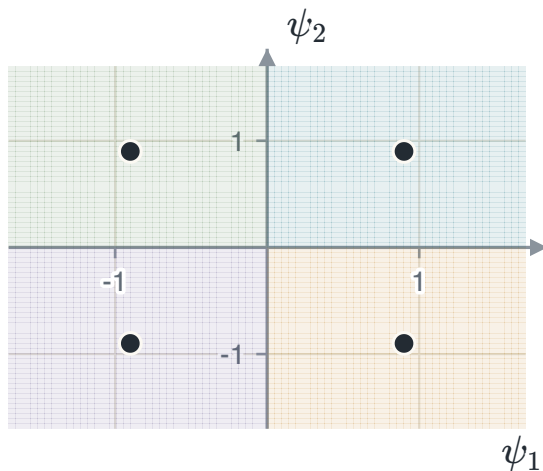
The M^2 in the denominator is the trouble. Doubling M divides the squared distance by roughly four. This is about 6 dB, and it does so at every doubling.

AVERAGE NEIGHBOR COUNT

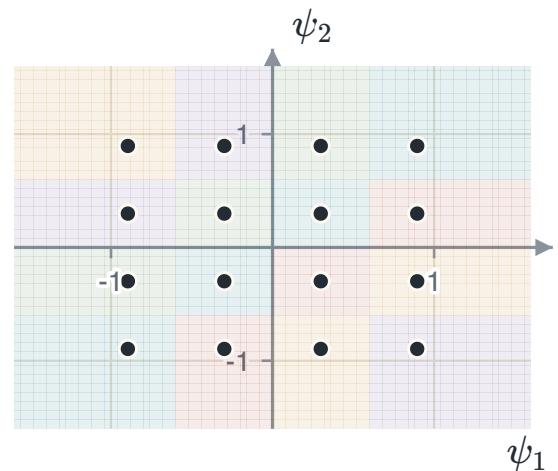
The two end points have one nearest neighbor each. The other $M - 2$ points have two. Averaging over all equally likely symbols gives $N_{\min} = 2(M - 1)/M$. Thus, $N_{\min} = 1.5$ for $M = 4$ and 1.75 for $M = 8$.

5.4 Quadrature amplitude modulation

Run one amplitude-modulated signal on the cosine and an independent one on the sine. The result is two $\sqrt{M}$ -level constellations at right angles. Therefore, M points on a square grid. The lecture material states this as M -QAM being two-dimensional M -ASK.



4-QAM: two levels a side. These are the four points of QPSK — at $M = 4$ the two schemes are the same scheme.



16-QAM: four levels a side, four bits a symbol. The corners have two neighbours, the edges three, the four in the middle four.

SQUARE M-QAM

$$E_{s,\text{avg}} = \frac{(M-1)d^2}{6} \implies d_{\min}^2 = \frac{6E_{s,\text{avg}}}{M-1}$$

$$N_{\min} = \frac{4(2) + 8(3) + 4(4)}{16} = 3 \quad \text{for 16-QAM}$$

Compare the denominators: $M-1$ for QAM and M^2-1 for PAM. Two dimensions reduce the rate at which minimum distance decreases as M increases.

EXAMPLE 5.2 – SIXTEEN POINTS ON A GRID AGAINST SIXTEEN ON A LINE

GIVEN	16-QAM and 16-PAM at the same average symbol energy. Both carry four bits a symbol.
FIND	The advantage of QAM in decibels.
METHOD	One formula each, then the ratio of the squared distances.
SOLUTION	QAM: $d_{\min}^2 = 6E_s/15 = 0.400E_s$. PAM: $d_{\min}^2 = 12E_s/255 = 0.0471E_s$. The ratio is 8.5, and $10 \log_{10} 8.5 = 9.29$ dB.
CHECK	Both methods carry the same number of bits at the same average power. Only their point geometry differs. The second dimension gives QAM a 9.3 dB distance advantage in this example.

5.5 M-ary frequency-shift keying

Use M frequencies, chosen so that the waveforms are orthogonal. Orthogonal signals need one basis function each. Therefore, the constellation lives in M dimensions with one point on each axis at distance $\sqrt{E_s}$ from the origin.

M-FSK

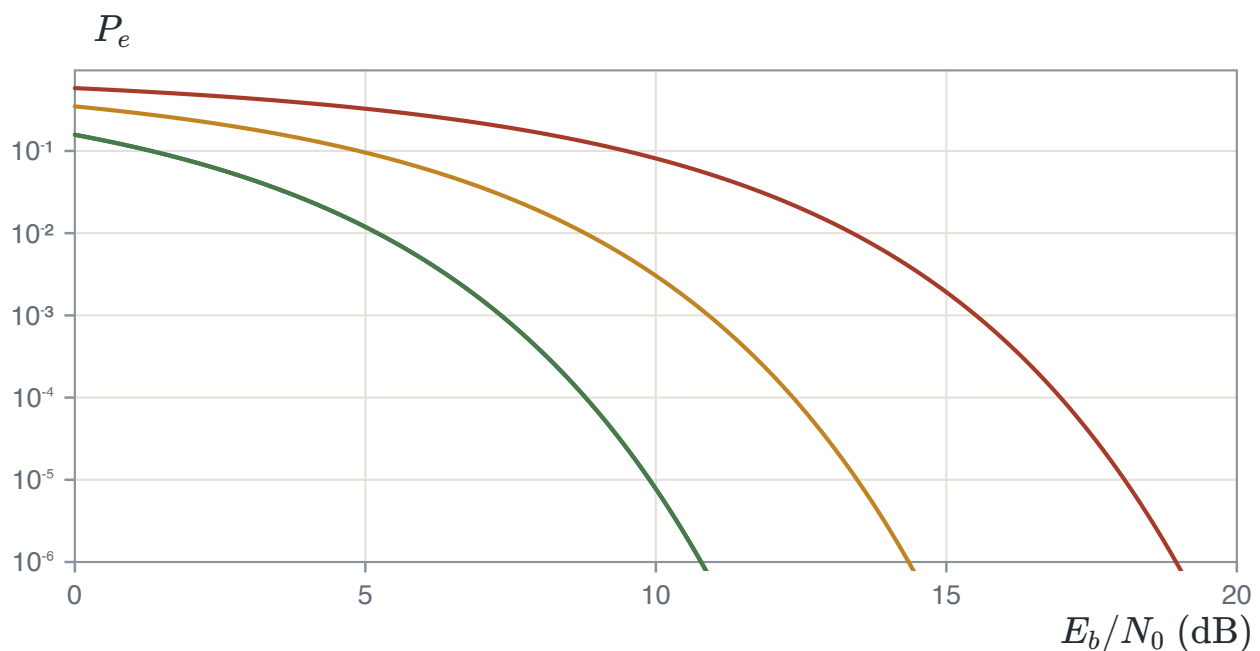
$$d_{jk} = \sqrt{2E_s} \quad \text{for every pair,} \quad N_{\min} = M - 1$$

$$P_e \approx (M-1) Q\left(\sqrt{\frac{E_s}{N_0}}\right)$$

The distance does not depend on M at all. Adding waveforms costs nothing in distance — but each new waveform needs its own frequency slot, so the price is paid in bandwidth instead.

THE TRADE, STATED PLAINLY

PSK and QAM hold the bandwidth fixed as M grows and pay in energy, because the points crowd together. FSK holds the distance fixed and pays in bandwidth. A deep-space link, with bandwidth to spare and no power, chooses FSK. A mobile phone, with the opposite problem, chooses QAM. Neither is better in the abstract — it depends entirely on which resource is scarce.



Symbol error probability against energy per bit for four sizes of PSK. $M = 2, 4, 8, 16$ from left to right. $M = 2$ and $M = 4$ lie almost on top of each other. This is why QPSK is everywhere. It carries twice the bits of BPSK for the same energy per bit.

5.6 Summary

SCHEME	$d_{\min}^2$	$N_{\min}$	ANCHOR
BPSK	$4E_b$	1	PS CH8.6.1
BFSK, BASK	$2E_b$	1	PS CH9.5
M -PSK	$4E_s \sin^2(\pi/M)$	2	PS CH8.6.3
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$	PS CH8.5.3
M -QAM	$6E_s/(M - 1)$	3 at $M = 16$	PS CH8.7.1
M -FSK	$2E_s$	$M - 1$	PS CH9.5

Every row was obtained the same way. Write the waveforms down, find the basis functions they need, place the points, measure the shortest distance, count how many pairs sit at it. The error probability then comes from chapter 4 without any further thought about waveforms. Convert once with $E_s = (\log_2 M)E_b$ when a question is stated per bit, and never convert twice.

What none of this says is how many bits a channel can carry at all. Every scheme here trades energy against bandwidth, and it is reasonable to ask whether there is a limit to the trade. Chapter 6 answers that, and the answer does not depend on which modulation is used.

CHAPTER 6

An introduction to information theory

The previous chapters started with a given bit stream and transmitted it through a channel. This chapter studies the source of those bits. Entropy measures the average information from the source. It also gives the minimum average rate for lossless source coding.

6.1 The information in one symbol

A **discrete memoryless source** emits one of K symbols each time, with fixed probabilities $p_1, \dots, p_K$, chosen independently of what came before. How much is learnt when one arrives? If it was certain, nothing. If it was almost impossible, a great deal.

SELF-INFORMATION

$$I(s_k) = \log_a \frac{1}{p_k} = -\log_a p_k$$

The base sets the unit: $a = 2$ gives **bits**, $a = e$ gives **nats**, $a = 10$ gives **Hartleys**. This course uses base two throughout.

THREE PROPERTIES, AND WHAT FORCES THE LOGARITHM

$I(s_k) \geq 0$, because a probability is at most one and its logarithm at most zero.

$I(s_k) \geq I(s_j)$ **when** $p_k \leq p_j$: rarer means more informative.

$I(s_k s_j) = I(s_k) + I(s_j)$ **for independent symbols**, because their probabilities multiply and the logarithm turns a product into a sum. That third property is the one that rules out every function except a logarithm.

ONE HALVING IS ONE BIT

$-\log_2 \frac{1}{2} = 1$, so every halving of a probability adds exactly one bit. A symbol of probability $1/1000$ is about ten halvings away from certainty, so it carries about 10 bits. Most estimates in this chapter can be done that way, without a calculator.

6.2 Entropy

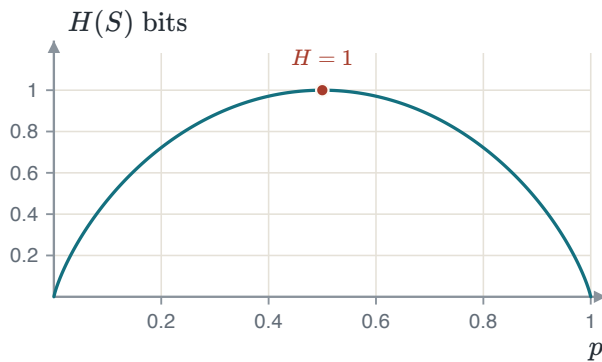
Self-information describes one symbol. Average it over the alphabet, weighting each symbol by how often it happens, and the result describes the source.

ENTROPY

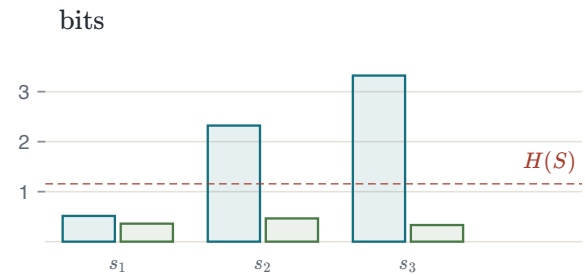
$$H(S) = E[I(s_k)] = -\sum_{k=1}^K p_k \log_2 p_k \quad \text{bits a symbol}$$

$$0 \leq H(S) \leq \log_2 K$$

Zero when one symbol has probability one — nothing is ever in doubt. At the top, $\log_2 K$, when all K are equally likely: putting $p_k = 1/K$ into the sum gives $\sum \frac{1}{K} \log_2 K = \log_2 K$.



The entropy of a binary source. One bit at $p = \frac{1}{2}$, nothing at either end, and flat near the top — a slightly unfair coin is almost as informative as a fair one.



The standard example. The left bar of each pair is what the symbol carries. The right bar is that weighted by how often it happens. The three right bars add to $H(S)$.

EXAMPLE 6.1 – THE ENTROPY OF A THREE-SYMBOL SOURCE

GIVEN $S = \{s_1, s_2, s_3\}$ with probabilities 0.7, 0.2, 0.1.

FIND The entropy.

SOLUTION $H(S) = -0.7 \log_2 0.7 - 0.2 \log_2 0.2 - 0.1 \log_2 0.1 = 0.3602 + 0.4644 + 0.3322 = 1.1568$ bits a symbol.

CHECK $\log_2 3 = 1.585$, and 1.1568 is below it, as it must be for a source that is not uniform. Numbering the three symbols would cost 2 bits each, so plain numbering wastes almost 0.85 bits every symbol. Recovering that is the whole of source coding.

Two numbers get confused here and should not be. $\log_2 K$ is the most a source of this size *could* carry. $\lceil \log_2 K \rceil$ is what a fixed-length code *costs*. The first gap is the source being uneven; the second is rounding.

6.3 Extended sources

Nothing forces a coder to work one symbol at a time. Group them in n s and treat each block as one symbol of a new alphabet. That is the **n -th extension**, written S^n , and it has K^n symbols.

EXTENSION

$$H(S^n) = n H(S)$$

The source is memoryless, so the symbols in a block are independent and their information adds. This is the third property of self-information, applied n times.

EXAMPLE 6.2 – THE SAME SOURCE, TWO AT A TIME

GIVEN	The source of Example 6.1, extended by two.
FIND	$H(S^2)$.
METHOD	S^2 has $3^2 = 9$ symbols with probabilities 0.49, 0.14, 0.07, 0.14, 0.04, 0.02, 0.07, 0.02, 0.01.
SOLUTION	Summing $-p \log_2 p$ over all nine gives 2.3136 bits a block.
CHECK	$2 \times 1.1568 = 2.3136$. The long way and the short way agree, and per symbol nothing has changed — 2.3136 over two symbols is 1.1568 each. What changes is how much of a whole bit gets wasted in rounding, and section 6.6 turns that into a bound.

ONLY BECAUSE IT IS MEMORYLESS

If the source had memory. As English does, where q is followed by u . The block probabilities would not be products, and $H(S^n)$ would be *less* than $nH(S)$. That difference is exactly the redundancy real compressors live on, and it is outside this course.

6.4 What a code costs

A source encoder turns each symbol into a string of bits, its **codeword**. The codewords need not all be the same length, and the good idea of this chapter is that they should not be. Short ones for common symbols, long ones for rare.

AVERAGE LENGTH AND EFFICIENCY

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{L_{\min}}{\bar{L}} = \frac{H(S)}{\bar{L}} \leq 1$$

The **source-coding theorem** says $\bar{L} \geq H(S)$ for any code from which the symbols can be recovered exactly. So $L_{\min} = H(S)$: the entropy is not merely a measure of information, it is a limit on how few bits will carry it.

SOURCE CODING IS NOT CHANNEL CODING

Source coding removes bits the message does not need. Channel coding adds bits back so that errors can be found and fixed. They pull in opposite directions, they are done in that order, and this course covers only the first.

Written English gives a concrete case. Its entropy is estimated at about 1.3 bits a letter, while a typical letter-by-letter variable-length code reaches $\bar{L} = 4.22$, so $\eta = 0.31$. Morse code is the same idea by hand: one dot for e , four symbols for z .

6.5 Codes that can be read back

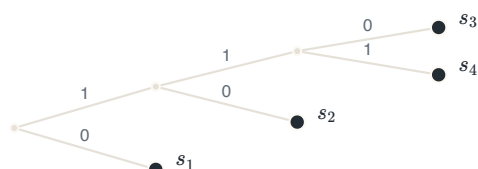
Short codewords are useful only when the receiver can separate them. A code is **uniquely decodable** if each coded bit string has one source-symbol sequence. A **prefix code** has no codeword that begins another codeword. This stronger condition permits immediate decoding when the last codeword bit arrives. A prefix code is also **instantaneous**.

SYMBOL	CODE I	CODE II	CODE III
s_1	0	0	0
s_2	1	10	01
s_3	00	110	011
s_4	11	111	0111

Code I is not a prefix code and is not uniquely decodable either: the string 00 is both s_3 and s_1s_1 . Code II is a prefix code. Code III is not a prefix code, since 0 begins all three of the others.

CODE III

Code III is uniquely decodable because each codeword starts with 0 and then contains only 1s. However, it is not instantaneous. After 011, the decoder must wait for the next bit to distinguish s_3 from the start of s_4 .



Code II. Every symbol is at a leaf, so no path to one passes through another. That is what the prefix property looks like.



Code III. Every symbol sits on one path, each hanging off the node before it. Nothing is at a leaf but the last, and that is why the decoder must wait.

6.6 The Kraft inequality, and how close a code can get

Before selecting codewords, determine whether their lengths can satisfy the prefix condition. Treat the binary tree as one unit. A codeword of length l uses a fraction 2^{-l} of the tree. These fractions cannot sum to more than one.

KRAFT INEQUALITY

$$\sum_{k=1}^K 2^{-l_k} \leq 1$$

The three Kraft sums are 1.5, 1, and 0.9375. Code I fails, so no prefix code has its lengths. Code II uses the tree completely. Code III satisfies the inequality but is not itself a prefix code. Thus, the inequality is necessary but not sufficient for given codewords.

HOW CLOSE A PREFIX CODE GETS

$$H(S) \leq \bar{L} < H(S) + 1$$

$$H(S) \leq \frac{L_n}{n} < H(S) + \frac{1}{n} \implies \lim_{n \rightarrow \infty} \frac{L_n}{n} = H(S)$$

The ideal length for symbol k is $-\log_2 p_k$, almost never a whole number. Therefore, each length is rounded up and the rounding costs under one bit. Applying the same bound to the n -th extension spreads that one bit over n symbols.

DYADIC PROBABILITIES

If every probability has the form $p_k = 2^{-l_k}$, the distribution is **dyadic**. No codeword length requires rounding. Then $\bar{L}$ and $H(S)$ equal the same sum. Therefore, $\bar{L} = H(S)$.

BLOCK-CODE SIZE

The coding gap decreases as $1/n$, but the block alphabet grows as K^n . A gap of 0.05 bit per symbol needs $n = 20$. For a three-symbol source, this block contains $3^{20} = 3.5 \times 10^9$ possible messages.

6.7 Huffman coding

The bound says a good code exists. Huffman coding builds one, and it builds the best one: no prefix code on single symbols has a smaller average length.

THE ALGORITHM

1. List the symbols in order of decreasing probability.
 2. Take the two least likely, label one 0 and the other 1, and merge them into one symbol whose probability is their sum. Re-sort.
 3. Repeat until two symbols remain, and label those 0 and 1.
- Each codeword is the labels along the path back to its symbol.

EXAMPLE 6.3 – THE STANDARD FIVE-SYMBOL SOURCE

GIVEN $p = 0.4, 0.2, 0.2, 0.1, 0.1$.

MERGING $0.1 + 0.1 = 0.2$. Then the two smallest are 0.2 and 0.2, giving 0.4. Then 0.2 and 0.4 give 0.6. Then 0.4 and 0.6 finish it.

SOLUTION Lengths 2, 2, 2, 3, 3, so $\bar{L} = 0.4(2) + 0.2(2) + 0.2(2) + 0.1(3) + 0.1(3) = 2.2$ bits a symbol.

EFFICIENCY $H(S) = 2.1219$ bits, so $\eta = 2.1219/2.2 = 0.9645$ — the code spends 3.68% more than the source strictly needs.

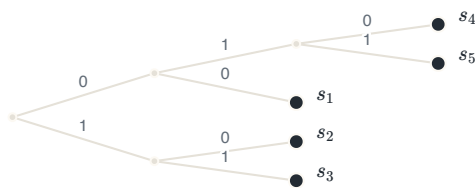
CHECK $2.1219 \leq 2.2 < 3.1219$: the two-sided bound holds. A fixed-length code would have cost $\lceil \log_2 5 \rceil = 3$ bits, so Huffman saved 0.8 bits a symbol.

Two Huffman choices are free. Either symbol in a merged pair can receive 0. A merged symbol can also occupy different positions during a probability tie. These choices give different codes with the same minimum average length. Their codeword-length variances can differ.

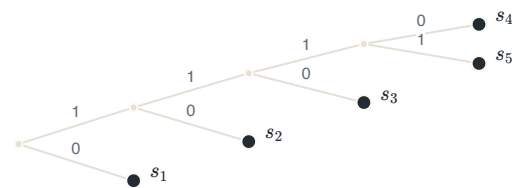
VARIANCE OF THE CODEWORD LENGTH

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

For the example: placing the merged symbol high gives lengths 2, 2, 2, 3, 3 and $\sigma^2 = 0.16$. Placing it low gives 1, 2, 3, 4, 4, the same $\bar{L} = 2.2$, and $\sigma^2 = 1.36$.



Merged symbol placed high. Lengths 2, 2, 2, 3, 3, average 2.2, variance 0.16. The tree is nearly balanced.



Merged symbol placed low. Lengths 1, 2, 3, 4, 4, the same average 2.2, variance 1.36. The tree is a ladder.

MINIMUM LENGTH VARIANCE

Variable codeword lengths make a buffer fill and empty at changing rates. Smaller length variance reduces these changes. During a probability tie, place the merged symbol as high as possible. This rule gives the minimum-variance Huffman code.

6.8 Lempel–Ziv coding

Huffman coding has one practical weakness, and it is not its length. It needs the probabilities *before* it can build the tree, and for a stream arriving over a wire nobody knows them in advance. A **universal** code is one that reaches the entropy without being told the source, and the Lempel–Ziv algorithm is the one in use everywhere.

THE ALGORITHM

Read the stream once. Each time, take the **shortest run of bits not seen before** and store it in a dictionary. Because it is the shortest new one, everything but its last bit is already stored. So it is sent as a **pointer** to that earlier entry followed by the one new bit, the **innovation**. The decoder builds the same dictionary from the same blocks in the same order. Therefore, the dictionary is never transmitted.

EXAMPLE 6.4 – PARSING AND SENDING A STREAM

GIVEN	The stream 0 1 00 011 1 . . . , with 0 and 1 held in the dictionary at positions 1 and 2.
METHOD	Parse left to right into pieces not seen before, then send each piece as (position of its start, new bit).
SOLUTION	00 takes position 3: its start 0 is at position 1 and its new bit is 0, so with four-bit blocks it goes as 001 0. Next 011 takes position 4: its start 01 is at position 4... so it is sent as 100 1.
DECODING	Receive 1101. The last bit is the innovation 1. The first three bits, 110, give pointer 6. Therefore, append 1 to dictionary entry 6. The decoder already contains this entry.
CHECK	Seven pieces at four bits each cost 28 bits where the raw stream was 18. The method <i>loses</i> on a short stream, and that is not a fault: the dictionary has to be paid for before it can pay back. In practice the block is 12 bits, giving $2^{12} = 4096$ entries, and a long file compresses to roughly two thirds of its size.

OPTIMAL AGAINST UNIVERSAL

Huffman is optimal but not universal: no prefix code on single symbols beats it, and it cannot start until the probabilities are known. Lempel–Ziv is universal but reaches the limit only in the long run. That trade is the whole difference between them, and it is why the second is what compresses an archive.

6.9 The discrete memoryless channel

The first part of this chapter studied the source. The second part studies the channel. A **discrete memoryless channel** maps one symbol from a finite input alphabet to a finite output alphabet. The current output depends only on the current input.

THE CHANNEL, AND WHAT THE TRANSMITTER ADDS

$$p(y_k | x_j) = P(Y = y_k | X = x_j), \quad \sum_{k=0}^{K-1} p(y_k | x_j) = 1 \text{ for every } j$$
$$p(x_j, y_k) = p(y_k | x_j) p(x_j), \quad p(y_k) = \sum_{j=0}^{J-1} p(y_k | x_j) p(x_j)$$

The transition probabilities collected into a matrix, one row per input, are the **channel matrix**. Every *row* sums to one, because something must come out when a symbol goes in. The columns do not, and expecting them to is the usual first mistake. The **input distribution** $p(x_j)$ is not part of the channel — it belongs to the transmitter, and it is chosen.

THE BINARY SYMMETRIC CHANNEL

Two inputs, two outputs, and one number. The probability p that a bit is flipped, called the **crossover probability**. Its matrix is $\begin{bmatrix} 1-p & p \\ p & 1-p \end{bmatrix}$. It is not an abstraction invented for this chapter. The binary receiver of chapter 2, seen from outside, *is* a binary symmetric channel with $p = Q\left(\sqrt{2E_b/N_0}\right)$. With equally likely inputs the output is equally likely whatever p is, even at $p = \frac{1}{2}$ where the channel is destroying everything. Therefore, a balanced output is no evidence that anything survived.

6.10 Mutual information

Before anything arrives, the uncertainty about the input is $H(X)$. Then one output symbol arrives, and it rarely settles the question but it does change the odds. What remains is an entropy like any other, averaged over which output actually came.

CONDITIONAL ENTROPY, AND THE INFORMATION THAT GOT THROUGH

$$H(X | Y) = \sum_k H(X | Y = y_k) p(y_k) = - \sum_k \sum_j p(x_j, y_k) \log_2 p(x_j | y_k)$$

$$I(X; Y) = H(X) - H(X | Y) = H(Y) - H(Y | X)$$

$$H(X, Y) = H(X) + H(Y) - I(X; Y)$$

Read the middle line in words: **how much doubt the arrival removed**, in bits, per use of the channel. The second form is the one used, because $H(Y | X)$ comes straight off the channel matrix while $H(X | Y)$ needs Bayes' rule first.

THREE PROPERTIES

Symmetry. $I(X; Y) = I(Y; X)$. Bayes' rule gives $p(x_j | y_k)/p(x_j) = p(y_k | x_j)/p(y_k)$, and substituting turns one sum into the other. What the output says about the input equals what the input says about the output — which is why the quantity is called *mutual*.

Non-negativity. $I(X; Y) \geq 0$, with equality exactly when X and Y are independent. An observation never leaves you knowing less than before.

Joint entropy. Adding $H(X)$ and $H(Y)$ counts the shared part twice, so it is subtracted once. That is the whole content of the third line.

EXAMPLE 6.5 – THE BINARY SYMMETRIC CHANNEL AT $p = 0.1$

GIVEN	A BSC with $p = 0.1$ and equally likely inputs.
FIND	$H(X)$, $H(Y X)$, $H(Y)$ and $I(X; Y)$.
METHOD	$H(Y X)$ comes off the matrix: whichever symbol goes in, the output distribution is $(0.9, 0.1)$ in some order. $H(Y)$ needs the output distribution first.
SOLUTION	$H(X) = 1$ bit. $H(Y X) = H(0.1) = -0.1 \log_2 0.1 - 0.9 \log_2 0.9 = 0.4690$ bits. The output is equally likely, so $H(Y) = 1$ bit, and $I(X; Y) = 1 - 0.4690 = 0.5310$ bits per use.
CHECK	One bit was offered and just over half a bit arrived. The missing 0.4690 bits were not delayed or corrupted. They were destroyed, and no receiver however clever recovers them. The joint entropy is $1 + 1 - 0.5310 = 1.4690$ bits, and at $p = 0.25$ the shared part falls to 0.1887 while the total rises to 1.8113. A noisier channel does not destroy uncertainty, it moves it out of the shared part.

THE TWO CONDITIONAL ENTROPIES ARE NOT THE SAME THING

$H(Y | X)$ is the uncertainty about the *output* given the input, and for the BSC it is $H(p)$ whatever the transmitter does. A property of the channel alone. $H(X | Y)$ is the uncertainty about the *input* given the output, and it depends on the input distribution too. $I(X; Y) = I(Y; X)$ is a statement about one number. It does not make the two ends interchangeable.

6.11 Channel capacity

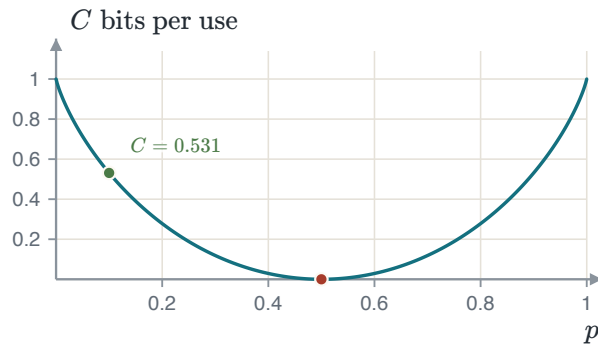
Mutual information measures one transmitter on one channel. Change how often each symbol is sent and the number moves. The channel is fixed and the input distribution is the engineer's to choose. Therefore, take the best choice. What comes out then depends on the channel matrix alone. This occurs because the one free thing has been optimised away. **capacity is a property of the channel, and mutual information is not.**

CAPACITY, AND THE CAPACITY OF THE BSC

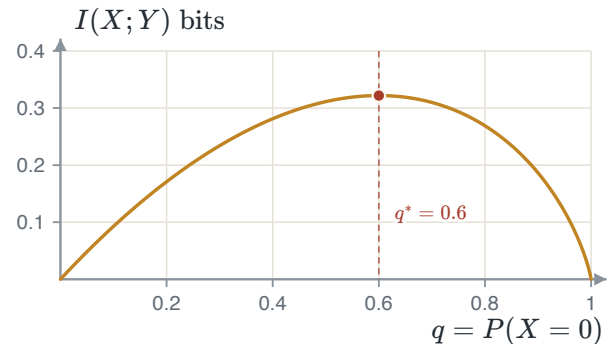
$$C = \max_{\{p(x_j)\}} I(X; Y) \quad \text{bits per channel use}$$

$$C_{\text{BSC}} = 1 - H(p), \quad H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

For the BSC, $H(Y | X) = H(p)$ whatever the transmitter does, and $H(Y) \leq 1$ bit with equality when the output is equally likely — which equally likely inputs deliver. So the maximum sits at $p(x_0) = \frac{1}{2}$ and the two pieces subtract. The unit is bits per *use* of the channel, not bits per second. Multiply by the number of uses a second to reach a rate.



Capacity of the binary symmetric channel. The capacity is one bit at $p = 0$ and $p = 1$. At $p = 1$, the receiver restores every bit by inversion. Capacity is zero at $p = \frac{1}{2}$, where the output is independent of the input.



The Z-channel of Example 6.6. Its mutual information peaks at $q = 0.6$, not at a half, which is what an asymmetric channel does: the transmitter should favour the symbol that survives.

The curve is flat near $p = 0$: a channel with one error in a thousand has capacity 0.9886 bits, so the first errors cost almost nothing. It then falls away steeply — at $p = 0.11$ the capacity is already 0.500. Halving the error probability of an already-good channel buys very little, and the effort belongs where the curve is steep.

EXAMPLE 6.6 – THE CAPACITY OF THE Z-CHANNEL

- GIVEN** A 0 always arrives as a 0. A 1 arrives correctly half the time and as a 0 otherwise. Write $P(X = 0) = q$.
- FIND** The capacity, and the input distribution that achieves it.
- METHOD** Nothing is symmetric, so the equally likely input is no longer the right guess. Form $I(X; Y)$ as a function of q and maximise it.
- SOLUTION** $P(Y = 0) = q + \frac{1}{2}(1 - q) = \frac{1}{2}(1 + q)$, and $H(Y | X) = q(0) + (1 - q)(1) = 1 - q$ because a transmitted 0 leaves nothing in doubt. Subtracting gives $I(X; Y) = q - \frac{1}{2}(1 + q) \log_2(1 + q) - \frac{1}{2}(1 - q) \log_2(1 - q)$. Setting the derivative to zero collapses the logarithmic terms to $\frac{1}{2} \log_2 \frac{1-q}{1+q} = -1$, so $\frac{1-q}{1+q} = \frac{1}{4}$ and $q^* = 0.6$.
- CHECK** $C = 0.3219$ bits per channel use, and $0.3219 = \log_2 \frac{5}{4}$ exactly — a check on the arithmetic that costs nothing. The optimum is not the balanced input: the transmitter should send the reliable symbol three times in five. A transmitter that guesses $q = 0.5$ still gets 0.3113 bits, losing about 3%.

THE CHANNEL CODING THEOREM

If the transmission rate satisfies $R_b < C$, there is a coding scheme whose probability of error is as small as required. If $R_b > C$, there is not — no scheme, however long or however clever. **An error-free data rate can never exceed the capacity.** The theorem says a code exists. It does not say what the code is, how long its blocks must be, or what the decoder costs. Finding codes that come close and can also be decoded took the fifty years after it was proved.

6.12 The bandlimited channel

Everything above counted symbols. A real channel is given instead as a bandwidth in hertz, an average transmitted power and a noise density. Its capacity is one of the most quoted results in engineering.

THE INFORMATION CAPACITY LAW

$$C = B \log_2 \left(1 + \frac{P}{N_0 B} \right) \quad \text{bits per second}$$

$$\frac{C}{B} = \log_2 \left(1 + \frac{E_b C}{N_0 B} \right) \implies \frac{E_b}{N_0} = \frac{2^{C/B} - 1}{C/B}$$

B is bandwidth, P is average transmitted power, and $N_0 B$ is in-band noise power. Use $P = E_b C$, where power equals energy per bit times bit rate. Divide by B to obtain the bandwidth efficiency C/B .

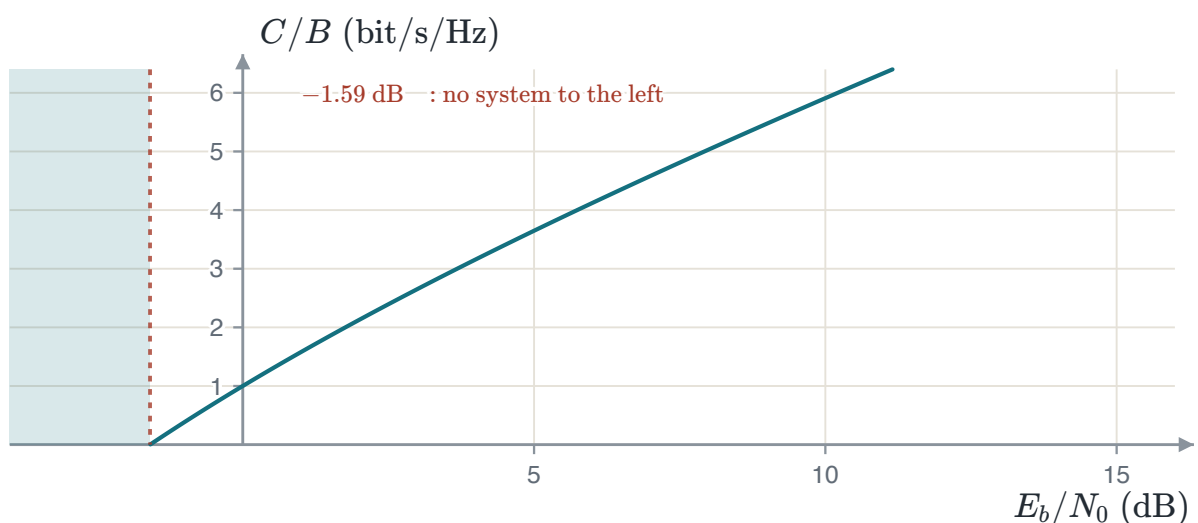
READ THE TWO KNOBS

C grows **linearly** with bandwidth and only **logarithmically** with power. Doubling the bandwidth at fixed noise density roughly doubles the rate. Doubling the power adds one bit per second per hertz at best and much less when the ratio is already large. Bandwidth is the better buy, and it is the one that is scarce and regulated. At $C/B = 1$ the requirement is $E_b/N_0 = 1$, which is 0 dB. At $C/B = 2$ it is 1.5, which is 1.76 dB.

THE SHANNON LIMIT

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{x} = \ln 2, \quad \left. \frac{E_b}{N_0} \right|_{\min} = \ln 2 = 0.693 = -1.59 \text{ dB}$$

Spend bandwidth freely — let $C/B \rightarrow 0$ — and the energy per bit a system needs falls, but not to nothing. No system of any kind communicates reliably below -1.59 dB: not a better code, not a better modulation, not a better receiver.



Bandwidth efficiency against energy per bit. Every working system sits below the curve and to the right of the line. The region to the left of -1.59 dB has never been occupied and never will be. Climbing from one bit per hertz to six costs about 14 dB, and the climb steepens the whole way.

A FLOOR THAT IS APPROACHED AND NEVER TOUCHED

Reaching the limit needs infinite bandwidth, and the rate per hertz goes to zero on the way. Coherent binary PSK needs about 9.6 dB for an error probability of 10^{-5} , so it sits some 11 dB above the floor. Closing that gap is what channel coding was invented for. This course stops at the uncoded schemes, which is where the gap is widest and easiest to see.

6.13 Summary

RESULT	STATEMENT	ANCHOR
Self-information	$I(s_k) = -\log_2 p_k$	PS CH12.1.1
Entropy	$H(S) = -\sum_k p_k \log_2 p_k$	PS CH12.1.1
Bounds	$0 \leq H(S) \leq \log_2 K$	PS CH12.1.1
Extension	$H(S^n) = nH(S)$	PS CH12.1.1
Average length	$\bar{L} = \sum_k p_k l_k, \eta = H(S)/\bar{L}$	PS CH12.2
Source-coding theorem	$\bar{L} \geq H(S)$, so $L_{\min} = H(S)$	PS CH12.2
Kraft	$\sum_k 2^{-l_k} \leq 1$: necessary, not sufficient	PS CH12.3
Prefix bound	$H(S) \leq \bar{L} < H(S) + 1$	PS CH12.2
Huffman	optimal, not unique. High placement gives least variance	PS CH12.3.1

Chapter 1 turned a waveform into bits. Chapters 2 to 5 got those bits across a channel and counted the errors. This chapter asked how few bits there needed to be in the first place, and answered with one number that the source itself decides. Everything between the two is engineering; the entropy is not.

APPENDIX A

Summary of formulas

Everything the course establishes, in the order it establishes it. Nothing here is derived — where a result needs an argument, the argument is in the chapter named beside it.

A.1 From an analog signal to bits — Chapter 1

SAMPLING AND RECONSTRUCTION

$f_s > 2W$ (the sampling theorem, W the highest frequency present)

$$x(t) = \sum_{n=-\infty}^{\infty} x(nT_s) \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right)$$

Below the Nyquist rate the replicas of the spectrum overlap and the overlap cannot be undone. $\operatorname{sinc}(x) = \sin(\pi x)/(\pi x)$ throughout this course.

UNIFORM QUANTIZATION

$$\Delta = \frac{2V}{L} = \frac{2V}{2^n}, \quad |e| \leq \frac{\Delta}{2}$$

$$\sigma_q^2 = \frac{\Delta^2}{12}, \quad \text{SQNR}|_{\text{dB}} = 6.02n + 1.76$$

The 1.76 assumes a full-scale sinusoid. The formula is an approximation at small n . At three bits it gives 19.82 dB where the measured value is 19.09 dB, and the gap halves with each extra bit.

RESULT	STATEMENT	CHAPTER
Companding	μ -law and A-law compress before quantizing so that the signal-to-noise ratio is flat across the range	1
PCM rate	$R = nf_s$ bits a second for n bits a sample	1

A.2 Baseband transmission — Chapter 2

THE MATCHED FILTER

$$h(t) = s(T - t), \quad \left(\frac{S}{N}\right)_{\max} = \frac{2E}{N_0}$$

The matched filter maximizes the signal-to-noise ratio at the sampling instant. The maximum depends on pulse energy, not pulse shape.

BINARY ERROR PROBABILITY AT BASEBAND

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \text{ (antipodal)}, \quad P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right) \text{ (on-off, orthogonal)}$$

NYQUIST CRITERION FOR ZERO INTERSYMBOL INTERFERENCE

$$\sum_k P\left(f + \frac{k}{T}\right) = T \quad \text{for all } f$$

$$B = \frac{1 + \alpha}{2T}, \quad 0 \leq \alpha \leq 1$$

The raised cosine meets the criterion for every roll-off α . At $\alpha = 0$ the bandwidth is the Nyquist minimum $1/2T$ and the pulse decays slowly. At $\alpha = 1$ it is twice that and the pulse decays quickly.

A.3 Signal space — Chapter 3

A SIGNAL AS A POINT

$$s_i(t) = \sum_{k=1}^N s_{ik} \psi_k(t), \quad s_{ik} = \int_0^T s_i(t) \psi_k(t) dt$$

$$E_i = \|\mathbf{s}_i\|^2 = \sum_{k=1}^N s_{ik}^2, \quad d_{ij} = \|\mathbf{s}_i - \mathbf{s}_j\|$$

The basis $\{\psi_k\}$ is orthonormal and comes from the Gram–Schmidt procedure applied to the signal set. Energy becomes a squared length and difference becomes a distance, which is the move the rest of the course rests on.

A.4 The optimal receiver — Chapter 4

THE DECISION RULE

$$\hat{s} = \arg \min_i \left\{ \|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P(\mathbf{s}_i) \right\} \quad (\text{MAP})$$

$$\hat{s} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2 \quad (\text{ML, equal priors})$$

With equal priors the rule is minimum distance. The equivalent correlation form is to maximise $\mathbf{r} \cdot \mathbf{s}_i - E_i/2$.

ERROR PROBABILITY

$$P(\mathbf{s}_k \rightarrow \mathbf{s}_j) = Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right)$$

$$P_e \leq \frac{1}{M} \sum_k \sum_{j \neq k} Q \left(\sqrt{\frac{d_{kj}^2}{2N_0}} \right), \quad P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right)$$

The first result is exact for two points. The second is the union bound. The third is its nearest-neighbor form. Here, $N_{\min}$ is the average number of nearest neighbors and can be noninteger. The loose bound needs only M and $d_{\min}$.

A.5 Digital modulation — Chapter 5

SCHEME	$d_{\min}^2$	$N_{\min}$
BPSK	$4E_b$	1
BFSK, BASK	$2E_b$	1
M -PSK	$4E_s \sin^2(\pi/M)$	2
M -PAM	$12E_s/(M^2 - 1)$	$2(M - 1)/M$
M -QAM (square)	$6E_s/(M - 1)$	3 at $M = 16$, 3.5 at $M = 64$
M -FSK	$2E_s$ for every pair	$M - 1$

READING THE TABLE

$$P_e \approx N_{\min} Q\left(\sqrt{\frac{d_{\min}^2}{2N_0}}\right), \quad E_s = (\log_2 M)E_b$$

Convert once between symbol and bit energy. BFSK and BASK need 3.01 dB more than BPSK for the same error probability. Doubling M costs approximately 6 dB in PAM. Moving from 4-PSK to 8-PSK costs 5.33 dB. At the same energy, 16-QAM gains 9.29 dB over 16-PAM.

A.6 Information theory — Chapter 6

INFORMATION AND ENTROPY

$$I(s_k) = -\log_2 p_k, \quad H(S) = -\sum_{k=1}^K p_k \log_2 p_k$$

$$0 \leq H(S) \leq \log_2 K, \quad H(S^n) = n H(S)$$

The upper bound is reached when all K symbols are equally likely. The extension result holds because the source is memoryless.

SOURCE CODING

$$\bar{L} = \sum_{k=1}^K p_k l_k, \quad \eta = \frac{H(S)}{\bar{L}} \leq 1$$

$$\sum_{k=1}^K 2^{-l_k} \leq 1, \quad H(S) \leq \bar{L} < H(S) + 1$$

The Kraft inequality is necessary but not sufficient for a prefix code. Over the n -th extension the upper bound becomes $H(S) + 1/n$, so $L_n/n \rightarrow H(S)$.

$$\sigma^2 = \sum_{k=1}^K p_k (l_k - \bar{L})^2$$

Huffman is optimal in average length and is not unique. Placing a merged symbol as high as possible on a tie gives the minimum-variance code among the optimal ones.

APPENDIX B

The laboratories

The course has four laboratories, and each one takes a result the chapters derive on paper and asks you to measure it instead. This appendix runs all four. Every curve, cloud and staircase on the pages that follow was computed when this page was drawn. The noise was generated, the filter was applied, the errors were counted. So what you see is the outcome of a run and not a drawing of one.

That matters for one reason above all. A formula tells you what should happen. A measurement tells you what did. Where the two agree you have learnt that the derivation is sound. Where they disagree you have learnt something better: which assumption in the derivation the experiment broke. Each of the four sections below ends on that disagreement, because it is the part of the laboratory worth carrying away.

Your numerical results will differ from these examples. Each group receives a different input number, so its waveform, bit pattern, and source statistics differ. Random noise also changes between runs. Compare curve shapes, gaps, trends, and expected directions instead of individual sample values.

LABORATORY SUMMARY

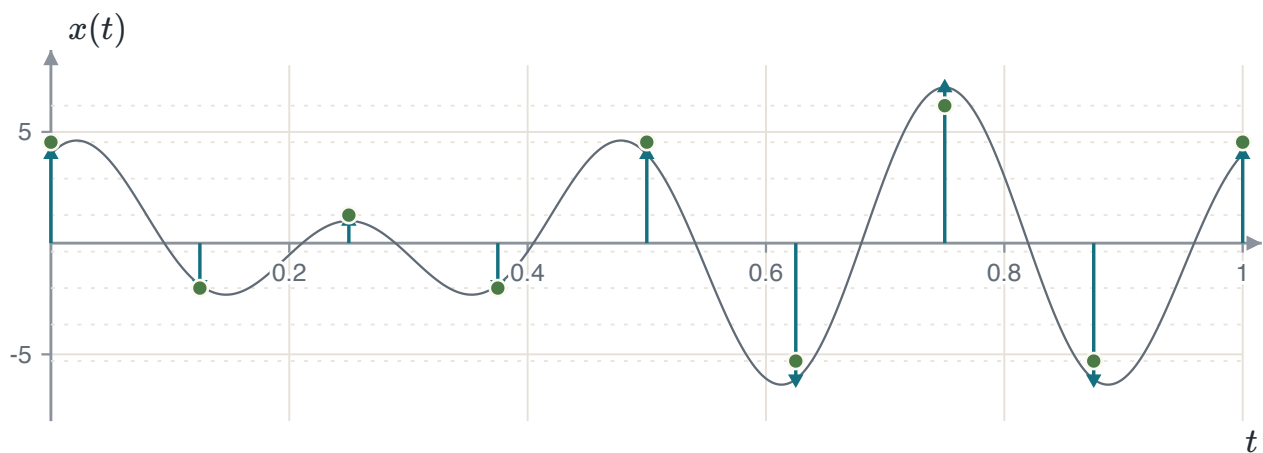
B.1 builds a uniform quantizer, measures SQNR, and encodes the samples as a polar-NRZ waveform. **B.2** measures the bit error rate of a matched-filter receiver and compares it with the Chapter 2 formula. **B.3** measures the symbol error rate of sixteen-point QAM and compares it with the Chapter 4 bound. **B.4** builds a Huffman code and measures its compression.

B.1 Quantization, the signal-to-quantization-noise ratio and PCM

The first laboratory applies the complete Chapter 1 chain. Start with a continuous waveform and sample it at the Nyquist rate. Quantize the samples with a uniform quantizer. Measure the ratio of signal power to quantization-error power. Encode each sample as a binary word and draw the polar-NRZ waveform. Then double the number of levels and repeat the measurements.

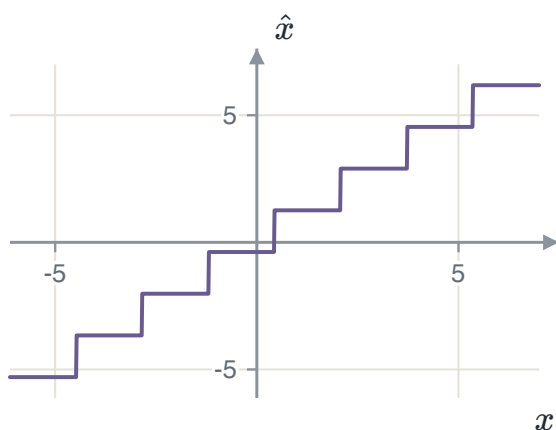
The waveform is $x(t) = f_1 \sin(2\pi f_1 t) + f_2 \cos(2\pi f_2 t)$. Each group receives integer values for f_1 and f_2 . Here, $f_1 = 3$ and $f_2 = 4$, so the Nyquist rate is 8 samples per second. A two-second interval then contains only seventeen samples.

The quantizer itself is four lines. Find the smallest and largest sample. Divide the distance between them into L equal cells. Therefore, the step is $\Delta = (x_{\max} - x_{\min})/L$. Decide which cell each sample falls in. Replace it by the middle of that cell. This is the value that minimises the error inside the cell when nothing is known about where in the cell the sample sits.

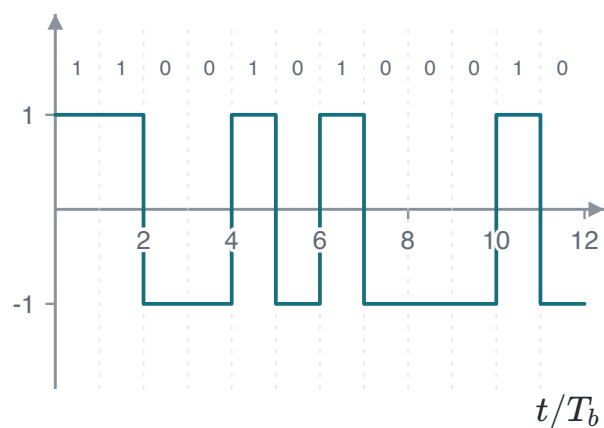


The first second of the run. The pale line is $x(t)$. Stems show samples x_n taken every $1/8$ s. Filled circles show quantized values $\hat{x}_n$ for $L = 8$. Dotted lines show the eight representation levels with spacing $\Delta = 1.640$ V.

With $L = 8$ each sample needs $R = \log_2 8 = 3$ bits, and at $f_s = 8$ samples a second the stream leaves the encoder at $R_b = Rf_s = 24$ bit/s. At $L = 16$ it is 32 bit/s. That is the price of the second experiment, and it is worth naming before the reward is measured: a third more bits a second.



The quantizer itself, drawn by feeding it a thousand values across its range. Eight treads, each $\Delta = 1.640$ V wide, each one tread high.



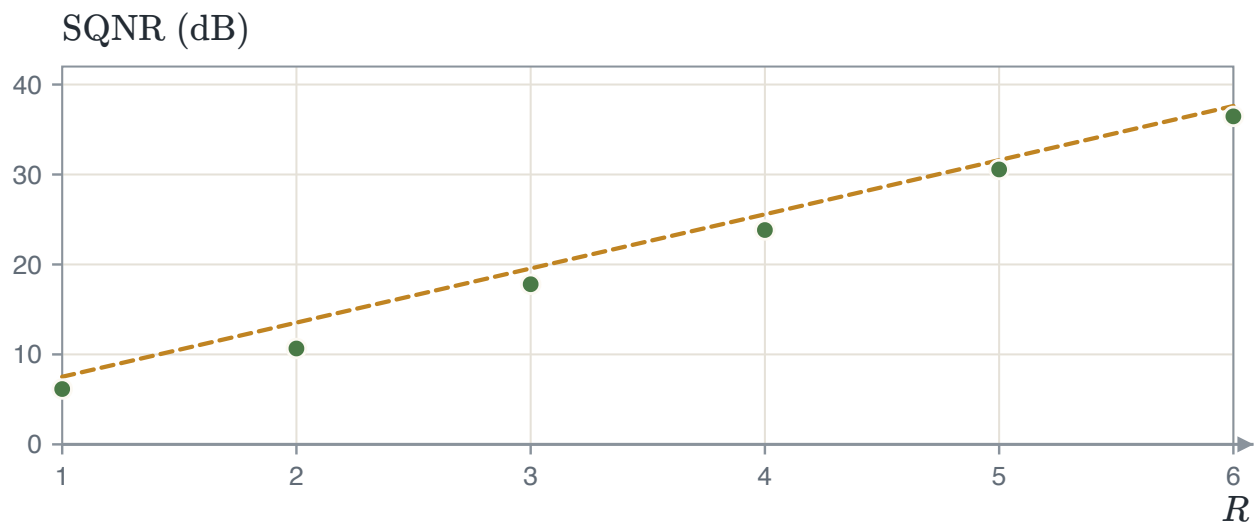
The first four samples encoded and sent. Each sample becomes a three-bit word for its cell, and each bit becomes one bit interval of a polar NRZ waveform. High for a one, low for a zero.

WHAT THE LABORATORY MEASURES

$$\text{SQNR} = \frac{E[X^2]}{E[(X - \hat{X})^2]} \longrightarrow \text{SQNR}|_{\text{dB}} = 10 \log_{10} \text{SQNR}$$

Both averages use the available samples. The numerator is the sample mean-square value. The denominator is the mean-square quantization error. Chapter 1 models this denominator as $\Delta^2/12$. Here, it is measured directly.

On this waveform the measurement gives 17.80 dB at $L = 8$ and 23.83 dB at $L = 16$. The difference is 6.03 dB. This is the six decibels a bit that Section 1.4 predicts, arrived at by counting. That agreement is the main result of the laboratory, and it holds for a reason that has nothing to do with this particular waveform. Doubling L halves Δ , halving Δ quarters $\Delta^2/12$, and a factor of four is 6.02 dB.



Measured against modelled, for one to six bits a sample, both computed from the seventeen samples the laboratory takes. The dashed line is the small-step model $10 \log_{10}(P_X/(\Delta^2/12))$ and the circles are the ratio actually measured. The circles rise about six decibels a bit, as they should. But they sit below the line by between 1.0 and 2.9 dB, and the gap neither closes nor settles as R grows. Something is wrong, and the next note says what.

SEVENTEEN SAMPLES IS NOT A POPULATION

The measured points and the modelled line disagree by up to 2.9 dB, and the gap does not shrink as L grows. It is tempting to blame the model, and Section 1.4 gives a real reason it could be blamed. The error is not quite uniform across a cell. But measure the same waveform with two thousand samples instead of seventeen and the gap falls to less than 0.6 dB everywhere. The model was right. The estimate was noisy. Both $E[X^2]$ and $E[(X - \hat{X})^2]$ are averages, and an average over seventeen numbers has a spread of its own. This is the error to name in your report. Do not draw a conclusion about a model from a sample too small to test it.

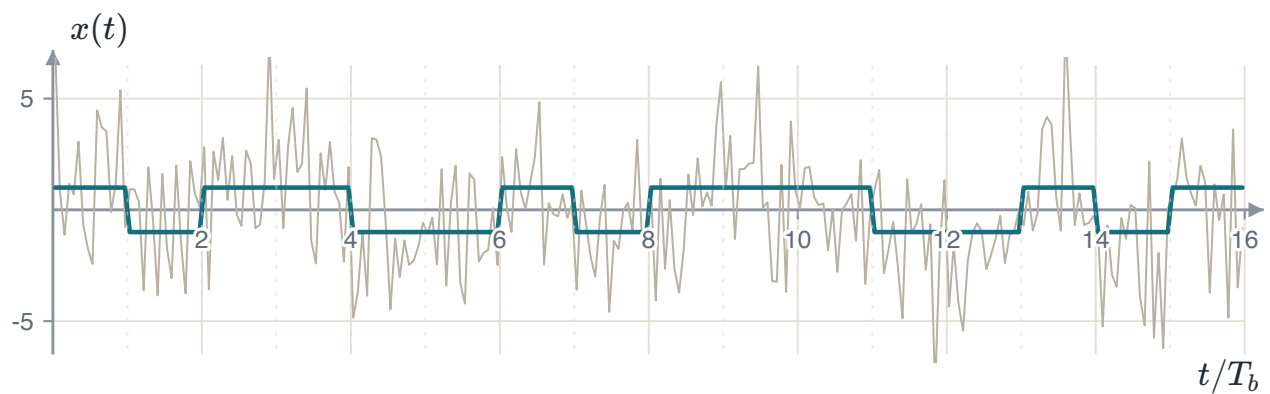
SAMPLING EXACTLY AT THE NYQUIST RATE

The waveform here has a 4 Hz component and the laboratory samples at exactly 8 Hz. This is the boundary the sampling theorem is stated at rather than inside. At exactly twice its frequency, a cosine is sampled at the same two points of every cycle. Therefore, that component contributes the same alternating ± 4 to every sample and none of its shape survives. The reconstruction is still correct here because the component is a cosine and lands on its peaks. Move the phase and it would not be. Chapter 1 asks for $f_s > 2W$ and means the strict inequality.

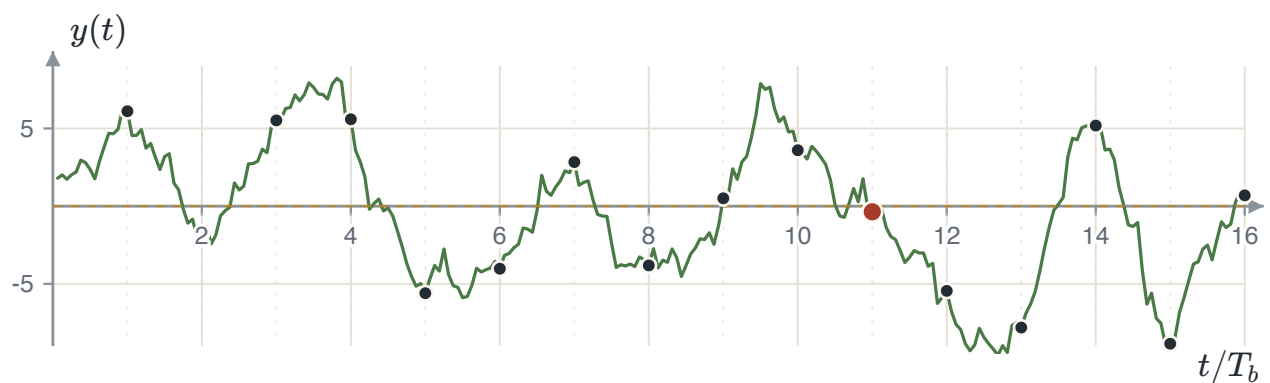
B.2 The matched filter and the bit error rate

The second laboratory builds the Chapter 2 receiver. Convert a bit stream into a waveform and add white Gaussian noise. Apply the matched filter and sample its output once per bit. Compare each sample with the decision threshold. Then compare the measured error rate with $Q(\sqrt{2E_b/N_0})$.

The signaling is antipodal. The pulse $\psi(t) = 1/\sqrt{T_b}$ has unit energy over one bit interval. A bit scales this pulse by $\pm A\sqrt{T_b}$. Thus, the transmitted waveform is constant at $+A$ or $-A$ during each interval. Its bit energy is $E_b = A^2 T_b$. Add sample noise with variance $N_0/2$ and use $10 \log_{10}(E_b/N_0)$.

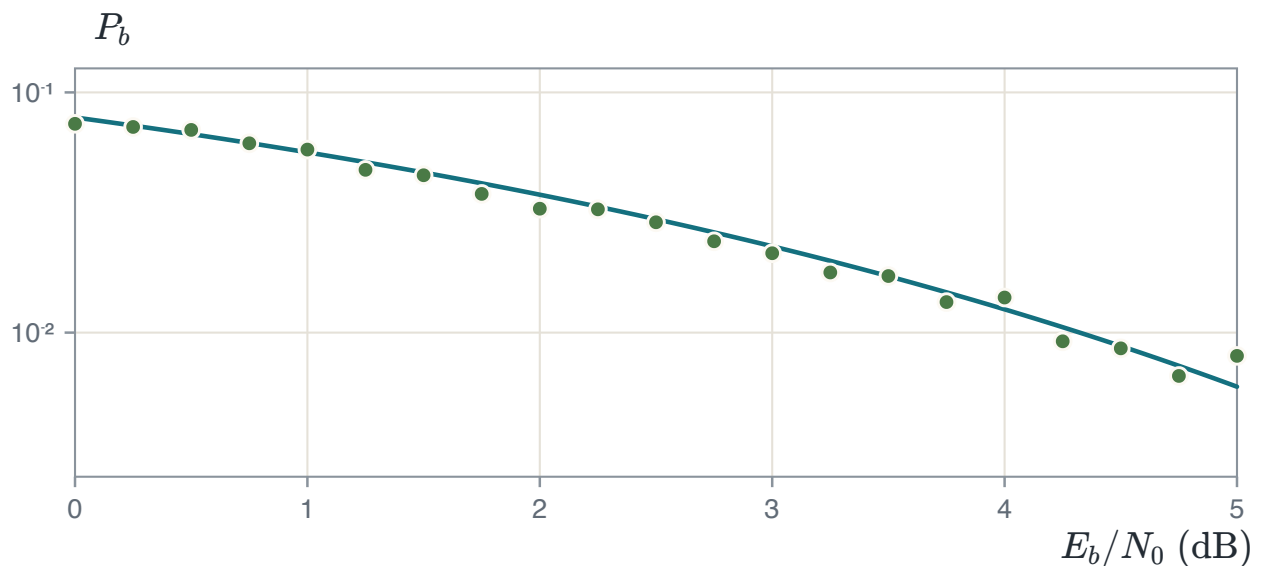


Sixteen bits at $E_b/N_0 = 2$ dB. The clean square wave is the transmitted $s(t)$. The hairline around it is $x(t) = s(t) + w(t)$. This is everything the receiver is given. Nothing about the square wave is visible in the hairline at this noise level, and that is the honest picture. The receiver does not see the signal, it computes with it.



The matched-filter output for the same sixteen bits, read once at the end of every bit interval. The dashed line is the threshold $\lambda = 0$. This is where equal priors put it. Fifteen of the sixteen readings are on the correct side of it. The eleventh is not, and it is drawn in red. At 2 dB the formula expects about one wrong bit in twenty-seven, and this run produced one in sixteen.

One run of sixteen bits settles nothing. One error could as easily have been none or three. The second part of the laboratory therefore drops the picture and repeats the experiment. Five thousand bits at each signal-to-noise ratio from 0 to 5 dB in quarter-decibel steps, counting errors and dividing by five thousand.



Measured bit error rate against the formula, five thousand bits a point. The line is $Q(\sqrt{2E_b/N_0})$ and the circles are counted errors. Nowhere on the sweep do the two differ by more than 0.005, and yet the circles clearly wander further from the line towards the right. The next note is about why those two statements are both true.

MONTE CARLO SCATTER

Counting k errors in N bits gives a standard spread of approximately $\sqrt{p(1-p)/N}$. At 0 dB, $p \approx 0.079$, and five thousand bits give a spread of 0.004. At 5 dB, the absolute spread decreases, but it becomes a larger fraction of p . A logarithmic plot shows this relative scatter. Measuring probabilities near 10^{-6} therefore requires millions of trials.

THREE-DECIBEL CONVENTION ERRORS

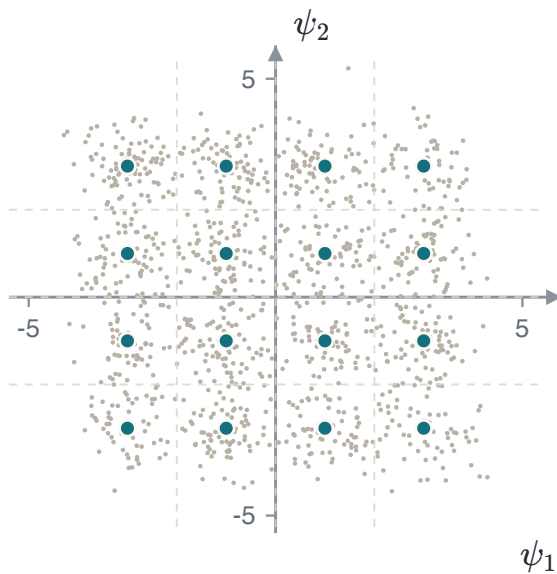
Two convention errors each give a factor of two. First, the sample-noise variance is $N_0/2$ for a two-sided density $N_0/2$. Using N_0 shifts the measured curve by 3 dB. Second, the formula $Q(\sqrt{2E_b/N_0})$ requires antipodal signaling. On-off signaling instead uses $Q(\sqrt{E_b/N_0})$. A parallel 3 dB shift usually indicates one of these errors.

B.3 Sixteen-point QAM and the union bound

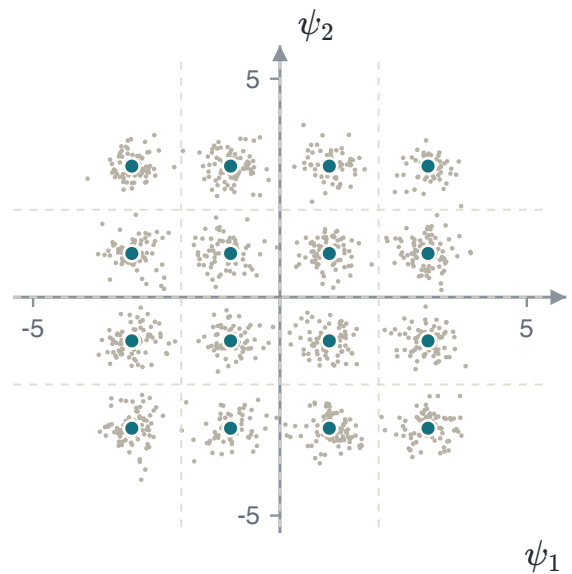
The third laboratory uses a two-dimensional sixteen-point square constellation. Add independent noise on each axis. The receiver selects the nearest point. Compare the measured symbol error rate with the nearest-neighbor approximation from Section 4.4.

Write each point as $s_I + js_Q$, with coordinates from $\{-3, -1, 1, 3\}$. These units give $E_s = 10$ and $d_{\min} = 2$. Each symbol carries four bits, so $E_b = E_s/4 = 2.5$. Section 5.4 gives $d_{\min}^2 = 6E_s/(M-1) = 4$. This result agrees with the point spacing.

The same constellation gives the optimal thresholds. Independent axes and equal level spacing produce three vertical and three horizontal boundaries. Each axis uses $\lambda \in \{-2, 0, 2\}$. Thus, nearest-point detection reduces to nearest-level detection on each axis.



$E_b/N_0 = 6$ dB. The clouds touch, so a large fraction of the symbols land across a boundary. The measured symbol error rate here is 0.108: about one symbol in nine is decided wrongly.



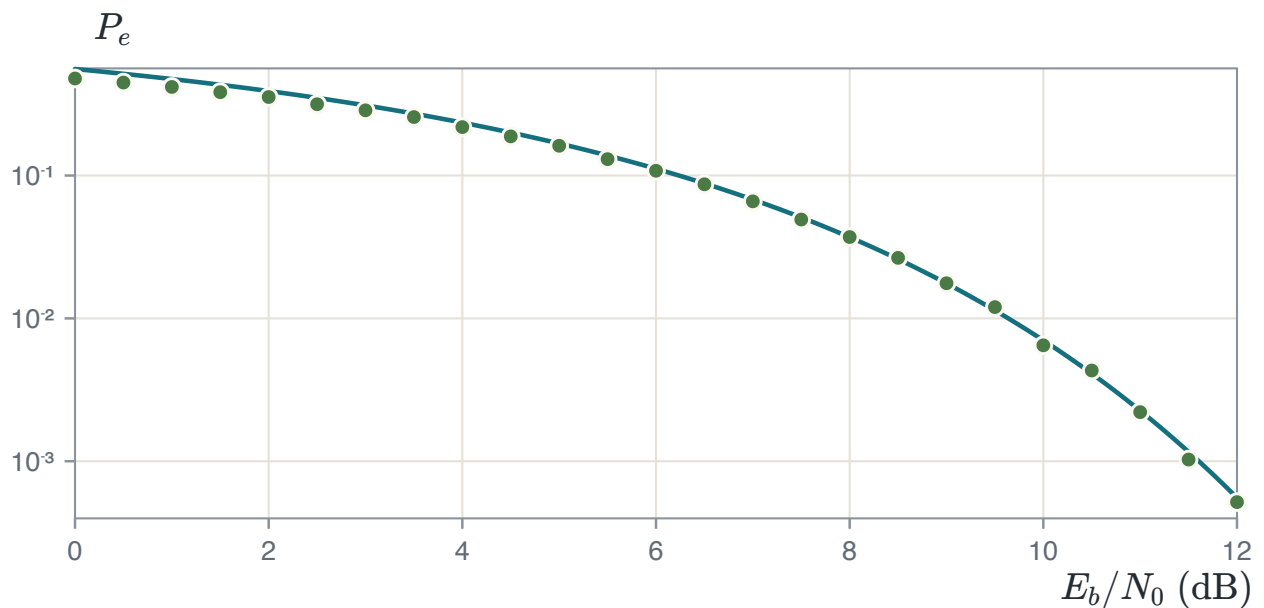
$E_b/N_0 = 11$ dB. Five decibels more energy a bit, and the clouds have pulled apart into sixteen separate blobs. The measured rate falls to 0.0022, about fifty times better.

Twelve hundred received symbols are drawn in each picture so that the clouds can be seen. The rates quoted beneath them do not come from those twelve hundred. A rate of 0.0022 measured on twelve hundred symbols would be three errors, and three is not a measurement. They come from the sweep below. This uses fifty thousand symbols at 6 dB and a quarter of a million at 11 dB. Choosing how many symbols to run is part of the experiment, and Section B.2 said why.

WHAT THE BOUND PREDICTS

$$P_e \approx N_{\min} Q \left(\sqrt{\frac{d_{\min}^2}{2N_0}} \right), \quad N_{\min} = 3, \quad d_{\min}^2 = 4$$

$N_{\min}$ is the average number of nearest neighbours: four corner points have two each, eight edge points three, and four inner points four, and $(4 \cdot 2 + 8 \cdot 3 + 4 \cdot 4)/16 = 3$. It is an average over the constellation and it does not have to be a whole number.



Measured symbol error rate against the nearest-neighbour approximation. The number of symbols is raised as the rate falls. Fifty thousand at the left, four hundred thousand at the right. So that every circle rests on a few hundred errors. The circles sit below the line at the left and join it by about 8 dB. That gap is not an error in either the simulation or the formula. It is what the approximation is.

MEASURED ERROR AND THE UNION BOUND

The union bound adds all pairwise error probabilities. These events overlap, so the sum can count one noise vector more than once. The bound is therefore larger than the exact union probability. At low signal-to-noise ratio, overlap makes the bound loose. At high ratios, nearest-neighbor errors dominate and overlap becomes small. Remaining differences between simulated points and the bound can come from finite trial counts.

SYMBOL AND BIT ENERGY

For sixteen-point QAM, $E_s = 4E_b$. Convert between these energies once when calculating N_0 . Do not calculate N_0 from one energy and use the other in the Q function. This error shifts the complete curve by $10 \log_{10} 4 = 6.02$ dB.

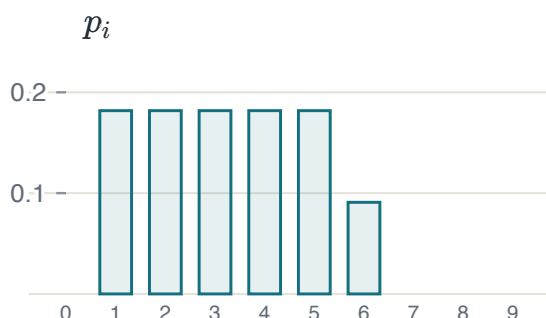
B.4 Huffman coding and how much it compresses

The fourth laboratory studies lossless source coding. A source emits symbols with known probabilities. Entropy gives the minimum average bits per symbol. Huffman coding gives the best single-symbol prefix code. The laboratory measures both values.

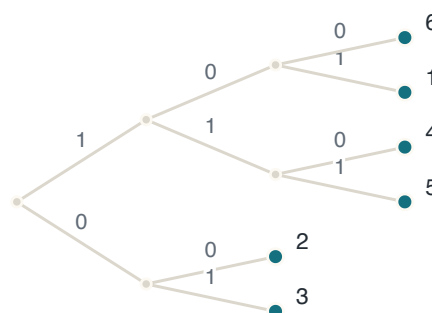
The source alphabet is the ten digits. Your group counts how often each digit occurs in a number it is given and divides by how many digits there are. This gives a probability for each. The case run here uses the eleven digits 1 2 3 4 5 6 5 4 3 2 1. Five digits occur twice and one occurs once. Therefore, five symbols have probability $2/11$, one has $1/11$, and four have probability zero and take no codeword at all.

DIVIDING BY THE WRONG TOTAL

The probabilities must come from dividing each count by the *number of symbols*. Dividing by anything else. The sum of the digit values, say. Gives a set of numbers that does not add up to one, and everything downstream is then quietly wrong. The entropy is not an entropy, the codebook is built for a source that does not exist, and the efficiency can come out above 100%. Print the sum of your probability vector and confirm it is one before you compute anything from it. It is one line, and it catches the only mistake in this laboratory that produces plausible numbers instead of an error message.



The source statistics p_i over the ten-digit alphabet. Four digits do not occur, and a symbol of probability zero is left out of both the entropy and the codebook.



The code the algorithm builds. Reading a path from the root, left is a 0 and right is a 1. The digit at each leaf is the symbol that path spells.

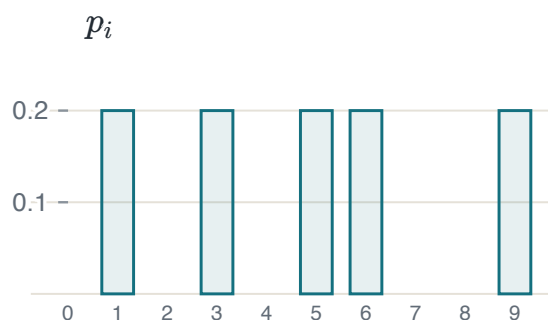
This source has entropy $H = 2.550$ bits per symbol. Its Huffman code has average length $\bar{L} = 2.636$, so $\eta = 96.7\%$. First, check the Section 6.6 bound $H \leq \bar{L} < H + 1$. Here, $2.550 \leq 2.636 < 3.550$. Second, compare $\bar{L}$ with the three-bit fixed length for six symbols and the four-bit length for ten symbols.

A TIE IN THE ALGORITHM IS NOT A MISTAKE

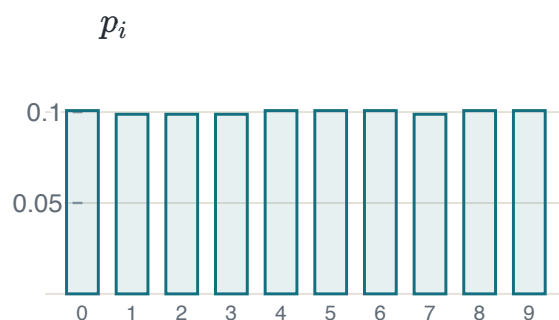
Whenever two nodes have the same probability, the algorithm may merge either, and different choices give different codebooks with the same average length. Your codewords will not match the ones drawn above and need not. What must match is $\bar{L}$, because Huffman is optimal and the optimal average length is unique even when the code is not. If two groups get different average lengths for the same statistics, one of them has an error. If they get different codebooks, neither of them does. Section 6.7 develops the tie-breaking rule that also minimises the variance of the codeword length.

The second part encodes one thousand symbols from the same distribution. Four-bit fixed-length coding uses 4000 bits. Huffman coding uses 2621 bits, so the measured compression is 34.5%. The average length predicts $1 - \bar{L}/4 = 34.1\%$. These values need not match exactly because the finite sequence is one random sample. A large difference can indicate an incorrect codebook.

The third part asks what the source statistics themselves are worth. Ten counts are drawn from a binomial trial with 50 attempts and success probability ρ , then normalised into a distribution. At $\rho = 0.01$ almost every attempt fails, so most counts come out zero and the surviving symbols are few. At $\rho = 0.99$ almost every attempt succeeds, so all ten counts come out near 50 and the distribution is nearly flat.



$\rho = 0.01$. Five symbols survive and the rest have probability zero. $H = 2.322$ bits, $\bar{L} = 2.400$ bits.



$\rho = 0.99$. All ten symbols occur about equally often. $H = 3.322$ bits, $\bar{L} = 3.395$ bits.

SOURCE	H	$\bar{L}$	η	COMPRESSION AGAINST 4 BITS
The digit source	2.550	2.636	96.7%	34.1%
$\rho = 0.01$	2.322	2.400	96.7%	40.0%
$\rho = 0.99$	3.322	3.395	97.8%	15.1%

EFFICIENCY AND COMPRESSION

The coding efficiency changes from 96.7% to 97.8%, while compression decreases from 40% to 15%. Efficiency measures the gap between the code length and the source entropy. Compression compares the code length with a four-bit fixed-length code. At $\rho = 0.99$, the source is nearly uniform and has entropy 3.322 bits. Therefore, little lossless compression remains possible.



Digital Communications · Lecture Notes

VERSION	DATE	CHANGES
v0	24 September 2026	First edition.

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